Graphical Abstract

A Stabilized Vector Nodal Meshless Method for $H(\mathbf{curl})$ Electromagnetic Problems on Irregular Point Clouds

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Highlights

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- Complete $\mathcal{P}^1$ basis overcomes curl stagnation of $\mathcal{L}^1$ VNMM.
- Dimensionless matrix $\widehat{\mathbf{A}}$ prevents support dilation as $h \to 0$.
- Local equivalence to Whitney 1-forms proved on conforming simplices.
- Dynamic meshless supports induce global non-conformity in $H(\mathbf{curl})$.
- Div-curl regularized weak form suppresses gradient leakage and spurious modes.

A Stabilized Vector Nodal Meshless Method for $H(\mathbf{curl})$ Electromagnetic Problems on Irregular Point Clouds

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Abstract

The Vector Nodal Meshless Method (VNMM) provides a mesh-connectivity-free framework for vector-field approximation on unstructured point clouds. However, the conventional formulation based on Nédélec's incomplete solenoidal polynomial space ($\mathcal{L}^1$) exhibits a persistent $\mathcal{O}(1)$ error plateau in curl approximation when applied to arbitrary node distributions. This work resolves this limitation through a two-level framework addressing both local polynomial consistency and global variational stability. Locally, the incomplete $\mathcal{L}^1$ basis is replaced by the complete linear vector space $\mathcal{P}_1 \times \mathcal{P}_1$, capturing all components of the first-order field Jacobian, removing derivative aliasing, and restoring first-order ($\mathcal{O}(h)$) curl convergence. Degenerate local stencils are rejected through scale-aware determinant qualification on a dimensionless placement matrix $\widehat{\mathbf{A}}$ combined with k-d tree searches. Globally, we demonstrate that while VNMM exhibits an exact constructive equivalence to lowest-order Whitney edge elements on conforming simplices, dynamic support selection on unconstrained point clouds produces a globally non-conforming approximation. Consequently, the approximation does not inherit the exactness of the discrete de Rham sequence, allowing discrete gra-

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dient fields to produce non-zero numerical curl residues and cause gradient nullspace leakage in eigenvalue problems. We resolve this by incorporating a div-curl regularized Ritz-Galerkin formulation that shifts spurious modes away from the operational spectrum, with no spurious modes detected in the tested operational bandwidth. The findings establish that integrating complete local vector polynomials, scale-invariant automated support selection, and variational div-curl regularization provides a robust, stabilized meshless framework for electromagnetic analysis on unstructured point clouds.

Keywords: Meshless methods, Vector Nodal Meshless Method (VNMM), $H(\mathbf{curl})$ approximation, Spurious modes, Resonant cavities, Electromagnetics

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1. Introduction

This section provides the contextual framework, fundamental motivations, and key scientific challenges that drive the development of advanced meshless methods for computational electromagnetics. First, the historical evolution and practical advantages of meshless paradigms over traditional grid-based methods are outlined. Subsequently, the mathematical bottlenecks of earlier vector meshless formulations—specifically derivative stagnation and support instability—are detailed, highlighting the necessity for the stabilized Vector Nodal Meshless Method (VNMM) proposed in this work.

1.1. Context and Motivation

The mathematical modeling of electromagnetic phenomena governed by Maxwell's partial differential equations plays a pivotal role in modern science and engineering applications [1, 2]. Because exact analytical solutions are strictly restricted to canonical geometries and homogeneous media, numerical discretization methods have become indispensable tools for simulating complex electromagnetic devices, waveguiding structures, and resonant cavities [1, 3].

Historically, spatial discretization in computational electromagnetics (CEM) originated with finite difference techniques, notably Yee's staggered Finite-Difference Time-Domain (FDTD) grid [4], and the conventional node-based scalar Finite Element Method (FEM) [1]. In node-based FEM formulations, vector field variables (such as the electric field $\vec{E}$ or magnetic field $\vec{H}$) are discretized by assigning independent scalar degrees of freedom

(DoFs) to each Cartesian component (E_x, E_y, E_z) at shared mesh vertices [1, 5]. While highly successful for scalar Helmholtz and Poisson problems, applying node-based scalar approximations directly to vector Maxwell equations leads to fundamental mathematical pathologies [5, 6]:

1. **Violation of Tangential Interface Continuity:** Physical vector fields across dielectric or magnetic material boundaries require continuous tangential components ($\mathbf{n} \times [\vec{E}] = \mathbf{0}$) while allowing normal components to exhibit a discontinuity [1, 3]. Node-based scalar shape functions enforce C^0 continuity across all vector components simultaneously, artificially locking normal jumps or under-specifying tangential continuity [5].
2. **Field Singularities at Sharp Edges:** Node-based approximations fail to correctly capture singular field behavior near sharp conducting re-entrant corners and edges [3, 6].
3. **Spectral Contamination by Spurious Modes:** Continuous scalar approximations do not satisfy the exact discrete de Rham complex property ($\mathrm{Im}(\nabla) \subset \ker(\nabla\times)$) [3, 5]. Consequently, the infinite-dimensional nullspace of irrotational gradient fields ($\vec{E} = \nabla\phi$) leaks into the physical discrete spectrum, generating non-physical, highly oscillatory *spurious modes* interspersed among true physical resonances [5, 6].

To overcome these severe flaws, Nédélec [7] introduced mixed vector finite elements—commonly designated as Whitney 1-forms or Edge Finite Element Methods (EFEM) [3, 8]. As extensively detailed by Bossavit [8], by associating degrees of freedom with line integrals of field circulation $e_k = \int_{E_k} \vec{E} \cdot d\boldsymbol{\ell}$ along element edges rather than scalar nodal points, EFEM constructs vector shape functions natively conforming in the Sobolev space $H(\mathbf{curl}; \Omega)$. EFEM naturally satisfies tangential interface continuity, models edge field singularities, and preserves the discrete de Rham sequence, completely eliminating spurious modes from electromagnetic spectrum calculations [3, 5, 8].

Despite the undeniable success of EFEM, all mesh-based formulations remain heavily burdened by the pre-requisite of generating high-quality 3D conformal meshes [9, 10]. For problems involving complex CAD geometries, moving interfaces (such as in rotating electrical machinery), severe spatial deformations, or adaptive mesh refinement, the computational overhead of mesh generation, remeshing, and element distortion repair represents a major bottleneck [9, 10, 11].

To liberate numerical solvers from topological mesh constraints, *Meshless (or Meshfree) Methods* emerged as a powerful computational paradigm in solid mechanics, fluid dynamics, and thermodynamics [10, 12, 13]. By constructing field approximations directly over arbitrary clouds of points without pre-arranged element connectivity, meshless methods eliminate mesh generation costs and facilitate seamless adaptive node insertion [10, 14]. Prominent weak-form meshless paradigms include the Element-Free Galerkin (EFG) method [14], based on Moving Least Squares (MLS) approximations over background quadrature cells; the Meshless Local Petrov-Galerkin (MLPG) method [15, 13], utilizing local quadrature subdomains to achieve a "truly meshless" formulation; the Point Interpolation Method (PIM) and Radial Point Interpolation Method (RPIM) [16, 12]; and the Reproducing Kernel Particle Method (RKPM) [10].

Meshless formulations have demonstrated significant advantages over traditional grid-based methods in low-frequency electromagnetics, particularly for modeling deforming domains, moving boundaries, and transient phenomena in electrical machines without requiring complex remeshing procedures [17, 18, 19]. Early attempts to adapt meshless techniques to computational electromagnetics—such as MLS-RKPM formulations [17], EFG schemes [20, 18], and MLPG vector formulations [13, 21, 22]—primarily constructed scalar meshless shape functions for individual vector components (E_x, E_y, E_z). However, treating vector components independently inherited all the historical pathologies of scalar node-based FEM: appearance of non-physical spurious modes, difficult imposition of tangential interface conditions across material discontinuities, and inflated system matrix dimensions requiring auxiliary divergence-free constraints or Lagrange multipliers [23, 24, 25].

To establish a truly $H(\mathbf{curl})$-conforming meshless vector framework, Lima and Mesquita [26] developed the *Edge Meshless Method* (EMM). The EMM distributes directed spatial line segments (edges) throughout the domain and constructs vector shape functions based on $H(\mathbf{curl})$ space properties [26, 24].

Serving as a crucial bridge toward higher-order vector meshless representations, Silva, Mesquita, and Lima [27] introduced $H(\mathbf{curl})$-conforming vector shape functions for the EMM based on Nédélec's incomplete solenoidal $\mathcal{L}^1$ polynomial space [7]. By refining local circulation projections along support edges, their formulation ensured exact tangential field continuity across material interfaces, divergence-free behavior, and complete suppression of spurious modes. Building upon these vectorial shape function concepts, Or-

tiz et al. [28, 29, 25] established higher-order expansions and investigated the asymptotic limit as physical edge lengths shrink to zero ($L_k \to 0$), giving rise to the Vector Nodal Meshless Method (VNMM). Subsequent developments also extended the VNMM framework to three-dimensional vector problems [30, 31] and periodic electromagnetic structures [32].

In VNMM, the 1D continuous edge degenerates into a point-collocated vector node $(P_i, \vec{t}_i)$, where each node P_i is assigned a unit vector director $\vec{t}_i$ [29, 25]. On physical boundaries and material interfaces, $\vec{t}_i$ is aligned strictly tangent to the boundary, enabling direct, unconstrained nodal collocation of essential PEC conditions ($e_i = \vec{E}(P_i) \cdot \vec{t}_i = 0$) and interface field continuity [29, 31]. By associating a single scalar directional projection DoF ($e_i = \vec{E}(P_i) \cdot \vec{t}_i$) with each node, VNMM combines the lightweight single-DoF-per-node efficiency of scalar meshless methods with the local $H(\mathbf{curl})$ vector regularity inspired by Whitney edge elements [29, 30, 31]. When nodes and directors align with the midpoints and edges of a conforming simplicial triangulation, the VNMM reproduces Whitney 1-forms exactly, inheriting global $H(\mathbf{curl})$ conformity. On arbitrary, unconstrained point clouds, however, the formulation operates as a non-conforming approximation composed of locally regular vector polynomials across dynamically chosen supports.

Table 1 summarizes the structural distinctions across prominent grid-based and meshless paradigms in computational electromagnetics. Compared to traditional scalar-component meshless schemes (such as EFG, MLPG, RPIM, and RBF-FD) that require auxiliary constraints or divergence cleaning to manage vector interfaces, the VNMM retains the intrinsic efficiency of a single scalar DoF per node while enforcing tangential continuity and PEC conditions directly at nodal boundaries through its director kinematics. Furthermore, by transitioning from the incomplete solenoidal $\mathcal{L}^1$ space to the complete affine vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$, the formulation proposed in this paper resolves the full local Jacobian, eliminating the $\mathcal{O}(1)$ derivative aliasing of earlier vector meshless implementations.

1.2. The Curl Stagnation Bottleneck in Arbitrary Supports

Despite the mathematical elegance of the original VNMM [29, 25], early formulations were fundamentally tied to structured support selection strategies. In foundational studies [29, 25, 30], candidate support triads $\{P_i, \vec{t}_i\}_{i=1}^3$ were constructed by referencing background spatial structures, such as macro-element planar subdivisions or Delaunay triangulations. While effective for

Table 1: Conceptual comparison of spatial discretization paradigms in computational electromagnetics.

Discretization Method	**Mesh Connectivity**	**Field Variable Representation**	**$H(\mathbf{curl})$ Interface Continuity**	**Spurious-Mode Control Mechanism**	**Curl Operator Convergence on Point Clouds**
Edge FEM (EFEM) [7, 3]	Conformal mesh	Edge circulation $e_k = \int_{E_k} \vec{E} \cdot d\boldsymbol{\ell}$	Natively conforming ($\mathbf{n} \times [\vec{E}] = \mathbf{0}$)	Exact discrete de Rham complex	$\mathcal{O}(h^1)$ (simplex-based)
Scalar EFG / MLS [14, 20]	None (Point cloud)	Component splitting (E_x, E_y)	Artificial (C^0 normal locking / penalty)	Auxiliary divergence penalty or projection	$\mathcal{O}(h^p)$ (high-order scalar MLS)
Vector MLPG [13, 22]	None (Local subdomains)	Component splitting (E_x, E_y)	Lagrange multipliers / penalty	Divergence-free projection / penalty	$\mathcal{O}(h^p)$ (local MLS integration)
RBF-FD / RPIM [12, 11]	None (Point cloud)	Component splitting (E_x, E_y)	Collocation / ghost node penalty	Divergence cleaning / penalty	$\mathcal{O}(h^p)$ (radial basis differentiation)
Original VNMM ($\mathcal{L}^1$) [29, 25]	Planar / Delaunay subdivision	Pointwise projection $e_k = \vec{E}(P_k) \cdot \vec{t}_k$	Natural tangential director alignment	Conditional (restricted to structured supports)	$\mathcal{O}(1)$ stagnation on arbitrary clouds
Proposed VNMM ($\mathcal{P}^1$)	None (Arbitrary point clouds)	Pointwise projection $e_k = \vec{E}(P_k) \cdot \vec{t}_k$	Natural tangential director alignment	Div-curl Ritz-Galerkin penalty ($s_{\text{div}} \in [4.0, 10.0]$)	Consistent $\mathcal{O}(h^1)$ via full Jacobian space

canonical geometries, reliance on underlying meshes undermines the core principle of meshless methods: unconstrained point-cloud discretization [10, 31].

When attempting to liberate the VNMM from spatial meshes by performing automated nearest-neighbor searches (e.g., k-d tree queries) over arbitrary unstructured point clouds, a severe mathematical limitation emerges: **for the arbitrary point-cloud collocation setting investigated here, the numerical approximation of the curl operator $(\nabla \times \vec{E}^h)_z$ fails to converge, exhibiting an $\mathcal{O}(1)$ relative error plateau under spatial grid refinement** [31].

While conforming edge elements enforce circulation cancellation along cell boundaries via Stokes' theorem [7, 3], point-collocated meshless schemes evaluate fields through pointwise projections without closed-contour integrals [31]. In the incomplete 3-node solenoidal space $\mathcal{L}^1$, the omission of dilatational gradient components ($\partial E_x/\partial x$, $\partial E_y/\partial y$) prevents local shape functions from capturing the complete first-order field variation. Under unstructured collocation, these unrepresented modes contaminate the rotational degree of freedom through the inverted placement matrix, producing a persistent $\mathcal{O}(1)$ modal aliasing noise floor under refinement. The mathematical origin and asymptotic scaling of this aliasing mechanism are analyzed in Section 2.2.5.

Furthermore, liberating the approximation from background meshes reveals a fundamental distinction in meshless vector electromagnetics:

$$\boxed{\text{local } H(\mathbf{curl}) \text{ regularity} \neq \text{global } H(\mathbf{curl}) \text{ conformity.}} \tag{1}$$

Although locally assembled vector shape functions are $H(\mathbf{curl})$-conforming over their support domains, active stencils must be queried dynamically across evaluation points. Consequently, tangential traces across adjacent support boundaries do not cancel exactly on arbitrary point clouds, rendering the global approximation space non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$). In electromagnetic eigenvalue problems, this loss of global conformity implies that the approximation does not inherit the exactness of the discrete de Rham sequence; discrete gradient fields fail to maintain an identically vanishing curl under numerical quadrature ($\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$), allowing irrotational gradient modes to leak into the physical spectrum (gradient leakage) and generate non-physical spurious modes. Overcoming both challenges therefore requires an integrated two-level methodology: completing the local polynomial space ($\mathcal{L}^1 \to \mathcal{P}^1$) to resolve the local Jacobian and eliminate derivative aliasing, coupled with a div-curl regularized Ritz-Galerkin weak formulation to control global gradient leakage.

1.3. Scope and Main Contributions of the Paper

This paper presents a stabilized Vector Nodal Meshless Method (VNMM) designed for stable vector field approximations on unconstrained point clouds without mesh connectivity. The core methodological contributions of this work are fourfold:

1. **Elimination of Modal Aliasing via Complete $\mathcal{P}^1$ Vector Basis:** Adopting the complete linear vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ resolves all four independent components of the local field Jacobian $\mathbf{J}^h = \nabla \vec{E}^h$ (Section 2.3), eliminating the truncation mechanism responsible for curl stagnation in unconstrained point clouds.
2. **Scale-Invariant Support Selection via Dimensionless Placement Matrix $\widehat{\mathbf{A}}$:** To enable stable, mesh-free operation on unstructured node distributions without scale-induced support dilation as $h \to 0$, we establish a geometric singularity classification in $\mathbb{R}^{6\times 6}$ (Sections 3.1 and 3.2) and eliminate the quartic spatial scaling dependence $\det(\mathbf{A}) \sim \mathcal{O}(h^4)$ by formulating the support selection criterion in terms of the scale-invariant, dimensionless placement matrix $\widehat{\mathbf{A}}$ ($|\det(\widehat{\mathbf{A}})| \geq \widehat{\mathrm{Tol}} > 0$). By normalizing local coordinates by the local nodal spacing h_{local}, this formulation decouples algebraic qualification from physical mesh dimensions, unifies regular and irregular grids under a single criterion, empirically bounds the local condition number in the tested stochastic configuration, and prevents scale-induced support dilation as $h \to 0$ (Section 3).
3. **Rigorous Local Constructive Equivalence Proof:** We establish the Local Dual Equivalence Theorem (Section 2.2.3), formally proving the local constructive equivalence under conforming midpoint/tangent placement between the local VNMM $\mathcal{L}^1$ shape functions and lowest-order Whitney 1-forms. This rigorously delineates the exact boundary where mesh-conforming topology ensures global $H(\mathbf{curl})$ conformity versus arbitrary point clouds where dynamic stencils induce global non-conformity.
4. **Variational Div-Curl Regularization for Non-Conforming Meshless Discretizations:** To address the gradient leakage stemming from global non-conformity ($V_h \not\subset H(\mathbf{curl};\Omega)$) on arbitrary point clouds, we incorporate a div-curl regularized Ritz-Galerkin weak formulation ($\mathbf{K} = \mathbf{K}_{\mathrm{curl}} + s_{\mathrm{div}}\mathbf{K}_{\mathrm{div}}$, Section 4), providing reliable spec-

tral stabilization and spurious-mode suppression without introducing auxiliary Lagrange multipliers or saddle-point indefiniteness.

To rigorously assess these contributions and delineate the operational performance of the stabilized VNMM, the numerical campaign in Section 5 is structured into three complementary stages:

- **Numerical Verification Stage (Sections 5.1–5.3):** First, single-simplex interpolation verifies the local constructive equivalence under conforming midpoint/tangent placement against Whitney edge elements down to machine precision ($\approx 10^{-16}$). Second, benchmarks on arbitrary point clouds demonstrate that the complete $\mathcal{P}^1$ basis eliminates derivative aliasing, restoring first-order asymptotic curl convergence ($\mathcal{O}(h^{0.97}) \approx \mathcal{O}(h^1)$) alongside an observed convergence rate of $\mathcal{O}(h^{2.17})$ for the electric field on the tested TE_{11} benchmark (19.8× precision gain and 25.9× curl error reduction on the finest mesh). Third, a representative stochastic stress test under 25% spatial jitter and random director orientations demonstrates that scale-invariant thresholding bounds placement matrix conditioning in the tested realization ($\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$, corresponding to unscaled $\kappa_2(\mathbf{A}) \leq 80$ at $\widehat{\mathrm{Tol}} = 0.80$, reaching $\kappa_2(\widehat{\mathbf{A}}) \approx 5.1$ at $\widehat{\mathrm{Tol}} = 1.50$) and restores monotonic error reduction along the evaluated benchmark sequence.
- **Physics-Based Spectral Validation Stage (Sections 5.4–5.6):** First, the formulation is applied to 2D transverse electric (TE_z) electromagnetic resonant cavity eigenvalue problems, showing that a Ritz-Galerkin weak form with div-curl penalty regularization ($s_{\mathrm{div}} = 6.0$) shifts spurious gradient nullspace modes outside the operational band in the tested setup ($\lambda > 50$), yielding clean modal pairing with an average wavenumber error of 1.00% under baseline $10 \times 10, 2 \times 2$ Gauss integration (assisted by under-integration softening relative to the true spatial discretization floor of 3.32%). Second, a convergence benchmark against canonical Edge FEM across six successive refinement levels evaluated under node-conforming 3×3 Gauss quadrature reveals an observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.80})$ over the tested refinement interval (compared to $\mathcal{O}(h_{\mathrm{DoF}}^{1.93})$ for Edge FEM), closing the coarse-grid accuracy gap with edge elements without volumetric mesh connectivity. Third, spectral validation is conducted directly on unconstrained, irregular point clouds (25% spatial jitter and random director orientations

$\theta_k \in [0, 2\pi)$) across six refinement levels, demonstrating strictly monotonic error reduction with an observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.20})$ across the evaluated sequence and spurious-free modal pairing down to 1.63% error.

- **Methodological Scope and Limitations (Section 5.7):** To establish transparent methodological boundaries, we analyze background quadrature dependencies, local stencil combinatorial complexity, point-cloud anisotropy, and penalty parameter sensitivity.

The remainder of this paper is organized as follows: Section 2 details the 2D VNMM vector shape function construction for $\mathcal{L}^1$ and $\mathcal{P}^1$ spaces and proves the Local Dual Equivalence Theorem; Section 3 analyzes local placement matrix singularities and formulates the support selection algorithm; Section 4 presents the div-curl regularized weak form and point-based EFG Gauss quadrature; Section 5 presents the numerical verification, spectral validation, and operational scope; and Section 6 concludes the paper with future research perspectives.

2. Mathematical Formulations of the 2D VNMM

This section establishes the foundational mathematical and functional framework of the two-dimensional Vector Nodal Meshless Method (VNMM 2D). It details the vector projection properties, field continuity enforcement, and the construction of local vector shape functions. First, the single-degree-of-freedom nodal projection scheme and the pointwise nodal imposition of physical boundary conditions are defined. Next, the traditional incomplete $\mathcal{L}^1$ solenoidal space is reviewed, proving its exact topological and algebraic local dual equivalence with classical Whitney 1-form edge finite elements on simplicial cells. Finally, the complete linear polynomial vector space $\mathcal{P}^1$ is introduced to resolve all components of the local Taylor Jacobian tensor and eliminate derivative modal aliasing in unconstrained node clouds.

2.1. Nodal Projection and Tangential Continuity

The VNMM [29, 25] constructs local $H(\mathbf{curl})$-regular vector approximations by distributing a set of M nodes throughout the computational domain Ω and its boundaries $\partial\Omega$. Unlike traditional scalar meshless formulations that assign separate degrees of freedom to individual spatial components (e.g.,

E_x and E_y), the VNMM associates a single physical unit vector director $\vec{t}_i = [t_{ix}, t_{iy}]^T$ with each node i.

The approximate vector field $\vec{E}^h(\mathbf{x})$ at any spatial location $\mathbf{x} = (x, y) \in \Omega$ is expressed as a linear combination of vector nodal shape functions $\vec{N}_i(\mathbf{x})$:

$$\vec{E}^h(x, y) = \sum_{i=1}^{M_{\text{supp}}} \vec{N}_i(x, y) e_i \tag{2}$$

where M_{supp} is the number of active support nodes in the local influence domain, and e_i represents the discrete scalar degree of freedom (DoF) associated with the i-th node. Physically, e_i corresponds to the pointwise directional projection of the electric field vector along the unit vector director $\vec{t}_i$ at the node coordinate $\mathbf{x}_i = (x_i, y_i)$:

$$e_i = \vec{E}(\mathbf{x}_i) \cdot \vec{t}_i \quad \text{[Volts/meter]} \tag{3}$$

To ensure physical consistency, local curl regularity, and direct enforcement of boundary conditions, the vector shape functions $\vec{N}_i(\mathbf{x})$ are required to satisfy the vector Kronecker delta projection property [26, 29]:

$$\vec{N}_i(\mathbf{x}_k) \cdot \vec{t}_k = \delta_{ik} = \begin{cases} 1, & \text{if } k = i \\ 0, & \text{if } k \neq i \end{cases} \tag{4}$$

for all nodes k belonging to the local support domain. Property (4) dictates that the shape function $\vec{N}_i$ exhibits a unit projection along its own nodal director $\vec{t}_i$ at position $\mathbf{x}_i$, while maintaining zero projection along the directors of all other surrounding support nodes.

2.1.1. Enforcement of PEC Boundary Conditions and Remarks on Material Interfaces

The vector Kronecker delta property (4) enables a direct, unconstrained nodal collocation of essential Dirichlet boundary conditions and tangential field continuity without requiring background auxiliary meshes, penalty parameters, or Lagrange multipliers [30, 29].

For nodes lying on Perfect Electric Conductor (PEC) boundaries Γ_{PEC}, where the physical boundary condition requires $\hat{n} \times \vec{E} = \mathbf{0}$ ($\hat{n}$ being the outward unit normal to $\partial\Omega$), the associated vector director $\vec{t}_k$ is aligned strictly tangent to the boundary contour:

$$\vec{t}_k \parallel \partial\Omega \implies \hat{n}(\mathbf{x}_k) \cdot \vec{t}_k = 0. \tag{5}$$

Consequently, setting the scalar nodal degree of freedom directly to zero:

$$e_k = 0 \quad \forall \mathbf{x}_k \in \Gamma_{\text{PEC}} \tag{6}$$

enforces the PEC requirement pointwise at every boundary node:

$$\vec{E}^h(\mathbf{x}_k) \cdot \vec{t}_k = 0 \quad \forall \mathbf{x}_k \in \Gamma_{\text{PEC}}, \tag{7}$$

by virtue of the vector Kronecker delta property (4).

Nodal Collocation versus Continuous Boundary Trace. From a functional analysis viewpoint, it is essential to distinguish the pointwise nodal collocation condition (7) from the continuous functional trace condition:

$$\hat{n} \times \vec{E}^h = \mathbf{0} \quad \text{everywhere on } \Gamma_{\text{PEC}}. \tag{8}$$

When the node cloud coincides with the edge midpoints of a conforming simplicial triangulation (as established by the Local Dual Equivalence Theorem in Section 2.2.3), the VNMM shape functions reproduce classical Whitney 1-forms identically. In that conforming simplicial configuration, internal degrees of freedom have identically vanishing tangential traces along boundary edges; thus, the nodal constraint (7) guarantees that the continuous trace condition (8) holds everywhere along Γ_{PEC}.

On arbitrary unconstrained point clouds, however, the assembled global approximation is non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$). For an evaluation point $\mathbf{x} \in \Gamma_{\text{PEC}}$ located between discrete boundary nodes, the dynamically queried local support stencil $\mathcal{S}(\mathbf{x})$ generally includes neighboring interior nodes whose degrees of freedom $e_i \neq 0$. Because the vector shape functions $\vec{N}_i(\mathbf{x})$ of interior nodes do not necessarily exhibit vanishing tangential components along the continuous boundary segment, setting $e_k = 0$ guarantees exact satisfaction of the tangential condition *strictly at the nodal coordinates* $\mathbf{x}_k$, while its continuous trace over the boundary is not guaranteed for arbitrary non-conforming point clouds.

This distinction becomes particularly prominent for curved or non-axis-aligned boundaries (such as circular cavities or cylindrical PEC shells). By assigning the unit director as the local unit tangent vector, $\vec{t}_k = \hat{\tau}(\mathbf{x}_k)$, the nodal condition $\vec{E}^h(\mathbf{x}_k) \cdot \hat{\tau}(\mathbf{x}_k) = 0$ is collocated directly on the smooth physical boundary without requiring geometric polygonization. Nevertheless, between discrete boundary nodes, the unit tangent vector $\hat{\tau}(\mathbf{x})$ rotates

continuously along the curve, and active stencils can include internal nodes. Hence, the boundary condition is enforced exactly at boundary nodes, while the continuous trace condition (8) over the entire boundary is not guaranteed for arbitrary non-conforming point clouds.

Extensions to Material Interfaces. For applications involving material interfaces Γ_{int} separating subdomains with distinct constitutive parameters ($\epsilon_{r1} \neq \epsilon_{r2}$ or $\mu_{r1} \neq \mu_{r2}$), assigning unit vector directors $\vec{t}_i$ oriented tangentially to Γ_{int} ($\vec{t}_i = \hat{\tau}(\mathbf{x}_i)$) enables direct nodal collocation of tangential continuity ($\vec{E}_1^h(\mathbf{x}_i) \cdot \vec{t}_i = \vec{E}_2^h(\mathbf{x}_i) \cdot \vec{t}_i = e_i$) without auxiliary Lagrange multipliers. Because degrees of freedom are associated solely with tangential projections, the normal component of the electric field is unconstrained across the interface, thereby allowing the normal field to undergo a step discontinuity ($E_{n1} \neq E_{n2}$) and avoiding the artificial normal locking characteristic of scalar C^0 meshless schemes.

It is important to emphasize, however, that while assigning tangential directors permits the normal field E_n to be discontinuous, it does *not* algebraically guarantee the physical continuity of the normal electric flux density:

$$D_{n1} = D_{n2} \iff \epsilon_{r1} E_{n1} = \epsilon_{r2} E_{n2}. \tag{9}$$

The continuity of normal induction is a natural interface condition governed variationally by the weak form and the divergence penalty constraint, rather than an automatic kinematic identity of the single-DoF vector shape functions. Furthermore, on arbitrary point clouds, support stencils that indiscriminately cross material boundaries would tend to unphysically smear out sharp constitutive parameter jumps across subdomains. A rigorous treatment of heterogeneous media therefore requires partitioned support stencils (restricting k-d tree candidate searches to individual homogeneous subdomains, following the multi-tree spatial data structure framework developed by Parreira et al. [33] where dedicated, separate k-d trees are constructed for each distinct material region) or domain-decomposition interface coupling along Γ_{int}. Because the present work focuses on validating the stabilized VNMM on homogeneous resonant cavity benchmarks ($\epsilon_r = 1, \mu_r = 1$), systematic implementation and benchmark validation of heterogeneous material interfaces are formally deferred to future works (Section 6.2).

2.2. The Incomplete $\mathcal{L}^1$ Basis (3 Support Nodes)

The construction of vector shape functions in the 2D VNMM directly inherits the mathematical foundations of $H(\mathbf{curl})$-conforming edge-based mesh-

less approximations [26, 27]. The incomplete 3-term solenoidal polynomial space

$$\mathcal{L}^1 = \text{span}\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} y \\ -x \end{bmatrix} \} \tag{10}$$

corresponds to the local two-dimensional reduction of Nédélec's first-type space [7], whose vector projection mechanisms were established for edge meshless line segments by Silva et al. [27] and later adapted to point-collocated nodes [29, 25].

Consequently, the local vector shape function $\vec{N}_i$ associated with the i-th support node in a 3-node local domain is defined as a linear combination of these basis vectors:

$$\vec{N}_i(x, y) = \beta_{1i} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta_{2i} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \beta_{3i} \begin{bmatrix} y \\ -x \end{bmatrix} \tag{11}$$

where $\boldsymbol{\beta}_i = [\beta_{1i}, \beta_{2i}, \beta_{3i}]^T \in \mathbb{R}^3$ represents the unknown local interpolation coefficient vector.

To determine these coefficients, the vector shape functions are constrained to satisfy the vector Kronecker delta projection property (4) for $i, k = 1, 2, 3$,

Applying constraint (4) to all three support nodes leads to the local 3×3 linear placement system:

$$\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i \tag{12}$$

where the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{3\times3}$ is explicitly given by:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & y_1 t_{1x} - x_1 t_{1y} \\ t_{2x} & t_{2y} & y_2 t_{2x} - x_2 t_{2y} \\ t_{3x} & t_{3y} & y_3 t_{3x} - x_3 t_{3y} \end{bmatrix} \tag{13}$$

and the canonical load vectors are $\mathbf{L}_1 = [1, 0, 0]^T$, $\mathbf{L}_2 = [0, 1, 0]^T$, and $\mathbf{L}_3 = [0, 0, 1]^T$.

Once the coefficients $\boldsymbol{\beta}_i = \mathbf{A}^{-1}\mathbf{L}_i$ are evaluated, the approximated electric field vector $\vec{E}^h(x, y)$ is constructed via nodal projection superposition:

$$\vec{E}^h(x, y) = \sum_{i=1}^{3} \vec{N}_i(x, y) e_i \tag{14}$$

where $e_i = \vec{E}(\mathbf{x}_i) \cdot \vec{t}_i$ represent the discrete scalar projection DoFs.

2.2.1. Mathematical Properties

The shape functions $\vec{N}_i$ of the $\mathcal{L}^1$ basis possess two distinct mathematical characteristics:

1. **Identical Solenoidal Property ($\nabla \cdot \vec{N}_i \equiv 0$):** Taking the divergence of Eq. (11) yields:

$$\nabla \cdot \vec{N}_i = \frac{\partial}{\partial x}(\beta_{1i} + \beta_{3i} y) + \frac{\partial}{\partial y}(\beta_{2i} - \beta_{3i} x) \equiv 0 \tag{15}$$

2. **Univariate Curl Representation:** Evaluating the scalar z-directed curl operator on $\vec{N}_i$ yields:

$$(\nabla \times \vec{N}_i)_z = \frac{\partial N_{iy}}{\partial x} - \frac{\partial N_{ix}}{\partial y} = \frac{\partial(-\beta_{3i} x)}{\partial x} - \frac{\partial(\beta_{3i} y)}{\partial y} = -2\beta_{3i} \tag{16}$$

Therefore, the approximated z-directed scalar curl field $(\nabla \times \vec{E}^h)_z$ is spatially constant across the 3-node support domain and depends exclusively on the third coefficient β_{3i}:

$$(\nabla \times \vec{E}^h)_z = -2 \sum_{i=1}^{3} \beta_{3i} e_i \tag{17}$$

2.2.2. Local Constructive Equivalence under Conforming Midpoint/Tangent Placement

The VNMM 2D was originally conceptualized as the zero-edge-length asymptotic limit of the Edge Meshless Method (EMM) [26, 29]. By examining this heritage, a fundamental local constructive equivalence under conforming midpoint/tangent placement between the VNMM point-collocation formulation and classic lowest-order Whitney edge finite elements (Nédélec 1-forms on triangular simplices [7, 3]) is formally established, as detailed in Fig. 1.

In conventional FEM (Fig. 1a), the degrees of freedom associated with the triangular cell Ω_e are line integrals of circulation $\alpha_k = \oint_{e_k} \vec{E} \cdot d\boldsymbol{\ell}$ along continuous 1D edges of finite physical length L_k. As demonstrated in the asymptotic collapse limit (Fig. 1c), when the physical edge length shrinks to zero ($L_k \to 0$), the 1D continuous edge degenerates into a point-collocated vector node $(P_k, \vec{t}_k)$, where the circulation DoF α_k naturally scales into a pointwise directional field projection $e_k = \vec{E}(P_k) \cdot \vec{t}_k$ [Volts/meter].

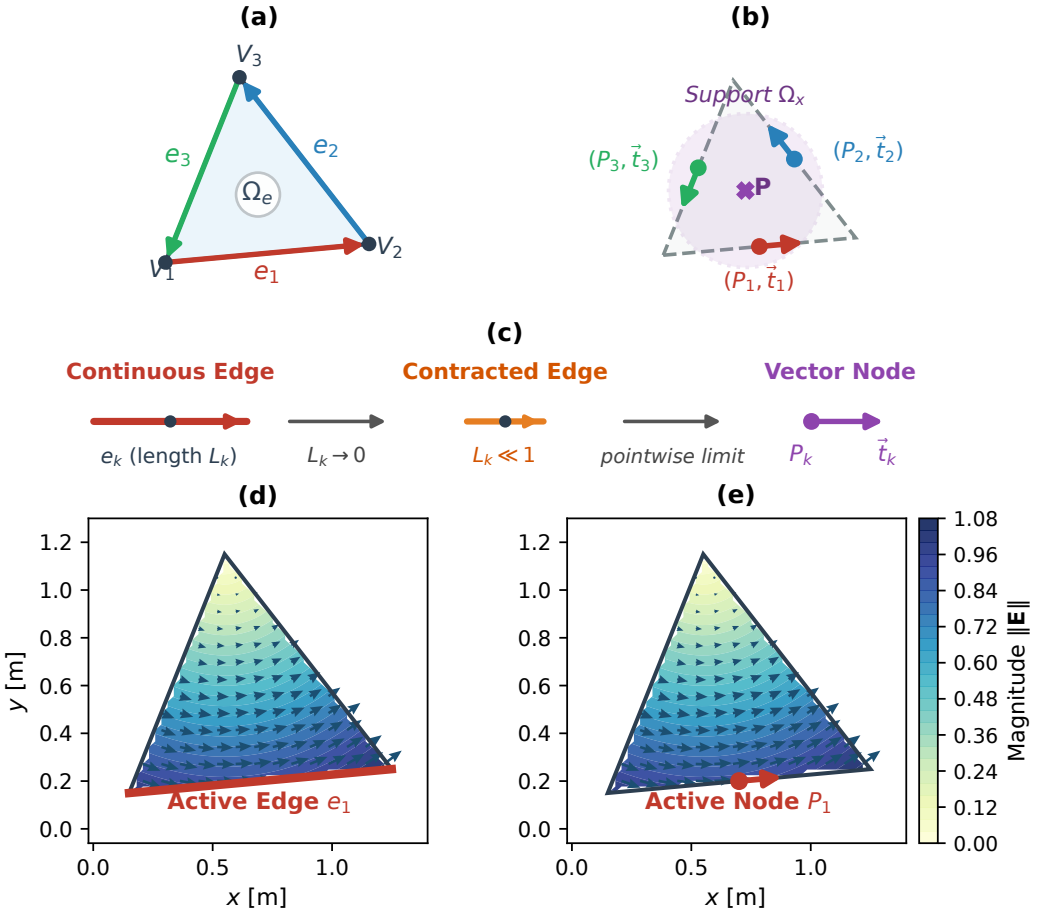


Figure 1: Local constructive equivalence under conforming midpoint/tangent placement between lowest-order Whitney edge elements (FEM 1-forms) and the VNMM 2D $\mathcal{L}^1$ vector formulation: (a) Primal triangular cell Ω_e with vertices V_1, V_2, V_3 and oriented edges e_1, e_2, e_3; (b) VNMM 2D support stencil with vector nodes $(P_k, \vec{t}_k)$ placed at the edge midpoints and compact support Ω_x; (c) Conceptual dual collapse limit as edge length contracts ($L_k \to 0$), identifying the discrete circulation integral $L_k^{-1} \int_{e_k} \vec{E} \cdot \vec{t}_k \, ds$ directly with the tangential projection at a point, $e_k = \vec{E}(P_k) \cdot \vec{t}_k$; (d) Vector field and magnitude contour of the Whitney 1-form $\vec{w}_1(\mathbf{x}) = \lambda_1 \nabla\lambda_2 - \lambda_2 \nabla\lambda_1$ associated with active edge e_1; (e) Vector field and magnitude contour of the normalized VNMM $\mathcal{L}^1$ vector basis function $\frac{1}{L_1}\vec{N}_1(\mathbf{x})$ associated with active node $(P_1, \vec{t}_1)$. Both fields are algebraically identical over Ω_e.

2.2.3. Formal Proof of the Local Dual Equivalence Theorem

Let $T \subset \mathbb{R}^2$ be an arbitrary non-degenerate triangular simplex with vertices V_1, V_2, V_3 ordered counter-clockwise and non-zero area $|T| > 0$. Let $e_k = (V_k, V_{k+1})$ be the directed boundary edges of length $L_k = \|V_{k+1} - V_k\| > 0$, with unit tangent vectors $\vec{t}_k = (V_{k+1} - V_k)/L_k$ and midpoints $M_k = \frac{1}{2}(V_k + V_{k+1})$ for $k \in \{1, 2, 3\}$ (with index wrap-around $V_4 \equiv V_1$).

Theorem 1 (Local Constructive Equivalence under Conforming Midpoint/Tangent Placement)**.** *Consider the support triad of vector nodes* $\{P_k, \vec{t}_k\}_{k=1}^3$ *defined at the edge midpoints* $P_k = M_k$ *with directors* $\vec{t}_k$*. The VNMM 2D* $\mathcal{L}^1$ *shape functions* $\vec{N}_k(\mathbf{x})$ *and the Whitney 1-forms* $\vec{w}_k(\mathbf{x}) = \lambda_i(\mathbf{x})\nabla\lambda_j - \lambda_j(\mathbf{x})\nabla\lambda_i$ *satisfy:*

$$\vec{N}_k(\mathbf{x}) \equiv L_k\, \vec{w}_k(\mathbf{x}) \quad \Longleftrightarrow \quad \vec{w}_k(\mathbf{x}) \equiv \frac{1}{L_k}\vec{N}_k(\mathbf{x}), \quad \forall \mathbf{x} \in \mathbb{R}^2,\ \forall k \in \{1, 2, 3\} \tag{18}$$

Proof. The proof proceeds in five constructive steps:

Step 1: Identity of Local Vector Polynomial Spaces. The lowest-order Whitney space $\mathcal{W}(T)$ on T is spanned by $\vec{w}_k(\mathbf{x}) = [a_1 + \alpha y, a_2 - \alpha x]^T$. By inspection of the spanning vectors, $\mathcal{W}(T) = \text{span}\{[1, 0]^T,\ [0, 1]^T,\ [y, -x]^T\} \equiv \mathcal{L}^1$. Thus, both approximation spaces share identical 3-dimensional vector polynomial spaces.

Step 2: Constancy of Tangential Field Projection Along Simplex Edges. Let $\vec{u}(\mathbf{x}) = [a_1 + \alpha y, a_2 - \alpha x]^T \in \mathcal{L}^1$. The directional projection of $\vec{u}(\mathbf{x})$ along unit director $\vec{t}_k = [\Delta x_k/L_k, \Delta y_k/L_k]^T$ is:

$$\vec{u}(\mathbf{x}) \cdot \vec{t}_k = \frac{1}{L_k}\left[a_1\Delta x_k + a_2\Delta y_k + \alpha(y\Delta x_k - x\Delta y_k)\right] \tag{19}$$

Since any point $\mathbf{x} = (x, y)$ on edge e_k satisfies $(x - x_k)\Delta y_k - (y - y_k)\Delta x_k = 0$, the rotational term $y\Delta x_k - x\Delta y_k = y_k\Delta x_k - x_k\Delta y_k$ is strictly constant along the entire edge length. Therefore, the tangential projection is invariant along e_k:

$$\vec{u}(\mathbf{x}) \cdot \vec{t}_k = \vec{u}(M_k) \cdot \vec{t}_k, \quad \forall \mathbf{x} \in e_k \tag{20}$$

Step 3: Reduction of Edge Circulation to Pointwise Midpoint Value. Using Eq. (20), the line integral defining the FEM circulation degree

of freedom along edge e_j simplifies to:

$$\oint_{e_j} \vec{u} \cdot d\boldsymbol{\ell} = \int_{-L_j/2}^{L_j/2} \left(\vec{u}(\mathbf{x}(s)) \cdot \vec{t}_j\right) ds = \left(\vec{u}(M_j) \cdot \vec{t}_j\right) \int_{-L_j/2}^{L_j/2} ds = L_j \left(\vec{u}(M_j) \cdot \vec{t}_j\right) \tag{21}$$

Substituting the k-th Whitney basis function $\vec{w}_k \in \mathcal{W}(T)$, which satisfies $\oint_{e_j} \vec{w}_k \cdot d\boldsymbol{\ell} = \delta_{jk}$ by construction, yields:

$$L_j \left(\vec{w}_k(M_j) \cdot \vec{t}_j\right) = \delta_{jk} \quad \Longrightarrow \quad (L_k \vec{w}_k(M_j)) \cdot \vec{t}_j = \delta_{jk} \tag{22}$$

Step 4: Uniqueness and Exact Identification with VNMM Shape Functions. By definition of the VNMM 2D $\mathcal{L}^1$ interpolation problem on the support triad $\{M_j, \vec{t}_j\}_{j=1}^3$, the shape functions satisfy $\vec{N}_k(M_j) \cdot \vec{t}_j = \delta_{jk}$. Consider the difference field:

$$\vec{d}_k(\mathbf{x}) = \vec{N}_k(\mathbf{x}) - L_k \vec{w}_k(\mathbf{x}) \in \mathcal{L}^1 \tag{23}$$

Evaluating its projection at each midpoint M_j:

$$\vec{d}_k(M_j) \cdot \vec{t}_j = \vec{N}_k(M_j) \cdot \vec{t}_j - L_k \left(\vec{w}_k(M_j) \cdot \vec{t}_j\right) = \delta_{jk} - \delta_{jk} = 0, \quad \forall j \in \{1, 2, 3\} \tag{24}$$

Because triangle T is non-degenerate, the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{3\times 3}$ is non-singular ($\det \mathbf{A} \neq 0$). The only vector field in $\mathcal{L}^1$ satisfying zero projections across all three support directors is the trivial field $\vec{d}_k(\mathbf{x}) \equiv \mathbf{0}$, which proves the exact identity $\vec{N}_k(\mathbf{x}) \equiv L_k \vec{w}_k(\mathbf{x})$ everywhere in $\mathbb{R}^2$.

Step 5: Verification of Differential Invariants. 1. *Divergence:* For any field $\vec{u} = [a_1 + \alpha y, a_2 - \alpha x]^T \in \mathcal{L}^1$, we have $\nabla \cdot \vec{u} = \partial(a_1 + \alpha y)/\partial x + \partial(a_2 - \alpha x)/\partial y \equiv 0$. Thus, $\nabla \cdot \vec{w}_k \equiv 0$ and $\nabla \cdot \vec{N}_k \equiv 0$ everywhere. 2. *Curl:* Applying Stokes' theorem over triangle T for Whitney 1-form $\vec{w}_k$:

$$\int_T (\nabla \times \vec{w}_k)_z d\Omega = \oint_{\partial T} \vec{w}_k \cdot d\boldsymbol{\ell} = \sum_{j=1}^{3} \delta_{jk} = 1 \implies (\nabla \times \vec{w}_k)_z = \frac{1}{|T|} \tag{25}$$

By the identity proved in step 4, $(\nabla \times \vec{N}_k)_z = L_k (\nabla \times \vec{w}_k)_z = L_k/|T|$. □

As graphically illustrated in Fig. 1d and 1e (and verified numerically in Section 5.1), the normalized VNMM $\mathcal{L}^1$ shape functions $\frac{1}{L_k}\vec{N}_k$ and canonical Whitney 1-forms $\vec{w}_k$ match identically across the simplex T down to floating-point machine precision ($\approx 10^{-16}$).

2.2.4. Local $H(\mathbf{curl})$ Regularity versus Global $H(\mathbf{curl})$ Conformity

Beyond establishing the mathematical connection to edge finite elements, this constructive identity clarifies a fundamental structural distinction in vector meshless approximations:

1. **Local $H(\mathbf{curl})$ Regularity on Virtual Simplices:** In conventional scalar meshless methods (e.g., Moving Least Squares or Element-Free Galerkin), evaluating field approximations outside the physical convex hull enclosed by the node coordinates $\mathbf{x}_k$ constitutes an unconstrained extrapolation, leading to rapid truncation error growth [26]. In contrast, because the VNMM vector basis reproduces the complete polynomial space of Whitney edge elements over the virtual macro-triangle Ω_x defined by the intersecting lines of action of vector directors $\vec{t}_k$ (Fig. 1b), the valid domain for local $H(\mathbf{curl})$ field approximation is not geographically restricted to the small physical convex hull of $\mathbf{x}_k$ [7, 3]. Under regular-support assumptions (well-conditioned local placement matrix $\mathbf{A}$ and non-degenerate intersection vertices), any field evaluation at $P \in \Omega_x$ preserves local $H(\mathbf{curl})$ regularity.
2. **Loss of Global $H(\mathbf{curl})$ Conformity on Unconstrained Point Clouds:** When nodes and directors form a conforming simplicial mesh (Section 2.2.3), tangential traces across adjacent element facets match identically, yielding global conformity $V_h \subset H(\mathbf{curl}; \Omega)$. On arbitrary, unconstrained point clouds, however, support stencils are selected dynamically as the evaluation point P moves across the domain. Because adjacent stencils differ in node composition and orientation, tangential traces do not cancel continuously across internal support boundaries. Consequently, the global approximation space is non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$). As developed in Section 4, this distinction is essential: local polynomial completeness (via $\mathcal{P}^1$) restores local Jacobian and curl consistency, but spectral stability in eigenvalue problems requires a div-curl regularized weak form to suppress gradient leakage arising from global non-conformity.

2.2.5. Mathematical Origin of Modal Aliasing and Curl Stagnation

While the incomplete solenoidal basis $\mathcal{L}^1$ exhibits exact dual equivalence to Whitney elements on conforming meshes (Section 2.2.2), its behavior diverges sharply on unconstrained point clouds, where curl approximations stagnate at an $\mathcal{O}(1)$ relative error plateau under refinement [31]. To uncover

the algebraic origin of this derivative stagnation, we examine the local Taylor series expansion of the exact electric field $\vec{E}$ centered at an evaluation point $P(x_P, y_P)$ [34, 10]:

$$\vec{E}(\mathbf{x}) = \vec{E}(P) + \mathbf{J}(P) \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \mathcal{O}(h^2) \tag{26}$$

where $\mathbf{J}(P) = \nabla \vec{E}$ represents the complete first-order field Jacobian tensor [10]:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial E_x}{\partial x} & \frac{\partial E_x}{\partial y} \\ \frac{\partial E_y}{\partial x} & \frac{\partial E_y}{\partial y} \end{bmatrix} \tag{27}$$

By expressing individual spatial derivatives in terms of fundamental vector calculus operators—namely, the field divergence $\nabla \cdot \vec{E} = \partial E_x/\partial x + \partial E_y/\partial y$ and the z-directed field curl $(\nabla \times \vec{E})_z = \partial E_y/\partial x - \partial E_x/\partial y$—the field Jacobian tensor is decomposed as:

$$\mathbf{J} = \frac{1}{2}(\nabla \cdot \vec{E})\mathbf{I} + \frac{1}{2}(\nabla \times \vec{E})_z \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} \delta_{\text{norm}} & -\delta_{\text{trans}} \\ \delta_{\text{trans}} & -\delta_{\text{norm}} \end{bmatrix} \tag{28}$$

where $\delta_{\text{norm}} = \frac{1}{2}(\partial E_x/\partial x - \partial E_y/\partial y)$ denotes the normal (longitudinal) field derivative difference, and $\delta_{\text{trans}} = \frac{1}{2}(\partial E_y/\partial x + \partial E_x/\partial y)$ represents the symmetric transverse field derivative component.

In the incomplete $\mathcal{L}^1$ basis, local shape functions provide only 3 degrees of freedom. As a result, normal field derivatives are artificially forced to vanish identically ($\partial E_x^h/\partial x \equiv 0$ and $\partial E_y^h/\partial y \equiv 0$), imposing $\nabla \cdot \vec{E}^h \equiv 0$ and $\delta_{\text{norm}} \equiv 0$ simultaneously.

This fundamental limitation is formalized in the following proposition:

Proposition 1 (Jacobian Incompleteness and Derivative Modal Aliasing Scaling)**.** *Let $\vec{E} \in [C^2(\Omega)]^2$ be an arbitrary twice continuously differentiable 2D vector field. The incomplete solenoidal space $\mathcal{L}^1 = \text{span}\{[1,0]^T, [0,1]^T, [y,-x]^T\} \subset \mathcal{P}_1 \times \mathcal{P}_1$ cannot reproduce the complete first-order Taylor expansion of $\vec{E}$. Specifically, while the complete space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ possesses $\dim(\mathcal{P}^1) = 6$ degrees of freedom to reconstruct the 2 translation components and all 4 independent entries of the Jacobian tensor $\mathbf{J} = \nabla \vec{E}$, $\mathcal{L}^1$ possesses only $\dim(\mathcal{L}^1) = 3$ degrees of freedom, enforcing the artificial linear constraints:*

$$\frac{\partial E_x^h}{\partial x} = 0, \quad \frac{\partial E_y^h}{\partial y} = 0, \quad \textit{and} \quad \frac{\partial E_x^h}{\partial y} + \frac{\partial E_y^h}{\partial x} = 0. \tag{29}$$

Under point-collocated projection on unstructured node clouds, unrepresented normal gradients ($\partial E_x/\partial x, \partial E_y/\partial y \neq 0$) generate a local projection remainder $\mathbf{r} \sim \mathcal{O}(h)$ that couples into the rotational coefficient through the inverted moment placement matrix row $\mathrm{Row}_3(\mathbf{A}^{-1}) \sim \mathcal{O}(h^{-1})$. In an asymptotic order-of-magnitude scaling sense, this induces an $\mathcal{O}(1)$ derivative aliasing error in the pointwise reconstructed curl under refinement ($h \to 0$) in the collocated setting:

$$|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z| \sim \|\,\mathrm{Row}_3(\mathbf{A}^{-1})\|\|\mathbf{r}\| = \mathcal{O}(h^{-1}) \cdot \mathcal{O}(h) = \mathcal{O}(1). \quad (30)$$

Proof. Let any discrete vector field $\vec{E}^h \in \mathcal{L}^1$ be expanded in terms of the solenoidal basis:

$$\vec{E}^h(x,y) = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c \begin{bmatrix} y \\ -x \end{bmatrix}, \quad a, b, c \in \mathbb{R}. \quad (31)$$

Direct differentiation with respect to spatial coordinates yields the discrete Jacobian tensor:

$$\mathbf{J}^h = \begin{bmatrix} \frac{\partial E_x^h}{\partial x} & \frac{\partial E_x^h}{\partial y} \\ \frac{\partial E_y^h}{\partial x} & \frac{\partial E_y^h}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 & c \\ -c & 0 \end{bmatrix}. \quad (32)$$

Inspect Eq. (32) establishes that $\partial E_x^h/\partial x \equiv 0$, $\partial E_y^h/\partial y \equiv 0$, and $\partial E_x^h/\partial y + \partial E_y^h/\partial x = c - c \equiv 0$, identically satisfying the linear constraints in Eq. (29).

Now consider a general smooth physical field $\vec{E} \in [C^2(\Omega)]^2$. Its Taylor expansion centered at evaluation point P is $\vec{E}(\mathbf{x}) = \vec{E}(P) + \mathbf{J}(P)\Delta\mathbf{x} + \mathcal{O}(h^2)$, where $\mathbf{J}(P)$ has non-zero diagonal entries ($\partial E_x/\partial x \neq 0$ or $\partial E_y/\partial y \neq 0$). Collocating $\vec{E}^h$ at support nodes $\mathbf{x}_k$ ($k = 1, 2, 3$) along unit directors $\vec{t}_k$ defines the linear placement system $\mathbf{A}[a, b, c]^T = \mathbf{e}$, where $e_k = \vec{E}(\mathbf{x}_k) \cdot \vec{t}_k$.

Subtracting the projection of the unrepresented linear components $(\mathbf{J}(P) - \mathbf{J}^h)\Delta\mathbf{x}_k \cdot \vec{t}_k$ yields a nodal projection residual $\mathbf{r} \in \mathbb{R}^3$ with asymptotic magnitude $\|\mathbf{r}\| \sim \mathcal{O}(h)$. In the 3×3 moment placement matrix $\mathbf{A}$ (Eq. (13)), translation columns 1–2 scale as $\mathcal{O}(1)$, while the rotational moment column 3 scales as $(\Delta y_k t_{kx} - \Delta x_k t_{ky}) \sim \mathcal{O}(h)$. Consequently, by Cramer's rule or Neumann inversion, the third row of $\mathbf{A}^{-1}$ (which maps nodal data to the rotational DoF $c = -\frac{1}{2}(\nabla \times \vec{E}^h)_z$) scales strictly as $\|\,\mathrm{Row}_3(\mathbf{A}^{-1})\| \sim \mathcal{O}(h^{-1})$.

The numerical error in the reconstructed curl at evaluation point P is

therefore governed by the propagation of the projection remainder:

$$|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z| = 2|c_{\text{exact}} - c| \\ \sim \|\operatorname{Row}_3(\mathbf{A}^{-1})\|\|\mathbf{r}\| = \mathcal{O}(h^{-1}) \cdot \mathcal{O}(h) = \mathcal{O}(1), \tag{33}$$

which is asymptotically independent of h and generates an $\mathcal{O}(1)$ error plateau under spatial refinement ($h \to 0$) in the collocated setting. □

Remark 1 (Mathematical Framing of Proposition 1)**.** *We emphasize that the derivation in Proposition 1 is framed as a local asymptotic order-of-magnitude scaling analysis ($\mathcal{O}(h^{-1}) \cdot \mathcal{O}(h) = \mathcal{O}(1)$) tracking the algebraic interaction between Taylor truncation remainders and the inverted moment placement matrix, rather than a strict uniform norm or functional analytic proof in Sobolev spaces (such as $L^\infty(\Omega)$ or $H(\mathbf{curl}; \Omega)$). It formally establishes the algebraic mechanism by which unrepresented gradient modes contaminate the point-collocated rotational degree of freedom, explaining the persistent $\mathcal{O}(1)$ curl noise floor observed empirically in arbitrary meshless point clouds (Section 5.2).*

In Edge Finite Element Methods (EFEM), this lack of explicit normal derivative representation does not cause stagnation because continuous line circulation integrals along element edges enforce closed-contour flux cancellation via Stokes' theorem [7, 3]. In the VNMM, however, continuous line integrals are replaced by discrete pointwise projections. Consequently, the $\mathcal{O}(1)$ derivative noise floor established in Proposition 1 cannot be eliminated by spatial mesh refinement alone ($h \to 0$). Adopting the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$, as formulated in Section 2.3, provides a direct and sufficient remedy by resolving the complete local Jacobian and eliminating this specific derivative-aliasing mechanism on unconstrained point clouds.

2.3. The Complete $\mathcal{P}^1$ Vector Polynomial Basis

To eliminate the derivative modal aliasing and curl stagnation inherent to the incomplete $\mathcal{L}^1$ space, we reformulate local vector interpolation using the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$. This transition draws a direct mathematical analogy to Nédélec's second family of mixed finite elements [35], in which vector field approximations are constructed over complete polynomial spaces $(\mathbb{P}_k)^d$ rather than solenoidal-restricted subspaces.

In two dimensions ($d = 2$), a complete first-order vector polynomial basis $\mathcal{P}^1$ provides 6 independent degrees of freedom ($M_{\text{supp}} = 6$ support nodes per evaluation point):

$$\mathcal{P}^1 = \text{span}\left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}x\\0\end{bmatrix}, \begin{bmatrix}y\\0\end{bmatrix}, \begin{bmatrix}0\\x\end{bmatrix}, \begin{bmatrix}0\\y\end{bmatrix} \right\} \tag{34}$$

Using this complete basis, the local vector shape function $\vec{N}_i(\mathbf{x})$ associated with the i-th support node is expressed as a linear combination of its six constituent terms:

$$\vec{N}_i(x, y) = \beta_{1i} \begin{bmatrix}1\\0\end{bmatrix} + \beta_{2i} \begin{bmatrix}0\\1\end{bmatrix} + \beta_{3i} \begin{bmatrix}x\\0\end{bmatrix} + \beta_{4i} \begin{bmatrix}y\\0\end{bmatrix} + \beta_{5i} \begin{bmatrix}0\\x\end{bmatrix} + \beta_{6i} \begin{bmatrix}0\\y\end{bmatrix} \tag{35}$$

where β_{ji} (for $j = 1, \ldots, 6$) represent the local interpolation coefficients to be determined.

Enforcing the vector Kronecker delta projection property [29]:

$$\vec{N}_i(\mathbf{x}_k) \cdot \vec{t}_k = \delta_{ik}, \quad \text{for } k = 1, \ldots, 6 \tag{36}$$

across all six support nodes yields the local linear placement system

$$\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i. \tag{37}$$

Evaluated in local translated coordinates centered at the evaluation point $P(x_P, y_P)$ (defined as $\Delta x_k = x_k - x_P$ and $\Delta y_k = y_k - y_P$), the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{6\times 6}$ is explicitly given by:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & \Delta x_1 t_{1x} & \Delta y_1 t_{1x} & \Delta x_1 t_{1y} & \Delta y_1 t_{1y} \\ t_{2x} & t_{2y} & \Delta x_2 t_{2x} & \Delta y_2 t_{2x} & \Delta x_2 t_{2y} & \Delta y_2 t_{2y} \\ t_{3x} & t_{3y} & \Delta x_3 t_{3x} & \Delta y_3 t_{3x} & \Delta x_3 t_{3y} & \Delta y_3 t_{3y} \\ t_{4x} & t_{4y} & \Delta x_4 t_{4x} & \Delta y_4 t_{4x} & \Delta x_4 t_{4y} & \Delta y_4 t_{4y} \\ t_{5x} & t_{5y} & \Delta x_5 t_{5x} & \Delta y_5 t_{5x} & \Delta x_5 t_{5y} & \Delta y_5 t_{5y} \\ t_{6x} & t_{6y} & \Delta x_6 t_{6x} & \Delta y_6 t_{6x} & \Delta x_6 t_{6y} & \Delta y_6 t_{6y} \end{bmatrix} \tag{38}$$

The corresponding coefficient vector $\boldsymbol{\beta}_i$ and canonical right-hand side

vector $\mathbf{L}_i$ are structured as:

$$\boldsymbol{\beta}_i = \begin{bmatrix} \beta_{1i} \\ \beta_{2i} \\ \beta_{3i} \\ \beta_{4i} \\ \beta_{5i} \\ \beta_{6i} \end{bmatrix}, \qquad \mathbf{L}_i = \begin{bmatrix} \delta_{1i} \\ \delta_{2i} \\ \delta_{3i} \\ \delta_{4i} \\ \delta_{5i} \\ \delta_{6i} \end{bmatrix} \tag{39}$$

Solving the linear system (37) via $\boldsymbol{\beta}_i = \mathbf{A}^{-1}\mathbf{L}_i$ completely defines the vector shape functions $\vec{N}_i(\mathbf{x})$. Consequently, the approximated electric field $\vec{E}^h(\mathbf{x})$ and its z-directed curl $(\nabla \times \vec{E}^h)_z$ within the support domain are expressed analytically as:

$$\vec{E}^h(x,y) = \sum_{i=1}^{6} \vec{N}_i(x,y) e_i \tag{40}$$

$$(\nabla \times \vec{E}^h)_z(x,y) = \frac{\partial E_y^h}{\partial x} - \frac{\partial E_x^h}{\partial y} = \sum_{i=1}^{6} (\beta_{5i} - \beta_{4i}) e_i \tag{41}$$

where $e_i = \vec{E}(P_i) \cdot \vec{t}_i$ denotes the scalar directional projection degree of freedom associated with the i-th node.

By providing 6 independent degrees of freedom, the complete $\mathcal{P}^1$ space resolves all four spatial derivative components of the local Jacobian tensor $\mathbf{J}^h = \nabla \vec{E}^h$ simultaneously:

$$\mathbf{J}^h = \begin{bmatrix} \frac{\partial E_x^h}{\partial x} & \frac{\partial E_x^h}{\partial y} \\ \frac{\partial E_y^h}{\partial x} & \frac{\partial E_y^h}{\partial y} \end{bmatrix} = \sum_{i=1}^{6} \begin{bmatrix} \beta_{3i} & \beta_{4i} \\ \beta_{5i} & \beta_{6i} \end{bmatrix} e_i \tag{42}$$

Unlike the incomplete $\mathcal{L}^1$ space, which artificially forces longitudinal (normal) field derivatives ($\partial E_x/\partial x$, $\partial E_y/\partial y$) to vanish, the complete $\mathcal{P}^1$ basis resolves both rotational and irrotational (longitudinal flux variation) components simultaneously. This shields the field curl calculation from longitudinal derivative truncation residues, removing the derivative modal aliasing mechanism and providing the polynomial consistency required for first-order curl convergence, while yielding an observed convergence rate of $\mathcal{O}(h^{2.17})$ for electric field interpolation in the investigated benchmark (Section 5.2).

3. Algebraic Conditioning and Support Selection

The construction of vector nodal shape functions $\vec{N}_i$ in the Vector Nodal Meshless Method (VNMM) relies on solving the local linear placement system $\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i$ (Eq. (37)). A fundamental prerequisite for the existence, uniqueness, and numerical stability of these vector shape functions is that the local moment placement matrix $\mathbf{A}$ remains strictly non-singular ($|\det(\mathbf{A})| > 0$).

Unlike scalar meshless formulations where matrix singularity is dictated purely by spatial nodal distributions (e.g., node co-location or spatial co-linearity), the conditioning of $\mathbf{A}$ in the VNMM is intrinsically coupled to both the spatial coordinates $\mathbf{x}_k = (x_k, y_k)$ and the orientation of the unit vector directors $\vec{t}_k = [t_{kx}, t_{ky}]^T$ assigned to each support node. In the following subsections, a detailed algebraic singularity analysis is conducted for both the incomplete $\mathcal{L}^1$ basis (Section 3.1) and the complete $\mathcal{P}^1$ basis (Section 3.2), establishing the mathematical foundation for the automated support selection strategy developed in Section 3.3.1.

3.1. Singularities and Geometric Constraints in the Incomplete $\mathcal{L}^1$ Basis (3×3 Matrix)

To establish a systematic classification of algebraic singularities in the original VNMM 2D formulation [29, 25], we revisit the 3×3 local moment placement matrix $\mathbf{A}$ generated by the 3-node solenoidal basis $\mathcal{L}^1$. Originally derived in Eq. (13), $\mathbf{A}$ is restated below to facilitate the singularity analysis:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & y_1 t_{1x} - x_1 t_{1y} \\ t_{2x} & t_{2y} & y_2 t_{2x} - x_2 t_{2y} \\ t_{3x} & t_{3y} & y_3 t_{3x} - x_3 t_{3y} \end{bmatrix} \tag{43}$$

For this matrix, algebraic singularity ($\det(\mathbf{A}) = 0$) occurs when the spatial-directional equations contributed by the candidate triad become linearly dependent. As classified visually in Figs. 2a–2d, these mathematical failures fall into four distinct pathological configurations:

1. **Global Vector Parallelism (Fig. 2a):** If all three unit directors are parallel or anti-parallel ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$), director components satisfy $t_{kx} = c \cdot t_{ky}$ for a constant scalar c. Thus, $\mathrm{Col}_1(\mathbf{A}) = c \cdot \mathrm{Col}_2(\mathbf{A})$, yielding $\det(\mathbf{A}) \equiv 0$ regardless of nodal spatial distribution.

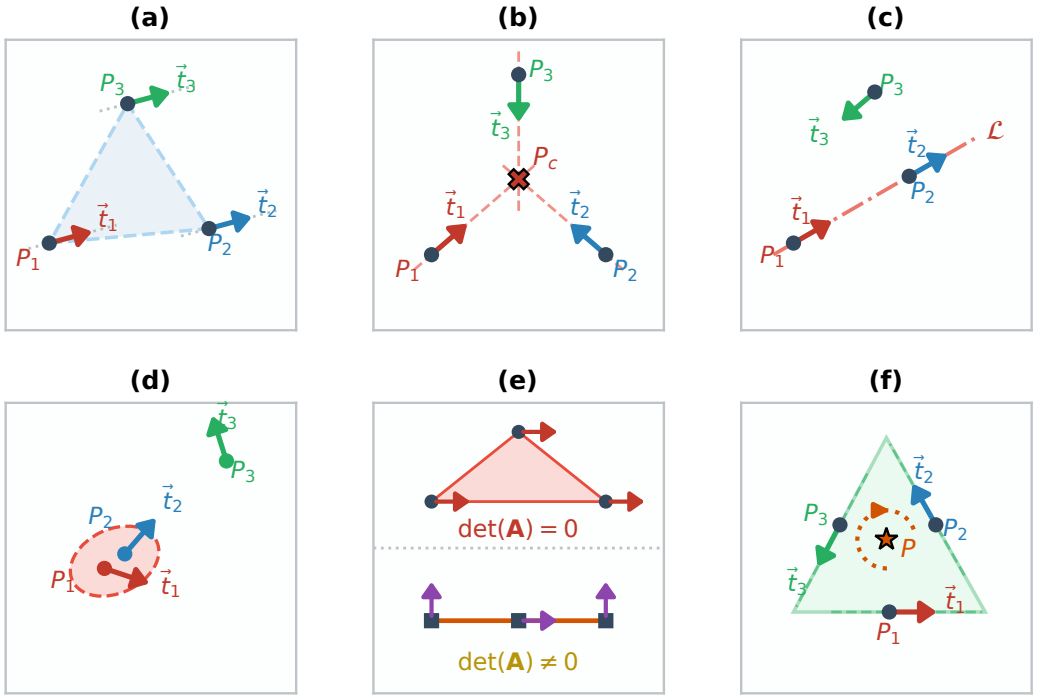


Figure 2: Geometric and directional singularity configurations of the VNMM 2D local placement matrix $\mathbf{A} \in \mathbb{R}^{3\times3}$ for the $\mathcal{L}^1$ solenoidal vector basis: (a) **Global vector parallelism** ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$): yields $t_{kx} = c\,t_{ky}$ for all k, causing linear dependency between the first two columns ($\mathrm{Col}_1(\mathbf{A}) = c\,\mathrm{Col}_2(\mathbf{A})$); (b) **Concurrent lines of action**: when the lines of action of all three directors intersect at a common focal point $P_c(x_c, y_c)$, the relation $t_{kx}(y_k - y_c) - t_{ky}(x_k - x_c) = 0$ renders the rotational column a linear combination of the translation columns, yielding $\mathrm{Col}_3(\mathbf{A}) = y_c\,\mathrm{Col}_1(\mathbf{A}) - x_c\,\mathrm{Col}_2(\mathbf{A})$, causing $\det(\mathbf{A}) = 0$; (c) **Shared line of action**: nodes P_1, P_2 lying on a common line $\mathcal{L}$ with $\vec{t}_1 = \vec{t}_2 \parallel (P_2 - P_1)$ share identical directional moments, $y_2 t_{2x} - x_2 t_{2y} = y_1 t_{1x} - x_1 t_{1y}$, producing duplicate rows $\mathrm{Row}_2(\mathbf{A}) = \mathrm{Row}_1(\mathbf{A})$; (d) **Nodal coincidence / spatial collapse**: as $\|P_1 - P_2\| \to 0$, if $\vec{t}_1 = \vec{t}_2$ identical rows are formed; if $\vec{t}_1 \neq \vec{t}_2$, two distinct vector degrees of freedom are colocated at a single spatial point, degenerating the 2D spatial sampling base into a 1D line; (e) **Decoupling of spatial geometry and vector alignment**: illustrates why geometric proximity algorithms alone fail—(top) spatially non-collinear nodes ($\mathrm{Area}(T) > 0$) with parallel directors yield $\det(\mathbf{A}) = 0$; (bottom) collinear nodes ($\mathrm{Area}(T) = 0$) with non-parallel directors can yield $\det(\mathbf{A}) \neq 0$ (algebraically invertible), yet fail to span 2D transverse derivatives; (f) **Admissible and stable support**: dual qualification requiring non-collinear spatial distribution ($\mathrm{Area}(T) > 0$) and non-vanishing circulation around evaluation point P, promoting bounded condition numbers for accurate field and curl reconstruction.

2. **Concurrency of Lines of Action (Fig. 2b):** If the infinite lines of action passing through $\mathbf{x}_k$ along $\vec{t}_k$ intersect at a single common point $P_c(x_c, y_c)$, every node satisfies $t_{kx}(y_k - y_c) - t_{ky}(x_k - x_c) = 0$. This forces column Col_3 to become a linear combination of the other two columns, $\mathrm{Col}_3 = y_c\,\mathrm{Col}_1 - x_c\,\mathrm{Col}_2$, causing $\det(\mathbf{A}) = 0$.
3. **Overlapping Lines of Action (Fig. 2c):** If two support nodes lie on a common line $\mathcal{L}$ with directors aligned parallel to $\mathcal{L}$, the second row becomes identical to the first ($\mathrm{Row}_2 = \mathrm{Row}_1$), causing rank deficiency ($\mathrm{rank}(\mathbf{A}) \leq 2$).
4. **Nodal Coincidence / Spatial Collapse (Fig. 2d):** Shared spatial coordinates ($\mathbf{x}_i \approx \mathbf{x}_j$) collapse spatial moment sampling, preventing 2D vector interpolation span.

3.1.1. Decoupling of Spatial Geometry and Vector Alignment

A crucial theoretical insight governing vector meshless approximations is the formal decoupling between spatial sampling quality and vector algebraic independence, as explicitly highlighted in Fig. 2e:

- **Spatially Ideal but Algebraically Singular (Fig. 2e, top):** A candidate triad can form an optimal 2D spatial distribution (e.g., an equilateral triangle enclosing a large physical area $|T| > 0$). A purely spatial geometric filter or distance-based k-d tree query would readily accept this support. However, if the assigned vector directors are globally parallel ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$), matrix $\mathbf{A}$ is strictly singular ($\det(\mathbf{A}) = 0$), rendering shape function construction impossible.

- **Algebraically Solvable but Spatially Degenerate (Fig. 2e, bottom):** Conversely, if three support nodes lie on a perfectly collinear 1D line in space (zero triangle area $|T| = 0$) but possess non-parallel directors, matrix $\mathbf{A}$ in the $\mathcal{L}^1$ basis can yield a non-zero determinant ($\det(\mathbf{A}) \neq 0$). While algebraically solvable, this configuration lacks 2D spatial sampling normal to the line of nodes, leading to unconstrained polynomial extrapolation and catastrophic derivative errors in field operators.

In contrast to the pathological or decoupled configurations depicted in Figs. 2(a)–(e), an admissible support—designated as an ideal support triad in Fig. 2(f)—requires a dual geometric-directional qualification. Geometrically, the candidate support nodes must form a non-degenerate 2D spatial

distribution ($|T| > 0$), shielding the local domain from nodal collapse or spatial collinearity. Directionally, the assigned vector directors $\vec{t}_k$ must provide coherent tangential circulation around the evaluation point $P(x_P, y_P)$, maximizing the directional moment $M_k = (\Delta\mathbf{x}_k \times \vec{t}_k)_z \gg 0$.

3.2. Singularity Analysis in the Complete $\mathcal{P}^1$ Vector Basis (6×6 Matrix)

Extending the singularity analysis to the complete vector basis $\mathcal{P}^1$ ($M_{\text{supp}} = 6$), we consider the corresponding 6×6 local placement matrix $\mathbf{A} \in \mathbb{R}^{6\times 6}$. Evaluated in local translated coordinates centered at the evaluation point $P(x_P, y_P)$ ($\Delta x_k = x_k - x_P, \Delta y_k = y_k - y_P$), matrix $\mathbf{A}$ (Eq. (38)) is restated below to facilitate the investigation:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & \Delta x_1 t_{1x} & \Delta y_1 t_{1x} & \Delta x_1 t_{1y} & \Delta y_1 t_{1y} \\ t_{2x} & t_{2y} & \Delta x_2 t_{2x} & \Delta y_2 t_{2x} & \Delta x_2 t_{2y} & \Delta y_2 t_{2y} \\ t_{3x} & t_{3y} & \Delta x_3 t_{3x} & \Delta y_3 t_{3x} & \Delta x_3 t_{3y} & \Delta y_3 t_{3y} \\ t_{4x} & t_{4y} & \Delta x_4 t_{4x} & \Delta y_4 t_{4x} & \Delta x_4 t_{4y} & \Delta y_4 t_{4y} \\ t_{5x} & t_{5y} & \Delta x_5 t_{5x} & \Delta y_5 t_{5x} & \Delta x_5 t_{5y} & \Delta y_5 t_{5y} \\ t_{6x} & t_{6y} & \Delta x_6 t_{6x} & \Delta y_6 t_{6x} & \Delta x_6 t_{6y} & \Delta y_6 t_{6y} \end{bmatrix} \tag{44}$$

By the unisolvence principle for vector degrees of freedom [3] (here termed the Null-Space Orthogonality Principle), a non-zero coefficient vector $\mathbf{p} = [a_1, a_2, b_1, b_2, b_3, b_4]^T \in \mathbb{R}^6$ belongs to $\ker(\mathbf{A}_{6\times 6})$ if and only if the corresponding affine vector field $\vec{E}(\mathbf{x}) \in \mathcal{P}^1$ (with components $E_x = a_1 + b_1\Delta x + b_2\Delta y$ and $E_y = a_2 + b_3\Delta x + b_4\Delta y$) satisfies the directional orthogonality condition $\vec{E}(\mathbf{x}_k) \cdot \vec{t}_k = 0$ across all six support nodes simultaneously. This characterizes exact singularity as the existence of a non-trivial affine vector field pointwise orthogonal to all six nodal directors.

While the complete $\mathcal{P}^1$ basis inherits the classical spatial-directional pathologies of $\mathcal{L}^1$ (such as global director parallelism and action-line concurrency), expanding the local placement space to 6 degrees of freedom creates new linear dependence modes. Specifically, algebraic analysis reveals four analytically identified singularity mechanisms governing $\mathbf{A}_{6\times 6}$:

1. **Partial Vector Parallelism (4-Node Threshold):** In contrast to $\mathcal{L}^1$ (which requires all 3 nodes to be parallel to collapse rank), $\mathbf{A}_{6\times 6}$ becomes strictly singular ($\det(\mathbf{A}) \equiv 0$, $\text{rank}(\mathbf{A}) \leq 5$) whenever **4 or more nodes** in a candidate sextet possess parallel vector directors,

with $\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3 \parallel \vec{t}_4$. Under coordinate rotation aligning the common direction with $\hat{x}$ ($t_{ky} = 0$), columns 2, 5, and 6 contain zeros in the first 4 rows, confining 3 vector columns to a 2-dimensional active row space and forcing linear dependence regardless of nodal spatial locations or the remaining 2 directors.

2. **Partial Spatial Collinearity (5-Node Threshold):** If **5 of the 6 support nodes** lie on a single straight line in 2D space ($\Delta y_k = m\Delta x_k$ for $k = 1, \ldots, 5$), columns 4 ($\Delta y_k t_{kx}$) and 6 ($\Delta y_k t_{ky}$) possess non-zero entries exclusively in the 6th row ($\text{Col}_4 \propto \text{Col}_6$). This direct column proportionality forces $\det(\mathbf{A}_{6\times6}) = 0$, regardless of the vector directors assigned to all six nodes.
3. **Total Spatial Collinearity Collapse (6 Nodes):** While 1D collinear nodes in $\mathcal{L}^1$ could yield a non-zero determinant if assigned non-parallel directors (becoming "algebraically solvable but spatially degenerate"), transitioning to $\mathcal{P}^1$ eliminates this spatial vulnerability. If all 6 support nodes are collinear in 2D space ($\Delta y_k = m\Delta x_k$), two independent column dependencies arise simultaneously ($\text{Col}_4 = m\,\text{Col}_3$ and $\text{Col}_6 = m\,\text{Col}_5$), collapsing the matrix rank to $\text{rank}(\mathbf{A}) \leq 4$, which implies $\det(\mathbf{A}_{6\times6}) \equiv 0$ unconditionally.
4. **Azimuthal Divergence Masking ($\vec{t}_k \perp \Delta\mathbf{x}_k$):** While pure concentric circulation ($\vec{t}_k \perp \Delta\mathbf{x}_k$) represented the optimal configuration for $\mathcal{L}^1$, it induces strict singularity in $\mathcal{P}^1$. Enforcing pure azimuthal alignment sets $\Delta x_k t_{kx} + \Delta y_k t_{ky} = 0$, yielding $\text{Col}_3 = -\,\text{Col}_6$ and rendering $\det(\mathbf{A}_{6\times6}) = 0$. Physically, pure azimuthal directors cannot sample the linear isotropic divergence mode $\vec{E}_{\text{div}} = [\Delta x, \Delta y]^T \in \mathcal{P}^1$, requiring the candidate support sextet to possess angular diversity containing both tangential and radial components.

3.3. Automated Support Selection Algorithm via Scale-Invariant Dimensionless Matrix

The mathematical classifications established in Sections 3.1 and 3.2 demonstrate that matrix singularity in the VNMM local placement system ($\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i$) stems from an intricate coupling between spatial node distributions ($\Delta x_k, \Delta y_k$) and unit vector director alignments (t_{kx}, t_{ky}). Furthermore, as established by the Null-Space Orthogonality Principle, the set of singular configurations forms an algebraic hypersurface: whenever the nodal directors happen to be pointwise orthogonal to any non-trivial

affine vector field $\vec{E} \in \mathcal{P}^1$ ($\vec{t_k} \propto R_{90}\vec{E}(\mathbf{x}_k)$), $\mathbf{A}_{6\times 6}$ becomes strictly singular, even in generic non-collinear point distributions with non-parallel directors. Consequently, attempting to construct a purely heuristic geometric selection scheme—relying on explicit conditional decision trees to test for individual geometric pathologies—is not only impractically complex and fragile, but fundamentally incomplete.

In contrast, evaluating the magnitude of the determinant $|\det(\mathbf{A}_{6\times 6})|$ of the local placement matrix serves as a unified algebraic filter against exact and near-singular configurations. Because $\mathbf{A}_{6\times 6}$ incorporates both spatial coordinates and vector director projections within a single operator, enforcing $|\det(\mathbf{A}_{6\times 6})| \geq \text{Tol}$ simultaneously screens out exact singular configurations (such as spatial collapse, partial vector parallelism, action-line concurrency, divergence masking, and arbitrary null-space orthogonality alignments) as well as candidate sextets in their immediate near-singular vicinity.

3.3.1. Scale-Invariant Support Selection via Dimensionless Matrix

In local translated coordinates centered at the evaluation point $P(x_P, y_P)$, the first two columns of the classical placement matrix $\mathbf{A} \in \mathbb{R}^{6\times 6}$ consist of dimensionless unit director components $t_{kx}, t_{ky} \sim \mathcal{O}(1)$, while columns 3–6 contain products of coordinates and directors, $\Delta x_k t_{kx}, \Delta y_k t_{kx}, \Delta x_k t_{ky}, \Delta y_k t_{ky} \sim \mathcal{O}(h)$ [m], where h denotes the local characteristic nodal spacing. Consequently, the standard placement matrix is dimensional, with mixed physical units ($[L]^0$ and $[L]^1$), and its determinant scales dimensionally as $[\det(\mathbf{A}_{6\times 6})] = [L]^4$:

$$\det(\mathbf{A}_{6\times 6}) \sim \mathcal{O}(h^4). \tag{45}$$

This quartic scaling introduces two fundamental difficulties if support qualification is applied directly to the dimensional matrix $\mathbf{A}$:

1. **Scale Coupling and Threshold Decay:** Because $\det(\mathbf{A}) \propto h^4$, the determinant decays rapidly under spatial refinement ($h \to 0$). For example, refining the grid spacing by a factor of 10 reduces $\det(\mathbf{A})$ by four orders of magnitude (10^4). Consequently, any criterion based directly on the dimensional determinant $\det(\mathbf{A})$ couples the algebraic qualification of stencils to the physical discretization scale, preventing the definition of a universal, scale-independent threshold across arbitrary domain sizes or multiscale meshes.
2. **Artificial Condition Number Inflation:** The mixed physical units create an artificial scaling imbalance between columns 1–2 ($\mathcal{O}(1)$) and

columns 3–6 ($\mathcal{O}(h)$). In standard 2-norm conditioning, the largest singular value is governed by the order-one columns ($\sigma_{\max}(\mathbf{A}) \sim \mathcal{O}(1)$), whereas the smallest singular value is constrained by the decaying spatial moment columns ($\sigma_{\min}(\mathbf{A}) \sim \mathcal{O}(h)$). Consequently, their ratio inflates the matrix condition number as $\kappa_2(\mathbf{A}) = \sigma_{\max}/\sigma_{\min} \propto \mathcal{O}(1/h)$ as $h \to 0$, purely as a metric artifact that obscures the genuine geometric and angular quality of the nodal arrangement.

To eliminate this $\mathcal{O}(h^4)$ scaling dependence and decouple algebraic qualification from physical dimensions, we formulate the support selection criterion and shape function calculation directly in terms of the **dimensionless placement matrix $\widehat{\mathbf{A}}$**.

Formulation of the Dimensionless Matrix $\widehat{\mathbf{A}}$ and Definition of h_{local}. To formulate a scale-invariant representation, local spatial coordinates must be normalized by a characteristic local length scale $h_{\text{local}} > 0$. For general point distributions without a background grid, we define the deterministic local length scale canonical to the evaluation point $\mathbf{x}_P$ as the mean Euclidean distance to its first six nearest spatial neighbors retrieved from the k-d tree:

$$h_{\text{local}}(\mathbf{x}_P) = \frac{1}{6} \sum_{k=1}^{6} \|\mathbf{x}_{(k)} - \mathbf{x}_P\|, \tag{46}$$

where $\mathbf{x}_{(k)}$ denotes the k-th nearest neighbor of $\mathbf{x}_P$ in Euclidean distance. In quasi-uniform point clouds where a nominal spacing h_{nom} is available, h_{local} can be chosen uniformly as $h_{\text{local}} = h_{\text{nom}}$, as discussed in Section 5.

Using h_{local}, we define normalized, dimensionless local coordinates:

$$\hat{x}_k = \frac{\Delta x_k}{h_{\text{local}}} = \frac{x_k - x_P}{h_{\text{local}}}, \quad \hat{y}_k = \frac{\Delta y_k}{h_{\text{local}}} = \frac{y_k - y_P}{h_{\text{local}}}. \tag{47}$$

For any candidate sextet $\{P_k, \vec{t}_k\}_{k=1}^{6}$, row k of the dimensionless matrix $\widehat{\mathbf{A}} \in \mathbb{R}^{6\times 6}$ is assembled as:

$$\widehat{\mathbf{A}}_{k,:} = \begin{bmatrix} t_{kx} & t_{ky} & \hat{x}_k t_{kx} & \hat{y}_k t_{kx} & \hat{x}_k t_{ky} & \hat{y}_k t_{ky} \end{bmatrix}, \quad k = 1, \ldots, 6. \tag{48}$$

Every entry of $\widehat{\mathbf{A}}$ is purely dimensionless and of order $\mathcal{O}(1)$. The exact algebraic relation connecting the dimensional placement matrix $\mathbf{A}$ and its dimensionless counterpart $\widehat{\mathbf{A}}$ is given by the diagonal scaling:

$$\mathbf{A} = \widehat{\mathbf{A}}\, \mathbf{D}_h, \quad \text{where } \mathbf{D}_h = \text{diag}(1,\ 1,\ h_{\text{local}},\ h_{\text{local}},\ h_{\text{local}},\ h_{\text{local}}). \tag{49}$$

Taking the determinant of Eq. (49) establishes the exact identity:

$$\det(\mathbf{A}) = \det(\widehat{\mathbf{A}})\det(\mathbf{D}_h) = h_{\text{local}}^4 \det(\widehat{\mathbf{A}}) \implies |\det(\widehat{\mathbf{A}})| = \frac{|\det(\mathbf{A})|}{h_{\text{local}}^4}. \quad (50)$$

Scale-Invariant Qualification and Preservation of Compact Supports. The support qualification criterion is thus formulated directly on the dimensionless determinant:

$$|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}} > 0, \quad (51)$$

where $\widehat{\text{Tol}}$ is a dimensionless scalar of order $\mathcal{O}(1)$ (e.g., $\widehat{\text{Tol}} \in [0.05, 1.50]$).

By operating directly on the balanced matrix $\widehat{\mathbf{A}}$, this criterion resolves the two fundamental difficulties identified at the beginning of Section 3.3.1:

1. **Resolution of Threshold Decay and Support Preservation:** By eliminating the $\mathcal{O}(h^4)$ scale coupling described in Difficulty 1, $|\det(\widehat{\mathbf{A}})|$ is strictly independent of the physical grid spacing $h \to 0$. As a result, geometrically well-proportioned configurations maintain $|\det(\widehat{\mathbf{A}})| \sim \mathcal{O}(1)$ down to arbitrarily fine discretizations, naturally preserving compact nearest-neighbor stencils without unphysical support dilation. In our numerical benchmarks under the calibrated operating threshold $\widehat{\text{Tol}} = 0.80$, the mean neighborhood query size retrieved by the k-d tree remains tightly bounded at $K_{\text{avg}} \approx 6.0$–$7.1$ on structured Cartesian lattices (Sections 5.4 and 5.5) and $K_{\text{avg}} \approx 10$–$11$ on stochastically perturbed clouds (Sections 5.2 and 5.3, Tables 3 and 4).
2. **Elimination of Artificial Condition Number Inflation:** By restoring dimensional parity across all columns, the condition number $\kappa_2(\widehat{\mathbf{A}})$ is cleared of the spurious $\mathcal{O}(1/h)$ penalty highlighted in Difficulty 2, reflecting purely the intrinsic geometric and angular arrangement of the nodes. As demonstrated empirically in the stochastic sensitivity analysis of Section 5.3 (Table 4), enforcing $\widehat{\text{Tol}} \geq 0.80$ under 25% spatial jitter and random directors effectively bounds the condition number to $\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$ across spatial refinement (in contrast to the unscaled dimensional matrix where $\kappa_2(\mathbf{A}) \approx 80$).

Determinant Sensitivity to h_{local} Scaling. Equation (50) also highlights an important scaling property regarding the choice of length scale h_{local}: because the four spatial columns of $\widehat{\mathbf{A}}$ (columns 3–6) scale inversely with h_{local}, any

alternate choice of normalization scale $h_{\text{local}} \to c\, h_{\text{local}}$ (with $c > 0$) induces a quartic transformation on the determinant:

$$|\det(\widehat{\mathbf{A}})| \longrightarrow c^{-4} |\det(\widehat{\mathbf{A}})|. \tag{52}$$

Consequently, while the threshold qualification criterion (51) is strictly invariant with respect to spatial domain refinement ($h \to 0$ for a consistent convention of h_{local}), the specific numerical values of $|\det(\widehat{\mathbf{A}})|$ (and hence calibrated numerical thresholds, such as $\widehat{\text{Tol}} = 0.80$) are conditioned on the chosen definition of h_{local}. The specific normalization conventions and their scaling behavior across our numerical benchmarks are detailed in Section 5.

Determinant-Based Empirical Support Qualification versus Condition Number. In analyzing stencil stability, it is essential to distinguish clearly between three algebraic levels:

- **Exact singularity ($\det(\widehat{\mathbf{A}}) = 0$):** Represents strict rank deficiency ($\text{rank}(\widehat{\mathbf{A}}) < 6$) and a non-trivial null space ($\ker(\widehat{\mathbf{A}}) \neq \{\mathbf{0}\}$), where inversion is mathematically impossible;
- **Near-singularity ($|\det(\widehat{\mathbf{A}})| \ll 1$):** Occurs when candidate support sextets lie in the close algebraic vicinity of the singular hypersurface in configuration space; and
- **Poor conditioning ($\kappa_2(\widehat{\mathbf{A}}) \gg 1$):** Measured by the ratio of extremal singular values $\kappa_2(\widehat{\mathbf{A}}) = \sigma_{\max}/\sigma_{\min}$, quantifying numerical error magnification under inversion.

From a linear algebra standpoint, it is important to formalize the relation between the determinant threshold (51) and the matrix condition number $\kappa_2(\widehat{\mathbf{A}}) = \sigma_{\max}(\widehat{\mathbf{A}})/\sigma_{\min}(\widehat{\mathbf{A}})$. Because the determinant equals the product of all six singular values ($|\det(\widehat{\mathbf{A}})| = \prod_{i=1}^{6} \sigma_i(\widehat{\mathbf{A}})$), the determinant governs the product of singular values rather than their ratio $\kappa_2(\widehat{\mathbf{A}}) = \sigma_{\max}/\sigma_{\min}$. A lower bound on $|\det(\widehat{\mathbf{A}})|$ therefore does not directly control the condition number in an arbitrary, unconstrained setting. Specifically, bounding $|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$ provides a lower bound on the minimal singular value:

$$\sigma_{\min}(\widehat{\mathbf{A}}) \geq \frac{|\det(\widehat{\mathbf{A}})|}{\sigma_{\max}^5(\widehat{\mathbf{A}})} \geq \frac{\widehat{\text{Tol}}}{\sigma_{\max}^5(\widehat{\mathbf{A}})}, \tag{53}$$

which translates into an upper bound on the 2-norm condition number:

$$\kappa_2(\widehat{\mathbf{A}}) \leq \frac{\sigma_{\max}^6(\widehat{\mathbf{A}})}{\widehat{\mathrm{Tol}}}. \tag{54}$$

To make inequality (54) effective, the maximal singular value $\sigma_{\max}(\widehat{\mathbf{A}})$ must be strictly controlled. Recalling that the spectral norm is bounded by the Frobenius norm ($\sigma_{\max}(\mathbf{M}) \leq \|\mathbf{M}\|_F$) [36], we evaluate from the definition of $\widehat{\mathbf{A}}$ in Eq. (48):

$$\sigma_{\max}^2(\widehat{\mathbf{A}}) \leq \|\widehat{\mathbf{A}}\|_F^2 = \sum_{k=1}^{6} \left(\|\vec{t}_k\|^2 + \|\hat{\mathbf{x}}_k\|^2 \|\vec{t}_k\|^2\right) = \sum_{k=1}^{6} \left(1 + \|\hat{\mathbf{x}}_k\|^2\right) \leq 6(1+C^2), \tag{55}$$

provided that candidate support nodes satisfy the **bounded spatial proximity hypothesis**:

$$\|\hat{\mathbf{x}}_k\| = \frac{\|\mathbf{x}_k - \mathbf{x}_P\|}{h_{\text{local}}} \leq C \quad \forall k = 1, \ldots, 6, \tag{56}$$

for a moderate constant $C > 0$. In the proposed selection algorithm (Section 3.3.2), hypothesis (56) is enforced by Phase 1, which queries the immediate K_{initial} spatial neighbors via the k-d tree, bounding candidate distances such that typically $C \leq 3$.

It is essential to stress, however, that without an explicit upper bound C on the candidate search radius, the condition $|\det(\widehat{\mathbf{A}})| \geq \widehat{\mathrm{Tol}}$ alone would not prevent $\sigma_{\min} \to 0$, as candidate nodes could be selected arbitrarily far away ($\|\hat{\mathbf{x}}_k\| \to \infty$) to artificially inflate the determinant while remaining ill-conditioned. Furthermore, even with C bounded, inequality (54) represents a conservative upper bound rather than an algebraic equivalence. Therefore, the criterion (51) is conceptually presented as a **determinant-based empirical support qualification** mechanism, rather than an absolute mathematical guarantee of condition number control. Its operational validity is demonstrated empirically in Section 5.3 and Table 4, where direct measurements of $|\det(\widehat{\mathbf{A}})|$ versus $\kappa_2(\widehat{\mathbf{A}})$ confirm that enforcing $\widehat{\mathrm{Tol}} \in [0.80, 1.50]$ within compact neighborhoods ($C \leq 3$) empirically bounds the local condition number to $\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$ in the tested stochastic configuration.

Computation of Vector Shape Functions and Differential Operators. Solving the linear collocation system $\mathbf{A}\boldsymbol{\beta} = \mathbf{I}_6$ via $\widehat{\mathbf{A}}$ is numerically straightforward.

Inverting the well-conditioned dimensionless matrix yields $\widehat{\boldsymbol{\beta}} = \widehat{\mathbf{A}}^{-1}$. From $\boldsymbol{\beta} = \mathbf{D}_h^{-1}\widehat{\boldsymbol{\beta}}$, the vector shape functions and their spatial derivatives at the evaluation point P ($\Delta x = 0, \Delta y = 0$) are obtained directly as:

$$\vec{N}_i(P) = \begin{bmatrix} \hat{\beta}_{1i} \\ \hat{\beta}_{2i} \end{bmatrix}, \tag{57}$$

$$(\nabla \times \vec{N}_i)_z = \frac{1}{h_{\text{local}}} \left(\hat{\beta}_{5i} - \hat{\beta}_{4i} \right), \tag{58}$$

$$\nabla \cdot \vec{N}_i = \frac{1}{h_{\text{local}}} \left(\hat{\beta}_{3i} + \hat{\beta}_{6i} \right). \tag{59}$$

Equations (57)–(59) show that the $\mathcal{O}(h^{-1})$ gradient scaling is factored out analytically through the scalar prefactor $1/h_{\text{local}}$, guaranteeing maximum numerical precision during matrix inversion.

3.3.2. The 4-Phase Selection Pipeline

To locate well-conditioned candidate sextets across unstructured point clouds without incurring exhaustive $\mathcal{O}(N^2)$ pairwise distance evaluations, the proposed selection framework combines orthogonal k-d tree spatial queries [37, 33, 38, 31, 39] with algebraic dimensionless qualification. The resulting algorithm operates through a 4-phase pipeline:

1. **Phase 1: Spatial Neighborhood Query via k-d Tree:** Using an orthogonal k-d tree binary spatial search structure constructed over all domain nodes ($\mathcal{O}(N \log N)$ pre-processing complexity), the algorithm queries the $K_{\text{initial}} \geq 12$ nearest physical neighbors surrounding the evaluation point $P(x_P, y_P)$ with an average query complexity of $\mathcal{O}(\log N)$ [37, 33, 38, 39]. The local spacing h_{local} is determined deterministically from the query distances via the canonical 6-neighbor mean (46) (or fixed to the nominal resolution h_{nom} for structured benchmark sequences).
2. **Phase 2: Combinatorial Candidate Generation with Progressive Anchor:** Candidate support sextets are generated systematically from the queried K_{initial} neighborhood. To prioritize geometric proximity to the evaluation point P, candidate sextets are assembled by fixing an anchor node (beginning with the node closest to P) and taking 5-combinations from subsequent nearest neighbors, ordered by increasing mean Euclidean distance.

3. **Phase 3: Local Coordinate Normalization and Dimensionless Matrix Assembly:** For each candidate sextet, spatial coordinates are translated and normalized to the local dimensionless evaluation frame ($\hat{x}_k = (x_k - x_P)/h_{\text{local}}, \hat{y}_k = (y_k - y_P)/h_{\text{local}}$), and the dimensionless placement matrix $\widehat{\mathbf{A}}_{6\times 6}$ is assembled using Eq. (48).
4. **Phase 4: Dimensionless Algebraic Qualification with Early Stopping:** The dimensionless determinant magnitude $|\det(\widehat{\mathbf{A}}_{6\times 6})|$ is computed. The first candidate sextet satisfying the scale-invariant criterion $|\det(\widehat{\mathbf{A}}_{6\times 6})| \geq \widehat{\text{Tol}}$ via Eq. (51) is immediately accepted as the active support, terminating the local search loop.

3.3.3. Computational Complexity and Combinatorial Cost Analysis

A critical aspect of practical meshless implementations concerns the algorithmic complexity of support selection [11, 33, 39]. In analyzing this pipeline, it is essential to distinguish clearly among three separate concepts: (i) the asymptotic complexity of the spatial neighborhood search, (ii) the theoretical worst-case combinatorial bound of candidate support generation, and (iii) the observed empirical cost achieved through early stopping.

The k-d tree provides logarithmic average neighborhood-query cost under standard assumptions ($\mathcal{O}(\log N)$ per evaluation point, preceded by an $\mathcal{O}(N \log N)$ one-off tree construction cost) [37, 33, 38, 39], whereas the subsequent support qualification has a worst-case combinatorial cost bounded by the binomial coefficient:

$$N_{\text{comb}}^{\max} = \binom{K_{\text{initial}}}{6}. \tag{60}$$

For the baseline initial neighborhood of $K_{\text{initial}} = 12$ nodes employed in this work, Eq. (60) yields $\binom{12}{6} = 924$ potential candidate sextets. If an exhaustive search were conducted across all candidate combinations, the algebraic qualification cost per evaluation point would scale as $C_{\text{eval}}^{\text{exhaustive}} \approx \binom{K_{\text{initial}}}{6} C_{\text{det}}$, where $C_{\text{det}} \sim \mathcal{O}(6^3) = \mathcal{O}(1)$ denotes the fixed floating-point cost of evaluating a 6×6 determinant. While $\binom{K_{\text{initial}}}{6}$ represents a local upper bound that is asymptotically independent of the global node count N, evaluating all combinations would still represent a substantial computational overhead across N_{eval} evaluation points.

Crucially, however, the proposed selection algorithm does *not* perform an exhaustive sweep. By coupling progressive proximity-ordered candidate

generation (Phase 2) with the **early-stopping qualification criterion** (Phase 4), the search evaluates candidate sextets sequentially and terminates immediately once an admissible support satisfying $|\det(\widehat{\mathbf{A}}_{6\times6})| \geq \widehat{\mathrm{Tol}}$ is identified.

This establishes a necessary distinction between the *neighborhood query size* (K_{avg}, the physical number of nearest neighbors retrieved from the k-d tree) and the *qualification count* ($\bar{N}_{\mathrm{tested}}$, the actual number of candidate sextets evaluated algebraically via $\det(\widehat{\mathbf{A}}_{6\times6})$ before an admissible support is found). While an initial candidate neighborhood of $K_{\mathrm{initial}} = 12$ nodes yields up to 924 potential combinations, early stopping drastically truncates the combinatorial tree, identifying an admissible configuration after evaluating only a minimal fraction of these possibilities.

To quantify the computational efficiency of early stopping in practice, Table 2 reports the empirical performance metrics measured across six refinement levels of the electromagnetic resonant cavity problem (formally formulated in Section 4 and investigated in Section 5.5), evaluated at $N_{\mathrm{eval}} = 400$ Gauss integration points.

Table 2: Computational cost of automated support selection across six cavity discretization levels ($N_{\mathrm{eval}} = 400$ Gauss evaluation points, $K_{\mathrm{initial}} = 12$).

Grid Nodes	**Mean Neighbors**	**Evaluated Sextets** (N_{tested})		**Support Time**	**Assembly Time**
N	K_{avg}	**Mean** ($\bar{N}_{\mathrm{tested}}$)	**Maximum**	t_{sup} [s]	t_{ass} [s]
81 (9×9)	6.6	**3.75**	29	0.012	0.025
169 (13×13)	6.6	**3.88**	29	0.012	0.025
289 (17×17)	6.6	**4.64**	29	0.014	0.027
441 (21×21)	7.1	**3.21**	9	0.011	0.023
625 (25×25)	6.7	**2.61**	8	0.009	0.023
841 (29×29)	6.6	**2.77**	8	0.010	0.022

As shown in Table 2, the actual number of candidate sextets evaluated per evaluation point remains remarkably small: for the tested cavity sequence, the mean qualification count $\bar{N}_{\mathrm{tested}}$ ranges from only 2.61 to 4.64 sextets (drastically below the combinatorial bound of 924), with the maximum evaluations never exceeding 29 across all 400 quadrature points. Support qualification consumes only 0.009–0.014 s, representing approximately 40% of the total global matrix assembly time ($t_{\mathrm{ass}} \approx 0.025$ s).

Even under severe 25% stochastic spatial jitter and random director orientations (Section 5.3), early stopping remains highly efficient across all scales: enforcing the stabilization threshold $\widehat{\mathrm{Tol}} \approx 0.80$ requires testing an average

of $\bar{N}_{\text{tested}} \approx 48$–$120$ candidate sextets (out of 924 possible combinations) to bypass near-parallel directors and near-collinear arrangements, with support qualification consuming merely 0.15–0.35 s across all 400 evaluation points (< 1 ms per evaluation point). When a more permissive threshold is employed ($\widehat{\text{Tol}} \approx 0.10$), $\bar{N}_{\text{tested}}$ drops to merely 1.8–3.8 evaluations per point (< 0.02 s total time), confirming that early stopping eliminates combinatorial bottlenecks across both regular and irregular discretizations.

Distinguishing the worst-case bound ($\binom{K_{\text{initial}}}{6} C_{\text{det}}$) from the observed empirical cost, the total operational runtime across N_{eval} evaluation points can be expressed as:

$$T_{\text{support}} = \underbrace{\mathcal{O}(N \log N)}_{k\text{-d tree construction}} + \sum_{p=1}^{N_{\text{eval}}} \left[\underbrace{\mathcal{O}(\log N)}_{\text{spatial query}} + \underbrace{\bar{N}_{\text{tested}}^{(p)} C_{\det(6\times 6)}}_{\text{observed algebraic qualification}} \right]. \tag{61}$$

While the worst-case qualification cost is bounded by $\binom{K_{\text{initial}}}{6}$, the empirical early-stopping count remains very small ($\bar{N}_{\text{tested}} \approx 2.6$–$4.6$ in unperturbed domains and moderate in perturbed clouds). Consequently, in the tested configurations, the observed support selection overhead is dominated by the logarithmic spatial neighbor query rather than by combinatorial matrix evaluations, reconciling robust support identification with practical computational efficiency.

4. Variational Formulation of the Resonant Cavity Problem

To evaluate the spectral accuracy and verify the suppression of non-physical spurious modes within the operational bandwidth in the Vector Nodal Meshless Method (VNMM 2D), we formulate the electromagnetic vector eigenvalue problem in a two-dimensional resonant cavity $\Omega = [0, \pi] \times [0, \pi] \subset \mathbb{R}^2$ with perfect electric conductor (PEC) walls $\partial\Omega$ [2].

4.1. The Transverse Electric (TE_z) Vector Eigenvalue Problem

In a homogeneous, isotropic, and lossless dielectric medium ($\epsilon_r = 1.0, \mu_r = 1.0$), time-harmonic Transverse Electric modes (TE_z) are governed

by the vector double-curl Helmholtz equation for the transverse electric field $\vec{E}(\mathbf{x}) = [E_x(x,y), E_y(x,y)]^T$ [3, 25]:

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \vec{E} \right) = \lambda \, \epsilon_r \vec{E}, \quad \text{in } \Omega \tag{62}$$

$$\hat{n} \times \vec{E} = \mathbf{0}, \quad \text{on } \partial\Omega \tag{63}$$

where $\lambda = k_0^2 = \omega^2 \mu_0 \epsilon_0$ denotes the target wavenumber eigenvalue, ω is the angular resonance frequency, and $\hat{n}$ is the outward unit normal vector to boundary $\partial\Omega$. In two dimensions, the curl operator acting on a 2D vector field yields a scalar normal component $(\nabla \times \vec{E})_z = \partial E_y/\partial x - \partial E_x/\partial y$, while the vector curl of a scalar field u_z returns a vector in the transverse plane $\nabla \times u_z = [\partial u_z/\partial y, -\partial u_z/\partial x]^T$.

The exact analytical eigenvalues for the square PEC cavity are given by [2]:

$$\lambda_{m,n}^{\text{exact}} = m^2 + n^2, \quad m, n \in \{0, 1, 2, \dots\}, \quad (m + n > 0) \tag{64}$$

4.2. Ritz-Galerkin Weak Form with Div-Curl Regularization

To solve the vector boundary-value problem (62)–(63) numerically, the strong-form double-curl Helmholtz equation must be recast into a variational weak formulation. In standard electromagnetic field theory, testing the Helmholtz equation against an admissible vector field $\vec{W}$ and applying vector Green's first identity yields the canonical curl-curl weak form. Let the underlying Hilbert space of square-integrable vector fields with square-integrable curl satisfying homogeneous tangential PEC boundary conditions be defined as [3, 8]:

$$\mathcal{S} \triangleq \left\{ \vec{E} \in H(\mathbf{curl}; \Omega) \;\middle|\; \hat{n} \times \vec{E} = \mathbf{0} \text{ on } \partial\Omega \right\}. \tag{65}$$

The classical, unregularized variational problem seeks eigenpairs $(\lambda, \vec{E}) \in \mathbb{R} \times \mathcal{S} \setminus \{\mathbf{0}\}$ satisfying:

$$\int_\Omega \frac{1}{\mu_r} (\nabla \times \vec{W})_z (\nabla \times \vec{E})_z \, d\Omega = \lambda \int_\Omega \epsilon_r \vec{W} \cdot \vec{E} \, d\Omega, \quad \forall \vec{W} \in \mathcal{S}. \tag{66}$$

While Eq. (66) is spectrally well-behaved when discretized using globally conforming Nédélec edge elements (where exact de Rham sequence preservation confines the infinite-dimensional static gradient nullspace strictly to $\lambda = 0$),

implementing this standard formulation directly within a meshless framework triggers severe spectral contamination: non-physical, highly oscillatory spurious modes infiltrate the low-frequency physical spectrum.

To understand and resolve this spectral pathology, this subsection is organized as follows:

- Section 4.2.1 analyzes the theoretical mechanism of spectral pollution, demonstrating how the global non-conformity of dynamically selected meshless supports destroys the discrete de Rham complex and causes discrete gradient nullspace leakage;
- Section 4.2.2 introduces a div-curl penalty regularized weak formulation that selectively penalizes irrotational gradient modes and shifts them outside the operational bandwidth.

4.2.1. Theoretical Origin: From Global Non-Conformity to Gradient Leakage

In standard Edge Finite Element Formulations (EFEM), the discrete space is globally $H(\mathbf{curl})$-conforming ($V_h \subset \mathcal{S}$) because shared tangential degrees of freedom across conforming element facets guarantee exact tangential trace continuity ($\hat{n} \times [\vec{E}^h] = \mathbf{0}$) [3, 5]. Under this global conformity, the discrete de Rham complex property $\mathrm{Im}(\nabla) \subset \ker(\nabla\times)$ holds exactly: irrotational gradient fields ($\vec{E} = \nabla\phi$) yield an identically zero curl ($\nabla \times (\nabla\phi) \equiv \mathbf{0}$) and map directly into the zero eigenvalue subspace ($\lambda = 0$), preventing spectrum contamination without auxiliary constraints [3, 6].

In contrast, on unconstrained point clouds, the assembled meshless VNMM vector approximation is globally non-conforming:

$$V_h \not\subset \mathcal{S} \subset H(\mathbf{curl}; \Omega). \tag{67}$$

Although each local vector shape function $\vec{N}_i(\mathbf{x})$ possesses local $H(\mathbf{curl})$ regularity within its local support stencil (Section 2.3), active support stencils are selected dynamically across the domain via k-d tree spatial queries. Consequently:

1. **Non-Inheritance of Discrete de Rham Exactness:** Because tangential traces suffer minor non-polynomial jumps across shifting support boundaries, the discrete approximation does not inherit the exactness of the discrete de Rham sequence. Specifically, discrete gradient fields fail to preserve the irrotational property during quadrature-based evaluation:

$$\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}. \tag{68}$$

2. **Discrete Gradient Nullspace Leakage:** Because $\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$ at background quadrature points, non-zero numerical curl residues leak the infinite-dimensional static gradient nullspace into the non-zero discrete spectrum, causing non-physical, highly oscillatory spurious modes to infiltrate the physical resonance band.
3. **The Methodological Chain:** This establishes the cohesive architectural chain of the proposed framework:

$$\boxed{\begin{array}{c} \mathcal{L}^1 \text{ (derivative defect)} \longrightarrow \mathcal{P}^1 \text{ (local completeness)} \\ \Downarrow \\ \text{dynamic supports} \longrightarrow V_h \not\subset H(\mathbf{curl}) \text{ (non-conformity)} \\ \Downarrow \\ \text{loss of de Rham exactness} \longrightarrow \text{gradient leakage (spurious modes)} \\ \Downarrow \\ \mathbf{K}_{\text{div}} \text{ regularization} \longrightarrow \text{spectral stabilization} \end{array}} \tag{69}$$

As outlined in Eq. (69), the methodological progression of this paper is following a direct causal chain: (i) upgrading from the incomplete 3-node basis $\mathcal{L}^1$ to the complete 6-node space $\mathcal{P}^1$ eliminates derivative modal aliasing and restores local curl consistency (Section 2.3); (ii) assembling well-conditioned 6-node stencils on arbitrary point clouds, however, inevitably relies on dynamically shifting nearest-neighbor supports (Section 3.3), which renders the global discrete space non-conforming ($V_h \not\subset H(\mathbf{curl})$); (iii) this global non-conformity breaks the discrete de Rham complex, producing numerical curl residuals ($\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$) that leak the static gradient nullspace into the non-zero discrete spectrum as spurious modes; and (iv) this unavoidable leakage can be mitigated by the divergence penalty matrix $\mathbf{K}_{\text{div}}$ developed next, which selectively adds numerical stiffness to irrotational gradient states and pushes them out of the operational frequency bandwidth.

4.2.2. Div-Curl Regularized Weak Formulation

To suppress these spurious modes without introducing saddle-point system indefiniteness or auxiliary Lagrange multipliers, we adopt a **Ritz-Galerkin weak formulation with div-curl penalty regularization**

[3, 8]:

$$\int_\Omega \frac{1}{\mu_r}(\nabla \times \vec{W})_z(\nabla \times \vec{E})_z \, d\Omega + s_{\text{div}} \int_\Omega (\nabla \cdot \vec{W})(\nabla \cdot \vec{E}) \, d\Omega = \lambda \int_\Omega \epsilon_r \vec{W} \cdot \vec{E} \, d\Omega, \quad \forall \vec{W} \in \mathcal{S} \tag{70}$$

where $s_{\text{div}} > 0$ is the non-dimensional divergence penalty parameter [3].

Upon discretizing the electric field as $\vec{E}^h(\mathbf{x}) = \sum_{j=1}^{M} e_j \vec{N}_j(\mathbf{x})$ using the scalar projection DoFs $e_j = \vec{E}(\mathbf{x}_j) \cdot \vec{t}_j$ (Eq. (3)) and test functions as $\vec{W}^h(\mathbf{x}) = \sum_{i=1}^{M} d_i \vec{N}_i(\mathbf{x})$ with test coefficients $d_i = \vec{W}(\mathbf{x}_i) \cdot \vec{t}_i$, the second bilinear term in Eq. (70) defines the **divergence penalty matrix** $\mathbf{K}_{\text{div}} \in \mathbb{R}^{M \times M}$, whose entries $K_{\text{div},ij}$ are explicitly defined by:

$$K_{\text{div},ij} \triangleq \int_\Omega (\nabla \cdot \vec{N}_i)(\nabla \cdot \vec{N}_j) \, d\Omega \tag{71}$$

Consequently, the global stiffness matrix $\mathbf{K}$ in the discrete generalized eigenvalue problem $\mathbf{Ke} = \lambda \mathbf{Me}$ decomposes into the sum of the primary curl-curl stiffness matrix $\mathbf{K}_{\text{curl}}$ and the scaled divergence penalty matrix $s_{\text{div}}\mathbf{K}_{\text{div}}$:

$$\mathbf{K} = \mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}} \tag{72}$$

with individual entries of the primary curl-curl stiffness matrix defined by:

$$K_{\text{curl},ij} \triangleq \int_\Omega \frac{1}{\mu_r}(\nabla \times \vec{N}_i)_z(\nabla \times \vec{N}_j)_z \, d\Omega. \tag{73}$$

The physical and algebraic role of the penalty matrix $\mathbf{K}_{\text{div}}$ scaled by s_{div} is twofold [3, 5]:

1. **Asymptotic Preservation of Physical Modes:** Solenoidal physical TE_z modes satisfy $\nabla \cdot \vec{E} \equiv 0$ analytically. At the continuous level, the divergence penalty term vanishes identically ($\int_\Omega (\nabla \cdot \vec{E})^2 d\Omega = 0$), leaving analytical physical eigenvalues unperturbed [3, 1]. In the discrete meshless formulation, because pointwise divergence is satisfied only approximately ($\nabla \cdot \vec{E}^h \sim \mathcal{O}(h)$), the penalty introduces a minor discrete perturbation that systematically decays under spatial refinement, maintaining the divergence-to-curl energy ratio $R_{\text{div}} \triangleq \frac{\mathbf{e}^T \mathbf{K}_{\text{div}} \mathbf{e}}{\mathbf{e}^T \mathbf{K}_{\text{curl}} \mathbf{e}} \ll 1$ for physical modes.

2. **Penalization of Gradient Modes:** Non-physical gradient fields satisfy $\nabla \cdot (\nabla\phi) = \Delta\phi \neq 0$. The penalty term $s_{\text{div}}\mathbf{K}_{\text{div}}\mathbf{e}$ adds artificial numerical stiffness to these states, shifting their spurious eigenvalues to high frequencies outside the operational bandwidth ($\lambda > 50$ in the tested cavity discretization) [3, 5, 6].

As demonstrated by the systematic parametric tuning in Section 5.4, the value $s_{\text{div}} = 6.0$ was selected as a representative value within the tested stable penalty range ($s_{\text{div}} \in [4.0, 10.0]$), achieving reliable spectral purification without inducing numerical locking.

4.3. Integration Strategy: Cell-Based vs. Point-Based (EFG-Style)

Numerical evaluation of weak form integrals in Eq. (70) requires background cell quadrature. In early VNMM implementations, support nodes and local moment placement matrices $\mathbf{A}$ were constructed once per background cell at its centroid P_c [29, 25]. However, as demonstrated by the background cell integration analysis in Section 5, expanding shape functions from P_c across large cells ($dx > h$) induces severe Taylor truncation degradation and matrix indefiniteness.

To obtain a mesh-connectivity-free field approximation, we employ a point-based Element-Free Galerkin (EFG) integration strategy over a background integration grid [14, 33]:

1. A regular background grid of $N_{cx} \times N_{cy}$ quadrilateral integration cells covers the computational domain Ω. In this work, two integration configurations are systematically employed depending on the numerical objective:
    - *Single-Mesh Engineering Baseline (Section 5.4):* An economical background grid of 10×10 cells ($N_c = 100$) integrated with a 2×2 Gauss-Legendre rule (4 points per cell, $N_g = 400$, providing $N_g/N_{\text{DoF}} \approx 1.11$ on the $N = 21 \times 21$ mesh), balancing integration fidelity and rapid assembly ($t_{\text{ass}} = 0.06\,\text{s}$).
    - *Multi-Level Asymptotic Convergence Refinement (Sections 5.5 and 5.6):* A node-conforming background grid where cell boundaries align with the nodal resolution ($N_{cx} = N_{cy} = N_v - 1$) combined with a high-order 3×3 Gauss-Legendre rule (9 points per cell, maintaining $N_g/N_{\text{DoF}} \approx 9.4$–$10.4$). As established by the quadrature sensitivity study in Section 5.4.4, this refined background

quadrature substantially reduces the quadrature-induced bias observed in the coarse baseline, ensuring that asymptotic convergence curves faithfully reflect the spatial discretization trend of the vector shape functions as $h_{\mathrm{DoF}} \to 0$.

2. **Pointwise Support Selection (Section 3.3.2):** At each Gauss integration point $P_g(x_g, y_g)$, an admissible support sextet $\{P_k, \vec{t}_k\}_{k=1}^{6}$ is determined on-the-fly using the 4-phase selection pipeline: a localized k-d tree query identifies nearest neighbors and sets the local scale h_{local}, followed by progressive proximity-ordered candidate generation and scale-invariant determinant qualification ($|\det(\widehat{\mathbf{A}}_{6\times 6})| \geq \widehat{\mathrm{Tol}}$). By coupling early stopping with scale invariance, this approach avoids combinatorial overhead while guaranteeing compact, well-conditioned supports without threshold decay under spatial refinement.
3. **Pointwise Shape Function Evaluation (Section 3.3.1):** Anchoring the coordinate frame directly at P_g ($\Delta x = 0, \Delta y = 0$) avoids expanding shape functions from distant cell centroids, preventing the severe Taylor truncation degradation observed in cell-based schemes. The vector shape functions $\vec{N}_i(P_g)$ and their spatial derivatives ($\nabla \times \vec{N}_i, \nabla \cdot \vec{N}_i$) are evaluated directly via Eqs. (57)–(59), with the $\mathcal{O}(h^{-1})$ gradient scaling factored out analytically through $1/h_{\mathrm{local}}$, ensuring optimal floating-point precision and algebraic stability during weak-form assembly.

Classification as a Meshless Formulation. In classical meshfree literature [14, 13, 10], a distinction is drawn between methods that construct approximations free of mesh connectivity (such as EFG, PIM, and the present VNMM) and methods that evaluate weak integrals without background cells (such as nodal integration [40] or MLPG [13]). In the present VNMM formulation:

- The field approximation and shape functions $\vec{N}_i(\mathbf{x})$ are strictly meshless and connectivity-free: no triangulation, polygonization, or element topological data structures are created or maintained between approximation nodes.

- Background quadrilateral cells serve solely as a geometric scaffolding for evaluating numerical integrals via Gauss-Legendre quadrature, exactly as in standard Element-Free Galerkin (EFG) schemes [14, 33].

- The background integration grid does not define or constrain the interpolation topology; support selection at any point is governed exclusively by k-d tree spatial proximity and algebraic determinant qualification.

4.4. Algebraic Discretization and Generalized Eigenvalue System

Under the discrete field representations $\vec{E}^h$ and $\vec{W}^h$ introduced in Section 4.2.2, substituting the vector shape functions into the regularized weak form (70) yields the global generalized algebraic eigenvalue problem $\mathbf{Ke} = \lambda\mathbf{Me}$ [25, 31]. The global regularized stiffness matrix $\mathbf{K} = \mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}} \in \mathbb{R}^{M\times M}$ is assembled directly from the curl-curl and divergence penalty blocks defined in Eqs. (71)–(73), whereas the entries of the global mass matrix $\mathbf{M} \in \mathbb{R}^{M\times M}$ are governed by the permittivity-weighted bilinear product on the right-hand side of Eq. (70):

$$M_{ij} \triangleq \int_{\Omega} \epsilon_r \, \vec{N}_i \cdot \vec{N}_j \, d\Omega. \tag{74}$$

Homogeneous PEC boundary conditions are collocated directly at boundary nodes by setting $e_k = 0$ for all boundary nodes whose unit directors $\vec{t}_k$ align tangent to $\partial\Omega$ ($\vec{t}_k \parallel \partial\Omega$), which enforces $\vec{E}^h(\mathbf{x}_k) \cdot \vec{t}_k = 0$ pointwise [29, 31]. Eliminating these constrained degrees of freedom yields the reduced algebraic eigenvalue system:

$$\mathbf{K}_{\text{red}}\mathbf{e}_{\text{red}} = \lambda\mathbf{M}_{\text{red}}\mathbf{e}_{\text{red}}, \tag{75}$$

of size $M_{\text{red}} \times M_{\text{red}}$ (equivalently designated as $N_{\text{DoF}} \equiv M_{\text{red}}$ in subsequent convergence comparisons).

While the unregularized Maxwell stiffness operator is symmetric positive-semidefinite due to the static gradient nullspace, the inclusion of the divergence penalty ($s_{\text{div}} > 0$) combined with essential PEC boundary constraints renders $\mathbf{K}_{\text{red}}$ strictly positive-definite on the discrete physical subspace, while the mass matrix $\mathbf{M}_{\text{red}}$ is natively positive-definite. This ensures that the discrete generalized eigenvalue problem involves well-behaved, sparse symmetric positive-definite (SPD) reduced matrices, allowing efficient and stable computation of the physical spectrum using Krylov-subspace shift-and-invert Arnoldi methods [41, 42], as implemented via the ARPACK solver in SciPy [43].

5. Numerical Verification and Validation

This section assesses the numerical performance of the proposed Vector Nodal Meshless Method (VNMM 2D). Following established guidelines in computational mechanics and electromagnetics, the numerical campaign is systematically organized into three complementary stages:

1. **Numerical Verification (Sections 5.1–5.3):** Focused on the mathematical properties of the discretized operators, including local constructive equivalence against Whitney 1-form finite elements under conforming midpoint/tangent placement on a single simplex, restoration of asymptotic curl convergence via the complete linear vector basis $\mathcal{P}^1$, and local algebraic placement matrix conditioning under stochastic spatial-directional jitter.
2. **Physics-Based Spectral Validation (Sections 5.4–5.6):** Focused on practical electromagnetic boundary-value problems, including Transverse Electric (TE_z) resonant cavity eigenvalue spectra under div-curl Ritz-Galerkin penalty regularization, asymptotic convergence benchmarking against canonical Edge FEM across six mesh refinement levels, and spectral validation directly on irregular point clouds with random director orientations across six refinement levels.
3. **Limitations and Scope of Applicability (Section 5.7):** Focused on establishing honest methodological boundaries, discussing background quadrature dependencies, extreme point-cloud anisotropy, penalty parameter sensitivity, and high-frequency dispersion.

Normalization Scale Conventions across Numerical Benchmarks. As established in Section 3.3.1, formulating the support selection criterion via the dimensionless matrix $\widehat{\mathbf{A}}$ requires specifying a local length scale h_{local}. By virtue of the quartic identity (52) ($h_{\text{local}} \to c\, h_{\text{local}} \implies |\det(\widehat{\mathbf{A}})| \to c^{-4} |\det(\widehat{\mathbf{A}})|$), while support qualification under a fixed threshold $\widehat{\text{Tol}}$ is strictly invariant with respect to spatial domain refinement ($h \to 0$), the calibrated numerical values of $|\det(\widehat{\mathbf{A}})|$ and chosen thresholds (such as $\widehat{\text{Tol}} = 0.80$) depend directly on the convention adopted for h_{local}. To guarantee exact reproducibility and transparency across all numerical benchmarks reported in this section:

- **Structured and conforming benchmarks (Sections 5.1, 5.4, and 5.5):** The normalization scale is set uniformly to the nominal grid spacing $h_{\text{local}} = h_{\text{nom}} = L/(N_x - 1)$;

- **Stochastically perturbed clouds and parametric sensitivity sweeps (Sections 5.2 and 5.3, Table 4):** The scale is set to the nominal boundary spacing $h_{\text{local}} = h_{\text{nom}} = 4\pi/n_{\text{front}}$ ($h_{\text{nom}} = 0.0654\,\text{m}$ on the ultra-dense grid $N = 4{,}192$). In such quasi-uniform perturbed distributions, adopting $h_{\text{local}} = h_{\text{nom}}$ aligns closely with the canonical 6-neighbor metric (46), which evaluates to $h_{\text{local}}(\mathbf{x}_P) \approx (0.90\text{–}1.02)h_{\text{nom}}$ throughout the domain interior;[1]

- **Arbitrary irregular point clouds without background discretization (Section 5.6):** In unconstrained point distributions without a predefined nominal grid scale, the canonical 6-neighbor mean $h_{\text{local}}(\mathbf{x}_P) = \frac{1}{6}\sum_{k=1}^{6} \|\mathbf{x}_{(k)} - \mathbf{x}_P\|$ (Eq. (46)) provides a deterministic, pointwise-adaptive scale.

5.1. Local Constructive Equivalence under Conforming Midpoint/Tangent Placement: VNMM $\mathcal{L}^1$ vs. Whitney EFEM

To establish a rigorous baseline connecting the proposed vector meshless formulation to established finite element theory, this first experiment evaluates the local vector interpolation capacity of the incomplete 3-node basis $\mathcal{L}^1$ against lowest-order Whitney 1-form edge elements (Nédélec 1st-kind) [7, 3].

5.1.1. Experimental Setup and Dual Topologies

The benchmark test considers the TE_{11} resonant mode over the square cavity domain $\Omega = [0, \pi] \times [0, \pi]$, governed by the exact analytical field [2, 1]:

$$\vec{E}_{\text{ex}}(x, y) = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \qquad (\nabla \times \vec{E}_{\text{ex}})_z(x, y) = -2\cos(x)\cos(y) \quad (76)$$

To compare both formulations under identical spatial sampling, a sequence of seven conforming triangular meshes (4×4 to 32×32) is generated with controlled stochastic vertex perturbation (10% jitter) to ensure arbitrary, non-equilateral simplices. All interpolation errors are evaluated across a fixed, invariant grid of 1,600 internal Cartesian points (40×40).

The geometric arrangement follows the dual configuration established in Section 2.2.3 and illustrated in Fig. 3. For each triangular cell T, VNMM

[1] Specifically, on a Cartesian grid of spacing h_{nom}, the 6-neighbor mean ranges analytically from $(4+\sqrt{2})h_{\text{nom}}/6 \approx 0.902\,h_{\text{nom}}$ at grid nodes to $(2\sqrt{2}+\sqrt{10})h_{\text{nom}}/6 \approx 0.999\,h_{\text{nom}}$ at cell centers, extending up to $\approx 1.02\,h_{\text{nom}}$ under 25% spatial jitter.

nodes are placed at the edge midpoints M_k with tangent directors $\vec{t}_k$, mapping discrete degrees of freedom according to the dual scaling law $e_{\text{fem},k} = L_k e_{\text{vnmm},k}$ (Eq. (18)).

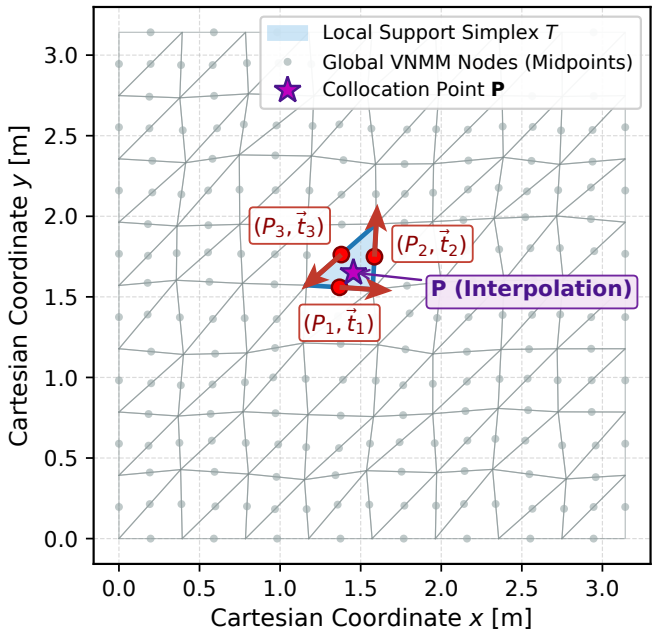


Figure 3: Conforming dual topology representation over the square domain $\Omega = [0, \pi] \times [0, \pi]$, displaying the underlying triangular finite element mesh alongside the dual VNMM $\mathcal{L}^1$ vector nodal cloud. Vector nodes are co-located at edge midpoints M_k with unit directional vectors $\vec{t}_k$ aligned along the triangle edges. This exact midpoint co-location guarantees that support triads $\{P_1, P_2, P_3\}$ form local triangles matching the EFEM cells, establishing pointwise local dual equivalence.

5.1.2. Numerical Identity and Asymptotic Convergence Rates

The asymptotic convergence behavior comparing the VNMM 2D $\mathcal{L}^1$ and Whitney EFEM formulations across the seven conforming mesh levels is presented in Fig. 4.

As demonstrated in Fig. 4a, the electric field RMS errors $\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$ for both formulations display an identical first-order asymptotic decay rate of $\mathcal{O}(h^{1.00})$. The point-by-point field difference between the two methods remains strictly bounded by double-precision floating-point machine precision, satisfying $\|\vec{E}^{\text{VNMM}} - \vec{E}^{\text{EFEM}}\|_{\text{RMS}} \leq 1.6 \times 10^{-16}$.

Similarly, Fig. 4b illustrates the curl operator RMS error $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\mathrm{RMS}}$, which exhibits an identical fitted convergence rate of $\mathcal{O}(h^{0.96})$ for both formulations. The mutual curl field difference is likewise bounded by machine precision, satisfying $\|(\nabla \times \vec{E})^{\mathrm{VNMM}} - (\nabla \times \vec{E})^{\mathrm{EFEM}}\|_{\mathrm{RMS}} \leq 2.5 \times 10^{-15}$. These log-log convergence curves empirically confirm the local constructive equivalence between the VNMM 2D $\mathcal{L}^1$ and Whitney 1-forms under conforming midpoint/tangent placement on triangular cells.

Crucially, this machine-precision equivalence holds strictly under conforming cell topologies where support triads match element midpoints with edge-tangent directors. On arbitrary point clouds, the VNMM ceases to be equivalent to a Whitney element, as arbitrary node coordinates, non-tangential director orientations, and dynamically shifting support stencils destroy simplex conformity, leading to the derivative stagnation examined next.

5.2. Vector Field Interpolation and Derivative Convergence in Arbitrary Meshless Clouds

To evaluate performance in practical meshless settings where nodes do not coincide with triangulation midpoints [30, 31], we assess convergence across unconstrained point distributions. This validates the modal aliasing analysis of Section 2.2.5 and confirms its resolution through the complete linear space $\mathcal{P}^1$.

5.2.1. Experimental Setup on Unconstrained Point Clouds

The interpolation performance is assessed for the analytical TE_{11} cavity mode specified in Eq. (76) over unstructured, non-conforming nodal distributions covering $\Omega = [0, \pi] \times [0, \pi]$. Boundary nodes are placed uniformly along the domain perimeter with counterclockwise unit tangent vectors, whereas internal nodes are generated via a stratified Cartesian grid with 25% stochastic spatial perturbation (*jitter*) and random directional vector orientations $\vec{t}_k = [\cos\theta_k, \sin\theta_k]^T$, where each angle θ_k is sampled independently from a uniform distribution over the interval $[0, 2\pi)$ ($\theta_k \sim \mathcal{U}(0, 2\pi)$).

To ensure strict statistical consistency, all field and derivative errors are evaluated across a fixed, invariant Cartesian grid of 1,600 internal sampling points (40×40) distributed uniformly in $[0.04\pi, 0.96\pi]^2$. The computational domain is discretized over six refinement levels, with node counts $N \in \{84, 186, 416, 884, 1{,}928, 4{,}192\}$ and average nodal spacings ranging from $h_{\mathrm{avg}} = 0.5236$ m down to 0.0654 m ($h_{\mathrm{avg}} \in \{0.52, 0.35, 0.22, 0.15, 0.10, 0.07\}$ m).

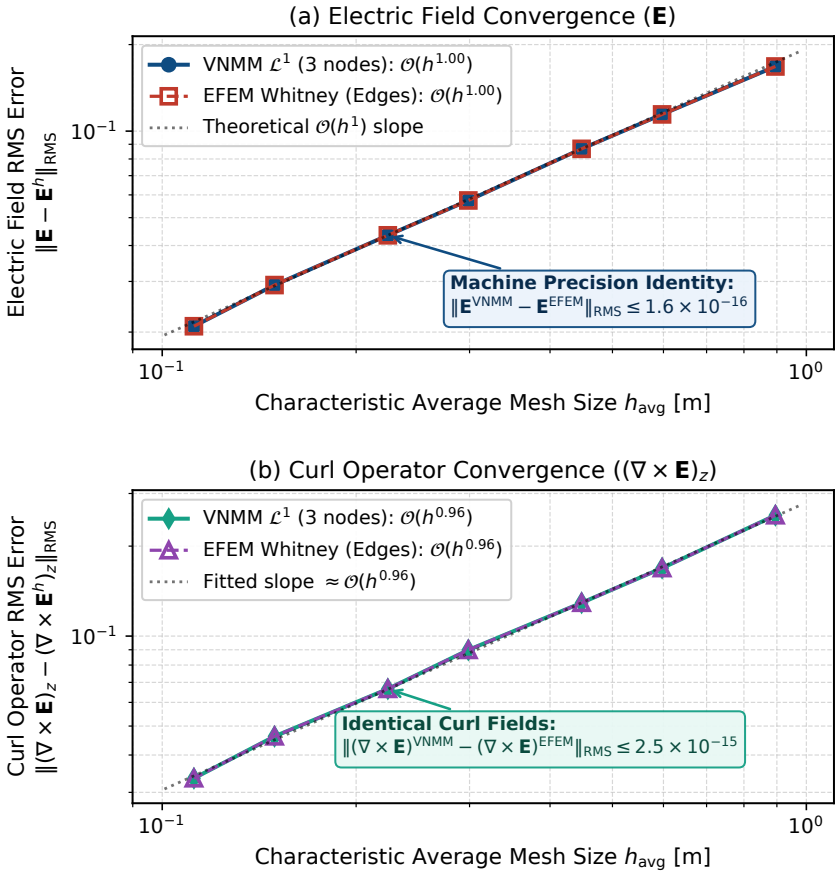


Figure 4: Asymptotic convergence rates comparing the Edge Finite Element Method (EFEM, Whitney 1-forms) and the Vector Nodal Meshless Method (VNMM $\mathcal{L}^1$, 3 support nodes) evaluated over the exact same conforming triangular mesh sequence: (a) electric field RMS error $\|\vec{E} - \vec{E}^h\|_{\mathrm{RMS}}$ exhibiting strictly identical $\mathcal{O}(h^{1.00})$ convergence with machine-precision difference $\|\vec{E}^{\mathrm{VNMM}} - \vec{E}^{\mathrm{EFEM}}\|_{\mathrm{RMS}} \leq 1.6 \times 10^{-16}$; (b) curl operator RMS error $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\mathrm{RMS}}$ displaying identical $\mathcal{O}(h^{0.96})$ convergence with mutual difference $\|(\nabla \times \vec{E})^{\mathrm{VNMM}} - (\nabla \times \vec{E})^{\mathrm{EFEM}}\|_{\mathrm{RMS}} \leq 2.5 \times 10^{-15}$, confirming the local constructive equivalence between Whitney edge elements and VNMM $\mathcal{L}^1$ under conforming midpoint/tangent placement.

5.2.2. Empirical Validation of Curl Stagnation in $\mathcal{L}^1$

When evaluated on unconstrained meshless clouds, the 3-node solenoidal basis $\mathcal{L}^1$ exhibits the fundamental breakdown in derivative convergence predicted theoretically in Section 2.2.5. As illustrated by the log-log curves in Fig. 5a, the electric field RMS error $\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$ under $\mathcal{L}^1$ decreases monotonically at a first-order rate of $\mathcal{O}(h^{0.97})$, dropping from 1.91×10^{-1} ($h = 0.5236$ m) down to 2.46×10^{-2} ($h = 0.0654$ m).

However, as revealed in Fig. 5b, the RMS error of the curl operator $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\text{RMS}}$ **stagnates at an $\mathcal{O}(1)$ error plateau** (≈ 0.85, varying between 8.33×10^{-1} and 8.88×10^{-1}, asymptotic rate $\mathcal{O}(h^{-0.01}) \approx \mathcal{O}(1)$), failing to converge under spatial refinement ($N = 4{,}192$). This persistent noise floor confirms the modal aliasing mechanism derived in Section 2.2.5, wherein unrepresented normal field derivatives leak into the rotational coefficient via the inverted placement matrix $\mathbf{A}^{-1}$.

5.2.3. Restoration of Asymptotic Convergence via Complete $\mathcal{P}^1$ Basis

Expanding the local approximation space to the complete linear polynomial vector space $\mathcal{P}^1$ introduced in Section 2.3 provides six independent degrees of freedom, capturing all four components of the local Jacobian tensor $\nabla\vec{E}^h$. Local support domains (averaging $K_{\text{avg}} \approx 11.0$ neighbors, ranging from 10.7 to 11.2) are selected via k-d tree spatial searches subject to the scale-invariant dimensionless determinant threshold developed in Section 3.3.

As demonstrated in Fig. 5, transitioning to $\mathcal{P}^1$ removes derivative modal aliasing and achieves two major performance gains:

1. **Observed Convergence for Field $\vec{E}^h$ (Fig. 5a):** Completing the linear polynomial space elevates the observed electric field convergence rate from $\mathcal{O}(h^{0.97})$ under $\mathcal{L}^1$ to an observed fitted slope of $\mathcal{O}(h^{2.17})$ under $\mathcal{P}^1$. At the finest discretization ($h = 0.0654$ m), the field RMS error drops from 2.46×10^{-2} ($\mathcal{L}^1$) down to 1.24×10^{-3} ($\mathcal{P}^1$), representing a $19.8\times$ **precision gain**. This empirical convergence acceleration highlights the substantial accuracy benefit of resolving the complete local Jacobian, in contrast to the incomplete $\mathcal{L}^1$ formulation whose missing longitudinal polynomial terms restrict error reduction.
2. **Restoration of Asymptotic $\mathcal{O}(h^1)$ Curl Convergence (Fig. 5b):** By accommodating normal derivatives in the Taylor expansion, the complete $\mathcal{P}^1$ basis restores first-order convergence $\mathcal{O}(h^{0.97})$ for the curl operator. The curl RMS error decreases monotonically from $2.55 \times$

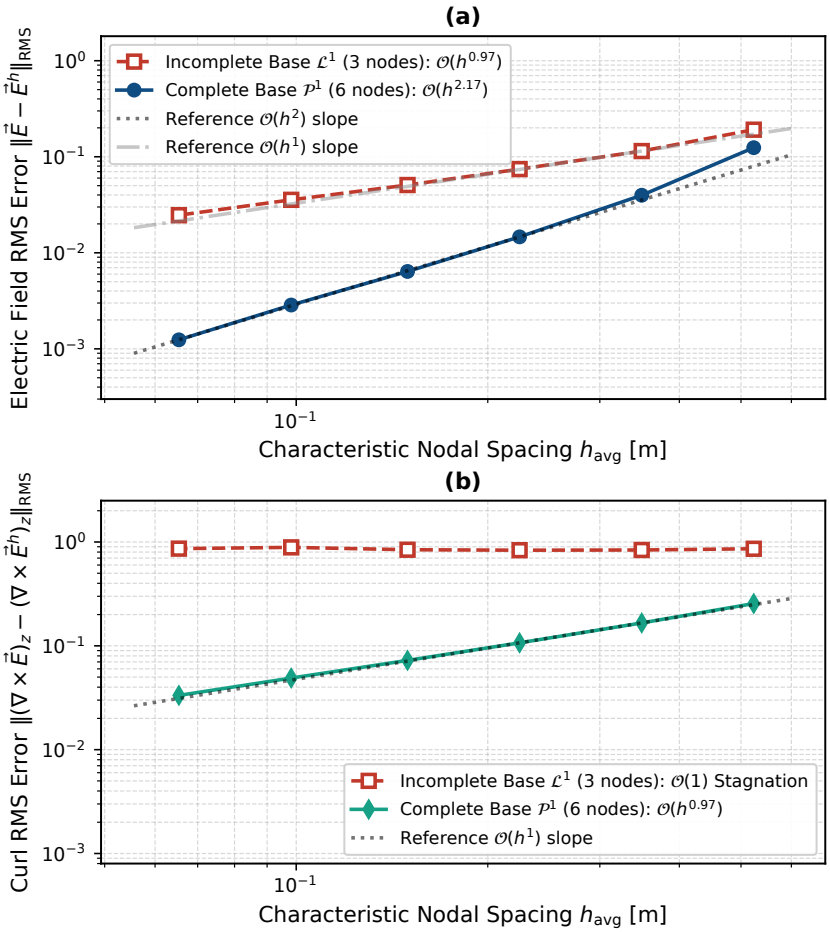


Figure 5: Log-log convergence comparison between the incomplete Whitney-like base $\mathcal{L}^1$ (3 support nodes) and the complete linear base $\mathcal{P}^1$ (6 support nodes) over $\Omega = [0, \pi] \times [0, \pi]$ evaluated across a fixed Cartesian grid of 1,600 internal points for the analytical TE_{11} cavity mode: (a) electric field RMS error displaying an observed convergence rate of $\mathcal{O}(h^{0.97})$ for $\mathcal{L}^1$ versus an observed rate of $\mathcal{O}(h^{2.17})$ for $\mathcal{P}^1$, delivering a 19.8× error reduction on the finest grid ($h = 0.0654$ m); (b) curl operator RMS error demonstrating $\mathcal{O}(1)$ error stagnation for $\mathcal{L}^1$ (plateau at ≈ 0.85 due to missing cross-derivatives), in contrast to asymptotic $\mathcal{O}(h^{0.97})$ convergence for $\mathcal{P}^1$, yielding a 25.9× error reduction.

10^{-1} down to 3.33×10^{-2} at $h = 0.0654$ m, delivering a 25.9× **error reduction** over the stagnated $\mathcal{L}^1$ formulation (8.63×10^{-1}).

Consequently, while $\mathcal{L}^1$ is suitable for conforming triangular meshes where circulation cancellation holds (Section 5.1), adopting the complete $\mathcal{P}^1$ vector space is sufficient to eliminate the derivative-aliasing mechanism and restore first-order curl convergence across unconstrained, arbitrary meshless point clouds. The quantitative performance comparison between the two bases across all six refinement levels is detailed in Table 3.

Table 3: Comparative vector interpolation error metrics and asymptotic convergence between the incomplete base $\mathcal{L}^1$ (3 support nodes) and complete base $\mathcal{P}^1$ (6 support nodes) over $\Omega = [0, \pi] \times [0, \pi]$ evaluated across a fixed grid of 1,600 internal points for the analytical TE_{11} cavity mode.

N_{total}	h_{avg} [m]	**Electric Field RMS Error**			**Curl Operator RMS Error**			K_{avg}
		$\mathcal{L}^1$	$\mathcal{P}^1$	**Gain**	$\mathcal{L}^1$	$\mathcal{P}^1$	**Gain**	($\mathcal{P}^1$)
84	0.5236	1.91×10^{-1}	1.24×10^{-1}	**1.5**×	8.61×10^{-1}	2.55×10^{-1}	**3.4**×	11.1
186	0.3491	1.15×10^{-1}	3.97×10^{-2}	**2.9**×	8.38×10^{-1}	1.66×10^{-1}	**5.0**×	11.2
416	0.2244	7.44×10^{-2}	1.47×10^{-2}	**5.1**×	8.34×10^{-1}	1.07×10^{-1}	**7.8**×	11.2
884	0.1496	5.08×10^{-2}	6.39×10^{-3}	**7.9**×	8.43×10^{-1}	7.24×10^{-2}	**11.6**×	10.9
1,928	0.0982	3.56×10^{-2}	2.86×10^{-3}	**12.5**×	8.88×10^{-1}	4.89×10^{-2}	**18.1**×	10.9
4,192	0.0654	2.46×10^{-2}	1.24×10^{-3}	**19.8**×	8.63×10^{-1}	3.33×10^{-2}	**25.9**×	10.7
Fitted Rate		$\mathcal{O}(h^{0.97})$	$\mathcal{O}(\mathbf{h^{2.17}})$	—	$\mathcal{O}(h^{-0.01})$	$\mathcal{O}(\mathbf{h^{0.97}})$	—	—

5.3. Numerical Robustness in the Tested Stochastic Realization Under Spatial-Directional Perturbations

While Section 5.2 demonstrated that the complete $\mathcal{P}^1$ space successfully eliminates derivative modal aliasing and restores asymptotic curl convergence, practical meshless computing on arbitrary unstructured point clouds demands evaluating the algebraic stability of the support qualification algorithm itself [11]. In unconstrained clouds, arbitrary spatial arrangements and director orientations can induce pathological local configurations—such as near-parallel directors (≥ 4 parallel nodes) or near-collinear nodes (≥ 5 collinear nodes)—that degrade the conditioning of the local placement matrix $\mathbf{A} \in \mathbb{R}^{6\times6}$ (Section 3.2).

To evaluate the stabilization mechanism provided by the scale-invariant dimensionless formulation ($|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$, Eq. (51), Section 3.3) within the complete $\mathcal{P}^1$ basis over $\Omega = [0, \pi] \times [0, \pi]$, we subject the point cloud sequence

to a representative benchmark scenario combining simultaneous stochastic spatial-directional distortions (25% spatial jitter and random director orientations):

1. **Stochastic Spatial Jitter:** Interior node coordinates are perturbed as $\mathbf{x}_k \leftarrow \mathbf{x}_k + \delta\mathbf{x}_k$, with independent random displacements δx_k, $\delta y_k \sim \mathcal{U}(-0.25\,\Delta x, 0.25\,\Delta x)$, introducing a 25% random spatial jitter relative to the base lattice spacing.
2. **Random Unit Director Orientations:** Unit vector directors assigned to interior nodes receive fully random angles $\theta_k \sim \mathcal{U}(0, 2\pi)$, defining $\vec{t}_k = [\cos\theta_k, \sin\theta_k]^T$, completely removing directional alignment symmetry.

Although extensive Monte Carlo characterization across multi-seed ensembles remains future work, this benchmark sequence provides a representative stochastic stress test for the automated qualification pipeline, assessing whether the scale-invariant determinant threshold can dynamically detect and bypass localized pathological candidate sextets without manual intervention. Figure 6 illustrates the resulting convergence behavior and parametric sensitivity analysis comparing an unconstrained permissive threshold against the stabilized formulation with scale-invariant thresholding.

5.3.1. Degradation Under Overly Permissive Thresholds ($\widehat{\text{Tol}} \ll 1$)

To demonstrate the critical role of scale-invariant support filtering, we first examine the behavior of the qualification algorithm when an excessively relaxed dimensionless threshold is applied (e.g., $\widehat{\text{Tol}} \leq 0.05$).

Under stochastic spatial-directional noise, an overly permissive threshold causes numerical stagnation:

- At $\widehat{\text{Tol}} \leq 0.05$, the early-stopping qualification accepts the immediate 6 nearest spatial neighbors ($K_{\text{avg}} \approx 6.3$).
- Because node coordinates and directors are stochastically perturbed, these immediate spatial neighbors frequently form ill-conditioned sextets approaching the singularity modes analyzed in Section 3.2 (such as 4 near-parallel directors or 5 near-collinear nodes), elevating the intrinsic condition number to $\kappa_2(\widehat{\mathbf{A}}) > 13$ (corresponding to unscaled $\kappa_2(\mathbf{A}) > 180$).

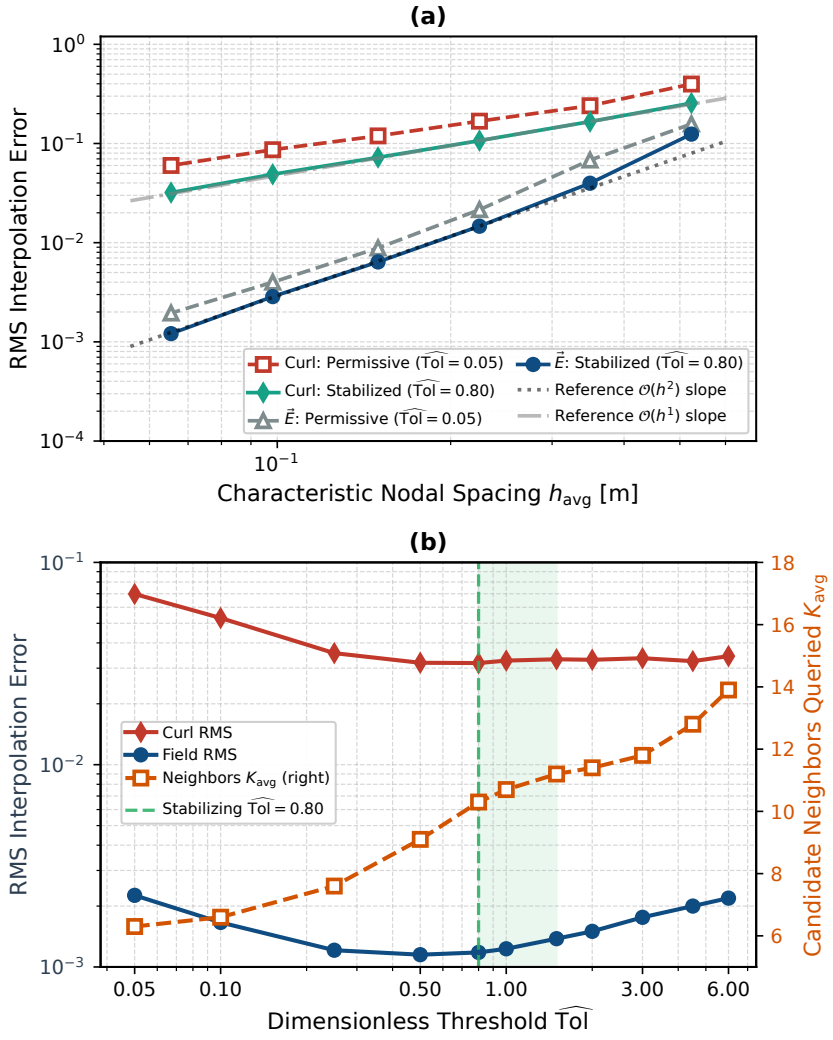


Figure 6: Numerical robustness and conditioning analysis in the tested stochastic realization for the 2D VNMM complete base $\mathcal{P}^1$ under spatial-directional perturbations (uniform nodal jitter of 25% and random director angles $\theta_k \in [0, 2\pi)$) over $\Omega = [0, \pi]^2$: (a) Convergence under stochastic mesh refinement comparing an overly permissive threshold ($\widehat{\text{Tol}} = 0.05$), highlighting curl degradation due to ill-conditioned sextets ($\kappa_2(\widehat{\mathbf{A}}) > 13$, unscaled $\kappa_2(\mathbf{A}) > 180$), against restored strict monotonic convergence when applying the scale-invariant threshold $\widehat{\text{Tol}} = 0.80$; (b) Interpolation error and candidate search neighborhood K_{avg} as a function of the dimensionless threshold $\widehat{\text{Tol}}$ for the fine grid ($N = 4{,}192$), identifying the stable operating regime for $\widehat{\text{Tol}} \in [0.80, 1.50]$.

- Because spatial derivatives scale analytically as $\mathcal{O}(h^{-1})$ via Eq. (58), the ill-conditioned inverted matrix $\widehat{\mathbf{A}}^{-1}$ amplifies local projection errors. As shown by the dashed crimson curve in Fig. 6a, this causes the curl operator RMS error to flatten and stagnate as $h \to 0.0654$ m, remaining elevated at an RMS error plateau of $\approx 6.01 \times 10^{-2}$ to 8.66×10^{-2} for $(\nabla \times \vec{E}^h)_z$.

5.3.2. Restoration of Monotonic Convergence via Scale-Invariant Thresholding

Enforcing a scale-invariant lower bound $\widehat{\mathrm{Tol}} \geq 0.80$ completely eliminates this derivative instability.

As demonstrated by the solid teal curves in Fig. 6a, requiring $|\det(\widehat{\mathbf{A}})| \geq 0.80$ systematically rejects candidate sextets afflicted by directional parallelism or near-collinear alignments. This prompts the candidate search to expand its local sampling radius moderately (K_{avg} expands from 6.3 to 10.3), automatically identifying well-conditioned support sextets with balanced angular diversity. By dynamically identifying an admissible support via the 4-phase selection algorithm (Section 3.3.2), the method restores strict monotonic error reduction across the tested refinement range for both field $\vec{E}^h$ and curl $(\nabla \times \vec{E}^h)_z$, reducing fine-mesh curl error to 3.18×10^{-2} on $N = 4{,}192$.

Crucially, because this criterion is formulated directly on the dimensionless placement matrix $\widehat{\mathbf{A}}$, the threshold $\widehat{\mathrm{Tol}}$ is intrinsically scale-invariant across all refinement levels. Unlike any dimensional determinant threshold, which decays as $\mathcal{O}(h^4)$ under grid refinement, the dimensionless criterion preserves compact supports ($K_{\text{avg}} \approx 6.0$ on regular grids, $K_{\text{avg}} \approx 10$–$11$ on perturbed clouds) down to arbitrarily fine grid scales without unphysical dilation.

5.3.3. Parametric Neighborhood Trade-off and Stable Operating Window

To identify the best-performing compromise between conditioning, curl error, and neighborhood size on $\Omega = [0, \pi]^2$, a systematic parametric sweep was conducted on the stochastically perturbed fine grid ($N = 4{,}192, h = 0.0654\,\text{m}$) across clean, round dimensionless thresholds ranging from $\widehat{\mathrm{Tol}} = 0.05$ up to $\widehat{\mathrm{Tol}} = 6.00$. Following the convention formalized in Section 3.3.1, h_{local} is set deterministically to the nominal characteristic spacing $h_{\text{nom}} = (4\pi)/n_{\text{front}} = 0.0654\,\text{m}$ across all evaluation points, guaranteeing exact numerical reproducibility of the placement matrices. As discussed in Section 3.3.1, because a lower bound on the determinant does not mathematically control

the condition number without a bounded proximity hypothesis ($\|\hat{\mathbf{x}}_k\| \leq C$), Table 4 provides the direct empirical qualification data evaluating $|\det(\widehat{\mathbf{A}})|$ versus $\kappa_2(\widehat{\mathbf{A}})$: it systematically reports how varying the threshold criterion $|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$ influences the condition number $\kappa_2(\widehat{\mathbf{A}})$ (and unscaled $\kappa_2(\mathbf{A})$), the candidate query size K_{avg}, and the resulting interpolation errors.

Table 4: Parametric error sensitivity, support neighborhood size K_{avg}, and placement matrix conditioning as a function of the scale-invariant dimensionless threshold $\widehat{\text{Tol}}$ for the stochastically perturbed fine grid ($N = 4{,}192, h = 0.0654\,\text{m}$) over $\Omega = [0, \pi]^2$.

Dimensionless Threshold	**Success**	**Neighbors**	**RMS Error**		**Conditioning**
$\widehat{\text{Tol}}$	**Rate**	K_{avg}	$\vec{E}^h$	$(\nabla \times \vec{E}^h)_z$	$\kappa_2(\widehat{\mathbf{A}})$ $[\kappa_2(\mathbf{A})]$
0.05 (Permissive)	100%	6.3	2.26×10^{-3}	6.98×10^{-2}	≈ 13.4 [186] (Sub-cond.)
0.10	100%	6.6	1.66×10^{-3}	5.31×10^{-2}	≈ 11.8 [162]
0.25	100%	7.6	1.21×10^{-3}	3.56×10^{-2}	≈ 9.0 [124]
0.50	100%	9.1	1.15×10^{-3}	3.19×10^{-2}	≈ 6.9 [94]
0.80 (Stable)	100%	**10.3**	$\mathbf{1.18 \times 10^{-3}}$	$\mathbf{3.18 \times 10^{-2}}$	$\approx$ **5.9** [**80**] **(Stable)**
1.00	100%	10.7	1.23×10^{-3}	3.27×10^{-2}	≈ 5.6 [75]
1.50 (Compromise)	100%	**11.2**	$\mathbf{1.38 \times 10^{-3}}$	$\mathbf{3.32 \times 10^{-2}}$	$\approx$ **5.1** [**66**] **(Stable)**
2.00	100%	11.4	1.50×10^{-3}	3.30×10^{-2}	≈ 4.6 [59]
3.00	100%	11.8	1.76×10^{-3}	3.36×10^{-2}	≈ 4.0 [50]
6.00 (Over-constrained)	100%	13.9	2.19×10^{-3}	3.44×10^{-2}	≈ 3.3 [39]

As illustrated in Fig. 6b:

1. **Stabilization Zone ($\widehat{\text{Tol}} \geq 0.80$):** Increasing $\widehat{\text{Tol}}$ from 0.05 to 0.80 produces a sharp, monotonic reduction in curl interpolation error (a 2.2× precision gain from 6.98×10^{-2} down to 3.18×10^{-2}), while bounding the intrinsic dimensionless conditioning to $\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$ (unscaled $\kappa_2(\mathbf{A}) \leq 80$) and decreasing to $\kappa_2(\widehat{\mathbf{A}}) \approx 5.1$ at the compromise threshold $\widehat{\text{Tol}} = 1.50$.
2. **Compact Support Locality and Trade-off ($K_{\text{avg}} \leq 11.2$):** This precision gain requires expanding the candidate search neighborhood from $K_{\text{avg}} = 6.3$ to only $K_{\text{avg}} = 10.3$ (at $\widehat{\text{Tol}} = 0.80$) and 11.2 (at the compromise threshold $\widehat{\text{Tol}} = 1.50$). While larger thresholds ($\widehat{\text{Tol}} \geq 3.00$) further reduce $\kappa_2(\widehat{\mathbf{A}})$ to ≈ 3.3–4.0, they do so at the cost of wider candidate search pools ($K_{\text{avg}} \to 13.9$ at $\widehat{\text{Tol}} = 6.00$) and elevated field

interpolation errors (2.19×10^{-3} vs. 1.18×10^{-3}, an 85% increase), establishing $\widehat{\text{Tol}} \in [0.80, 1.50]$ as the best-performing compromise between conditioning, curl error, and neighborhood size. Crucially, this compact neighborhood size matches that of unperturbed clouds ($K_{\text{avg}} \approx 11.0$, Table 3), demonstrating that scale-invariant thresholding effectively suppresses directional collinearities without scale-induced support dilation, thereby preserving spatial search efficiency and compact support locality without combinatorial overhead (Section 3.3.3).

While this benchmark demonstrates the mechanical action of scale-invariant thresholding across an individual perturbed refinement sequence, a full Monte Carlo characterization involving ensembles of tens of stochastic realizations per refinement level constitutes an important avenue for future statistical validation of generalized unstructured cloud generators.

5.4. Spectral Validation: Resonant Cavity Analysis and Div-Curl Regularization

Having established the variational framework for the electromagnetic vector eigenvalue problem in Section 4, this experiment evaluates the spectral accuracy of the Vector Nodal Meshless Method (VNMM 2D) and verifies the suppression of non-physical spurious modes across the operational frequency band. Numerical simulations are conducted for Transverse Electric (TE_z) modes over the square PEC cavity using the complete $\mathcal{P}^1$ basis, comparing discrete spectra against the analytical reference eigenvalues defined in Eq. (64).

Figure 7 presents the discrete eigenvalue spectra ($\lambda = k_c^2$), contrasting the unregularized formulation against the div-curl regularized formulation ($s_{\text{div}} = 6.0$) developed in Eq. (70).

Numerical evaluations are conducted on the base-case regular discretization with $N_x = 21, N_y = 21$ ($N_{\text{total}} = 441$ nodes, yielding $M_{\text{active}} = 361$ internal degrees of freedom after enforcing homogeneous tangential PEC boundary conditions $e_k = 0$ on $\partial\Omega$). The regular Cartesian grid over $\Omega = [0, L_x] \times [0, L_y] = [0, \pi]^2$ is defined by nodal coordinates:

$$\mathbf{x}_{i,j} = \left(i\frac{L_x}{N_x - 1},\ j\frac{L_y}{N_y - 1} \right), \quad i = 0, \ldots, N_x - 1, \quad j = 0, \ldots, N_y - 1. \tag{77}$$

Boundary nodes ($i \in \{0, N_x - 1\}$ or $j \in \{0, N_y - 1\}$) are aligned tangentially to $\partial\Omega$ ($\vec{t}_{i,j} = [1, 0]^T$ on horizontal walls $j \in \{0, N_y - 1\}$ and $\vec{t}_{i,j} = [0, 1]^T$

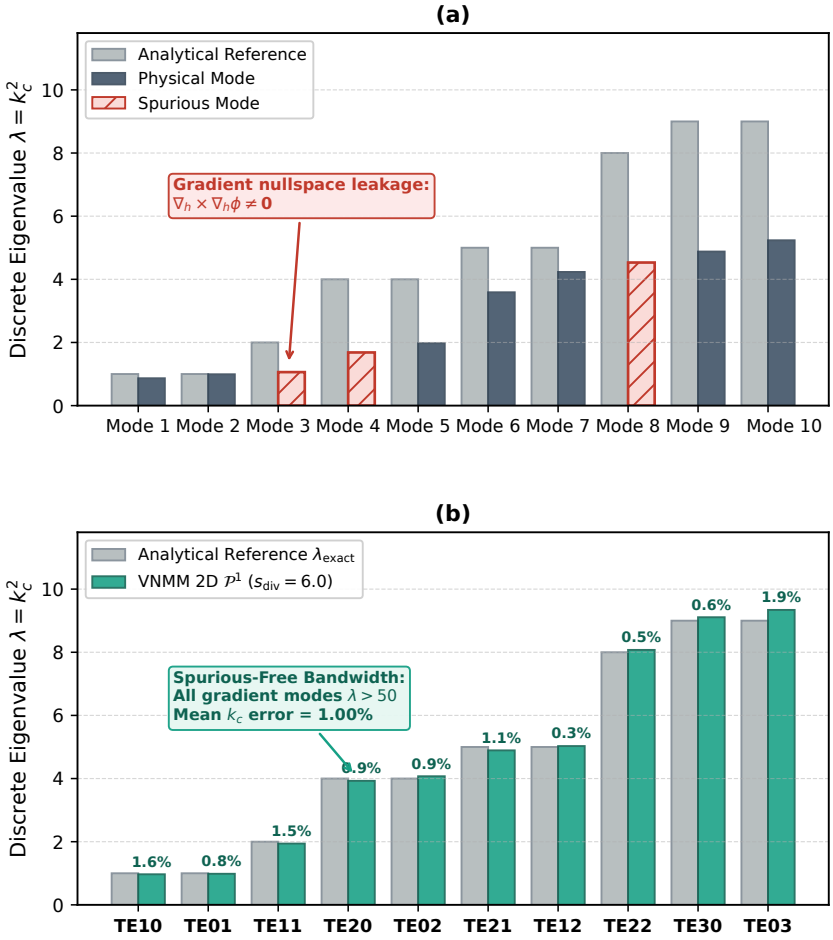


Figure 7: Discrete eigenvalue spectrum $\lambda = k_c^2$ for the TE_z square PEC cavity: (a) Unregularized curl-curl formulation ($s_{\text{div}} = 0.0$), illustrating the severe spectral contamination caused by discrete gradient nullspace leakage ($\nabla \times (\nabla \phi) \neq \mathbf{0}$), where non-zero spurious modes infiltrate the physical sequence; (b) Div-curl regularized formulation ($s_{\text{div}} = 6.0$), demonstrating effective spectral purification across the useful physical bandwidth with no spurious modes observed among the tested resonances, shifting gradient modes to $\lambda > 50$ for the tested cavity discretization and yielding a mean cutoff wavenumber error of 1.00% under baseline $10 \times 10, 2 \times 2$ Gauss integration (which benefits from pre-asymptotic under-integration softening relative to the true spatial discretization floor of 3.32%, as investigated in Section 5.4.4).

on vertical walls $i \in \{0, N_x - 1\}$), enabling direct nodal collocation of homogeneous PEC conditions ($\vec{E}^h(\mathbf{x}_k) \cdot \vec{t}_k = 0$) via $e_k = 0$. For interior nodes ($i \in \{1, \ldots, N_x - 2\}$, $j \in \{1, \ldots, N_y - 2\}$), unit director vectors $\vec{t}_{i,j}$ are assigned following an alternating orientation pattern, governed by the exact mathematical rule:

$$\vec{t}_{i,j} = \begin{cases} \begin{bmatrix} \cos(45^\circ) \\ \sin(45^\circ) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}, & \text{if } (i+j) \text{ is even,} \\ \begin{bmatrix} \cos(135^\circ) \\ \sin(135^\circ) \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}, & \text{if } (i+j) \text{ is odd.} \end{cases} \tag{78}$$

This checkerboard alternating pattern guarantees local orthogonal angular diversity: every internal node is surrounded by nearest neighbors whose assigned directors are mutually orthogonal ($\vec{t}_{45^\circ} \cdot \vec{t}_{135^\circ} = 0$), preventing directional parallelism and ensuring that any candidate 6-node stencil contains balanced projection moments.

The weak form integrals are computed over a background grid of 10×10 quadrilateral cells ($N_c = 100$) using 2×2 Gauss-Legendre quadrature (400 Gauss points), maintaining a suitable quadrature-to-DoF ratio ($N_g/N_{\text{DoF}} \approx 1.11$) [14, 12, 10]. At each Gauss integration point P_g, active support sextets are determined via the scale-invariant qualification criterion $|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$ (Section 3.3.1), with the threshold set to the stabilized operating value $\widehat{\text{Tol}} = 0.80$ identified in Section 5.3. On this structured Cartesian lattice with the alternating unit directors defined in Eq. (78), the geometric symmetry and orthogonal angular diversity of immediate neighbors naturally generate well-conditioned placement matrices with $|\det(\widehat{\mathbf{A}})| \geq 1.00$ across all 400 quadrature points (mean $|\det(\widehat{\mathbf{A}})| \approx 3.44$). Consequently, the selection algorithm identifies compact nearest-neighbor stencils ($K_{\text{avg}} = 7.10$) without support dilation, ensuring that the discrete system matrices $\mathbf{K}$ and $\mathbf{M}$ remain identically invariant across the entire stable threshold range $\widehat{\text{Tol}} \leq 1.00$ (such as $\widehat{\text{Tol}} \in [0.10, 0.80]$).

5.4.1. Spectral Contamination in the Unregularized Curl-Curl Formulation ($s_{\text{div}} = 0.0$)

As analyzed theoretically in Section 4.2, the point-collocated meshless approximation does not inherit the exactness of the discrete de Rham sequence. Unlike Edge Finite Element Methods (EFEM) that achieve closed-contour cir-

culation cancellation, discrete gradient fields evaluated in unregularized meshless weak forms generate non-zero numerical curl residues ($\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$ at Gauss quadrature points).

The empirical impact of this nullspace leakage is clearly observed in Fig. 7a. Setting $s_{\text{div}} = 0.0$ transforms analytical zero-eigenvalues belonging to $\ker(\nabla\times)$ into small, positive non-zero states ($0.8 < \lambda < 5.0$). These non-physical spurious gradient modes infiltrate the physical TE_z sequence, corrupting modal ordering and elevating the mean relative cutoff wavenumber error to unacceptable levels (19.75% to 90.0%).

5.4.2. Spectral Purification via Div-Curl Penalty Regularization ($s_{\text{div}} = 6.0$)

To eliminate the non-physical spurious gradient modes identified in Section 5.4.1 without introducing auxiliary Lagrange multipliers or expanding the system matrix, the div-curl variational penalty parameter is activated with $s_{\text{div}} = 6.0$ in Eq. (70).

As established in Section 4.2, the divergence penalty matrix $\mathbf{K}_{\text{div}}$ operates selectively between physical and spurious subspaces. While physical TE_z cavity modes are divergence-free at the continuous level, discrete approximation residues ($\nabla \cdot \vec{E}^h \sim \mathcal{O}(h)$) generate only a minor eigenvalue perturbation that decreases with refinement, with the divergence-to-curl energy ratio remaining bounded at $R_{\text{div}} = \frac{\mathbf{e}^T \mathbf{K}_{\text{div}} \mathbf{e}}{\mathbf{e}^T \mathbf{K}_{\text{curl}} \mathbf{e}} \sim 10^{-3}$–$10^{-2}$ for the first 10 physical modes (with R_{div} decreasing approximately linearly with h over the tested refinement range). In contrast, for non-physical gradient states ($\vec{E} = \nabla\phi$), this ratio reaches $R_{\text{div}} \sim 10^3$–10^4 in the unregularized system; for the tested cavity discretization and choice of $s_{\text{div}} = 6.0$, this imparts immense artificial numerical stiffness that shifts the discrete non-physical gradient states outside the operational frequency window of the tested setup ($\lambda > 50$).

The experimental confirmation of this selective purification is demonstrated in Fig. 7b. Activating $s_{\text{div}} = 6.0$ ensures that no spurious modes are detected among the first ten computed resonances within the tested operational spectral window ($\lambda \leq 10$), removing non-physical gradient branches from the low-frequency discrete spectrum and restoring 1-to-1 modal correspondence with the analytical reference eigenvalues defined in Eq. (64).

5.4.3. Modewise Validation and Cutoff Wavenumber Precision

The numerical performance of the regularized VNMM 2D ($\mathcal{P}^1$, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$) for the first 10 physical TE_{nm} cavity modes is detailed in Table 5.

Table 5: Comparison of the first 10 physical TE_{nm} resonant modes in the square PEC cavity $\Omega = [0, \pi]^2$ between analytical reference values and the proposed regularized VNMM 2D formulation ($\mathcal{P}^1$, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$, 361 active DoFs).

Mode (TE_{nm})	$\lambda_{\text{analytical}}$	$k_{c,\text{analytical}}$	λ_{VNMM}	$k_{c,\text{VNMM}}$	**Error** λ (%)	**Error** k_c (%)
TE_{10}	1.00	1.000	0.9679	0.984	3.21%	**1.62**%
TE_{01}	1.00	1.000	0.9836	0.992	1.64%	**0.82**%
TE_{11}	2.00	1.414	1.9409	1.393	2.95%	**1.49**%
TE_{20}	4.00	2.000	3.9275	1.982	1.81%	**0.91**%
TE_{02}	4.00	2.000	4.0724	2.018	1.81%	**0.90**%
TE_{21}	5.00	2.236	4.8925	2.212	2.15%	**1.08**%
TE_{12}	5.00	2.236	5.0287	2.242	0.57%	**0.29**%
TE_{22}	8.00	2.828	8.0743	2.842	0.93%	**0.46**%
TE_{30}	9.00	3.000	9.1089	3.018	1.21%	**0.60**%
TE_{03}	9.00	3.000	9.3409	3.056	3.79%	**1.88**%
Global Mean	—	—	—	—	**2.01**%	**1.00**%

The regularized VNMM 2D achieves a global mean cutoff wavenumber error of 1.00% (with a maximum error of 1.88% for the TE_{03} mode), strictly matching analytical degeneracies (TE_{10}/TE_{01}, TE_{20}/TE_{02}, TE_{21}/TE_{12}, TE_{30}/TE_{03}) without spurious mode infiltration. Importantly, parametric sensitivity evaluations confirm that this spectral purification remains effective across a stable penalty window $s_{\text{div}} \in [4.0, 10.0]$, verifying that the formulation does not require delicate, problem-dependent parameter tuning.

5.4.4. Sensitivity to Background Cell Density and Gauss Quadrature Order

To address the dependency on background integration scaffolding and assess the sensitivity of spectral predictions to quadrature resolution, we conduct a parametric sensitivity analysis over the base cavity discretization ($N = 21 \times 21$, 361 active DoFs). Table 6 details the resulting mean and maximum cutoff wavenumber errors, relative Frobenius matrix differences against an ultra-refined reference integration ($\mathbf{K}_{\text{ref}}, \mathbf{M}_{\text{ref}}$ computed on 40×40 cells with 4×4 Gauss points, $N_g = 25{,}600$), and global assembly times.

The parametric results in Table 6 reveal an apparent paradox that warrants careful physical and numerical interpretation: while refining the background quadrature reduces the relative Frobenius errors in both the stiffness matrix $\mathbf{K}$ and mass matrix $\mathbf{M}$ toward zero (e.g., from $\approx$ 55–77% at $10 \times 10, 2 \times 2$ down to $\approx$ 3–8% at $40 \times 40, 2 \times 2$), the global cutoff wavenumber error increases from 1.00% to an asymptotic plateau of $\approx 3.32\%$.

Table 6: Sensitivity of resonant cavity spectral accuracy, discrete operator convergence, and assembly cost to background cell discretization and Gauss quadrature order ($N = 21 \times 21$, 361 active DoFs, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$, reference grid: 40×40 cells, 4×4 Gauss).

Background Grid	**Gauss Rule**	**Gauss Pts**	**Wavenumber Error**		**Relative Matrix Difference**		**Assembly**
($N_{cx} \times N_{cy}$)	($p \times p$)	(N_g)	**Mean** k_c (%)	**Max** k_c (%)	$\frac{\|\mathbf{K}-\mathbf{K}_{\text{ref}}\|_F}{\|\mathbf{K}_{\text{ref}}\|_F}$	$\frac{\|\mathbf{M}-\mathbf{M}_{\text{ref}}\|_F}{\|\mathbf{M}_{\text{ref}}\|_F}$	**Time (s)**
10×10 (100 cells)	2×2 (4 pts)	400	**1.00**%	**1.88**%	5.49×10^{-1}	7.68×10^{-1}	**0.06** s
10×10 (100 cells)	3×3 (9 pts)	900	2.08%	6.30%	6.28×10^{-1}	3.10×10^{-1}	0.10 s
10×10 (100 cells)	4×4 (16 pts)	1600	2.55%	5.91%	5.16×10^{-1}	2.51×10^{-1}	0.14 s
20×20 (400 cells)	2×2 (4 pts)	1600	2.98%	7.18%	4.11×10^{-1}	1.63×10^{-1}	0.12 s
20×20 (400 cells)	4×4 (16 pts)	6400	3.33%	6.73%	1.09×10^{-1}	4.08×10^{-2}	0.44 s
30×30 (900 cells)	4×4 (16 pts)	14400	3.21%	6.38%	7.83×10^{-2}	5.36×10^{-2}	0.90 s
40×40 (1600 cells)	2×2 (4 pts)	6400	3.34%	6.67%	7.91×10^{-2}	3.31×10^{-2}	0.42 s
40×40 (1600 cells)	4×4 (16 pts)	25600	3.32%	6.39%	Reference	Reference	1.61 s

This behavior is governed by four coupled mechanisms:

1. **True Spatial Discretization Error Floor:** On a fixed nodal discretization ($N = 21 \times 21$, $h = \pi/20 = 0.157\,\text{m}$), the spatial polynomial interpolation and the div-curl penalty parameter ($s_{\text{div}} = 6.0$) render the continuous variational formulation slightly over-stiff. In the fully converged quadrature limit ($N_c \geq 30, p \geq 4$), the computed numerical eigenvalues are systematically higher than the analytical reference eigenvalues across all modes (e.g., $\lambda_1 = 1.011 > 1.000$, $\lambda_3 = 2.056 > 2.000$, and $\lambda_{10} = 10.186 > 10.000$), unmasking an intrinsic spatial discretization error floor of 3.32% for cutoff wavenumbers $k_c = \sqrt{\lambda}$.
2. **Fortuitous Error Cancellation and Pre-Asymptotic Softening:** At the coarse background grid of 10×10 cells with 2×2 Gauss points ($N_g = 400$, $N_g/N_{\text{DoF}} \approx 1.11$), the cell size $dx = 2h$ exceeds the nodal spacing. In accordance with classical under-integrated Galerkin theory [14, 44], numerical under-integration across support boundaries exerts an artificial softening effect on the discrete operators, shifting the numerical eigenvalues downward (e.g., $\lambda_1 = 0.968 < 1.000$, $\lambda_3 = 1.941 < 2.000$, and $\lambda_{10} = 9.341 < 10.000$). Because the positive spatial discretization bias ($\lambda > \lambda_{\text{exact}}$) and the negative quadrature under-integration bias ($\lambda < \lambda_{\text{exact}}$) have opposing signs, they partially cancel each other at $10\times10, 2\times2$, producing an artificially low, pre-asymptotic global error of 1.00%. As quadrature is refined, this softening vanishes, uncovering the genuine 3.32% spatial discretization error.
3. **Point-Based Support Stencil Variation:** Because the point-based EFG formulation queries the k-d tree at each individual Gauss evalu-

ation point P_g, refining the quadrature points alters the spatial locations at which stencils are constructed. For $10 \times 10, 2 \times 2$, exactly 285 unique support sextets are formed across the 400 Gauss points, whereas $10 \times 10, 4 \times 4$ samples 812 unique stencils across 1,600 points.

4. **Non-Polynomial Integrand Jumps and Intra-Cell Support Discontinuities:** In point-based meshless methods, support node sextets $\{P_k, \vec{t}_k\}_{k=1}^{6}$ are selected independently at each evaluation point based on k-d tree spatial proximity and algebraic determinant qualification. Crucially, support changes do not merely occur across background cell boundaries; rather, they occur whenever the evaluation point shifts sufficiently to alter the set of nearest admissible nodes. Consequently, different Gauss integration points within the *very same* background cell frequently employ distinct support sextets (as evidenced by Mechanism 3, where 400 Gauss points across 100 cells activate 285 unique stencils, averaging nearly 3 distinct stencils per cell). This means that the weak form integrands inherently exhibit non-polynomial jumps and gradient discontinuities cutting directly through the interior of background cells [14, 44]. Gauss-Legendre quadrature achieves its nominal convergence behavior for sufficiently smooth integrands within each integration cell; internal support transitions reduce this regularity and can therefore degrade quadrature accuracy, behaving as an under-integrator that softens the discrete operators. While this softening fortuitously balances spatial stiffness at $10 \times 10, 2 \times 2$, background cells aligned with the nodal density ($N_c = N_v - 1$) combined with higher-order rules (3×3) substantially reduce the quadrature-induced bias observed in the coarse baseline. Crucially, however, because stencils can still change internally across integration points within a single cell, aligning cells with nodes does not guarantee that cells conform to regions of integrand smoothness; thus, the 3×3 Gauss rule substantially mitigates, but does not mathematically eliminate, the quadrature error. A rigorous, mathematically exact elimination of background integration errors would require specialized schemes such as stabilized conforming nodal integration (SCNI) [40] or meshless local Petrov-Galerkin (MLPG) subdomains [13].

Consequently, the baseline choice of 10×10 cells with 2×2 Gauss points ($N_g = 400$, $N_g/N_{\mathrm{DoF}} \approx 1.11$) represents an economical engineering compromise ($t_{\mathrm{ass}} = 0.06\,\mathrm{s}$) for single-mesh calculations, preserving clean spectral

pairing without spurious mode infiltration. However, because the low error (1.00%) in this coarse configuration is pre-asymptotic and assisted by under-integration softening, it is unsuited for multi-level convergence studies: freezing a coarse 10×10 cell grid while refining the nodal discretization would cause background quadrature errors to dominate and distort the observed convergence rates. Therefore, for all subsequent systematic refinement and convergence analyses (Sections 5.5 and 5.6), the background quadrature scaffolding is systematically refined in tandem with the nodal density ($N_{cx} = N_{cy} = N_v - 1$) and integrated using a high-order 3×3 Gauss-Legendre rule (9 points per cell). This substantially reduces the quadrature-induced bias observed in the coarse baseline, ensuring that the empirical convergence behavior faithfully reflects the spatial discretization properties of the meshless vector basis.

5.5. Comparative Convergence: Pure VNMM 2D vs. Edge FEM

To validate the asymptotic convergence of the proposed Vector Nodal Meshless Method against the canonical standard in computational electromagnetics, this section evaluates the resonant frequencies of the Transverse Electric (TE_z) cavity across six successive discretization levels over $\Omega = [0, \pi]^2$:

1. **Pure Edge FEM (Whitney 1-Forms):** Canonical conforming finite element solver on structured triangular meshes [3], associating exactly one tangential circuital degree of freedom with each interior triangle edge ($N_{\text{DoF}} = N_{\text{edges,int}}$).
2. **Pure VNMM 2D ($\mathcal{P}^1$ Complete Basis):** Full meshless point-collocation solver over Ω using the 6-node linear polynomial basis with div-curl regularization parameter $s_{\text{div}} = 6.0$ and scale-invariant support selection threshold $\widehat{\text{Tol}} = 0.80$ (Eq. (51), Section 3.3.1). The regular nodal grid and unit directors across all six refinement levels ($N_v \in \{15, 22, 29, 36, 43, 49\}$) are configured identically to the protocol defined in Section 5.4: tangential alignment on $\partial\Omega$ and the checkerboard alternating orientation pattern ($\theta_{i,j} \in \{45°, 135°\}$ according to Eq. (78)) for all interior nodes. Each active internal node supports exactly one scalar directional projection degree of freedom ($N_{\text{DoF}} = (N_v - 2)^2$). In accordance with the methodology established in Section 5.4.4, weak form integrals across all six refinement levels are evaluated on background cells conforming to the nodal resolution

($N_{cx} = N_{cy} = N_v - 1$) using a high-order 3×3 Gauss-Legendre quadrature rule (9 points per cell, yielding $N_g/N_{\mathrm{DoF}} \approx 9.4$–$10.4$), substantially reducing the quadrature-induced bias observed in the coarse baseline and ensuring that the benchmark captures the spatial discretization convergence trend. As established in Section 5.4, on this structured Cartesian lattice with alternating unit directors, immediate candidate neighbors satisfy $|\det(\widehat{\mathbf{A}})| \geq 1.00$ (mean 3.00–3.69) across all quadrature points, ensuring that compact nearest-neighbor stencils ($K_{\mathrm{avg}} \approx 6.5$) are selected identically for any $\widehat{\mathrm{Tol}} \leq 1.00$.

5.5.1. Resolution Metric and Degree-of-Freedom Equivalence

In Sections 5.2 and 5.4, the notation $N_x \times N_y$ denoted the underlying geometric Cartesian grid of points. However, when comparing edge-based finite elements against node-based meshless formulations, a direct identification of geometric vertex count introduces an artificial discrepancy.

In a two-dimensional structured triangular mesh generated by bisecting rectangular cells, the number of edges E scales with the number of vertices V according to Euler's planar graph formula [45]:

$$E \approx 3V. \tag{79}$$

Consequently, an Edge FEM mesh of $N \times N$ vertices generates approximately $N_{\mathrm{DoF}} \approx 3(N-1)^2$ active degrees of freedom. Conversely, a node-based meshless method generates exactly one degree of freedom per internal node, $N_{\mathrm{DoF}} = (N-2)^2 \approx V$. Comparing both methods using identical vertex grids would compare an Edge FEM problem with three times as many degrees of freedom as the meshless solver, skewing both the linear system size and the spatial resolution.

To establish a physically and computationally fair benchmark, we define an edge-equivalent nodal discretization parameter N_v for the meshless domain such that the VNMM nodal density $\rho_{\mathrm{node}} = N_{\mathrm{nodes}}/|\Omega|$ matches the active Edge FEM DoF density $\rho_{\mathrm{edge}} = N_{\mathrm{edges}}/|\Omega|$:

$$N_v = \mathrm{round}\big(\sqrt{3}\,(N-1)\big) + 1. \tag{80}$$

Furthermore, to eliminate geometric grid-topology bias across both methods, convergence is tracked as a function of the characteristic degree-of-freedom spacing:

$$h_{\mathrm{DoF}} = \sqrt{\frac{|\Omega|}{N_{\mathrm{DoF}}}}. \tag{81}$$

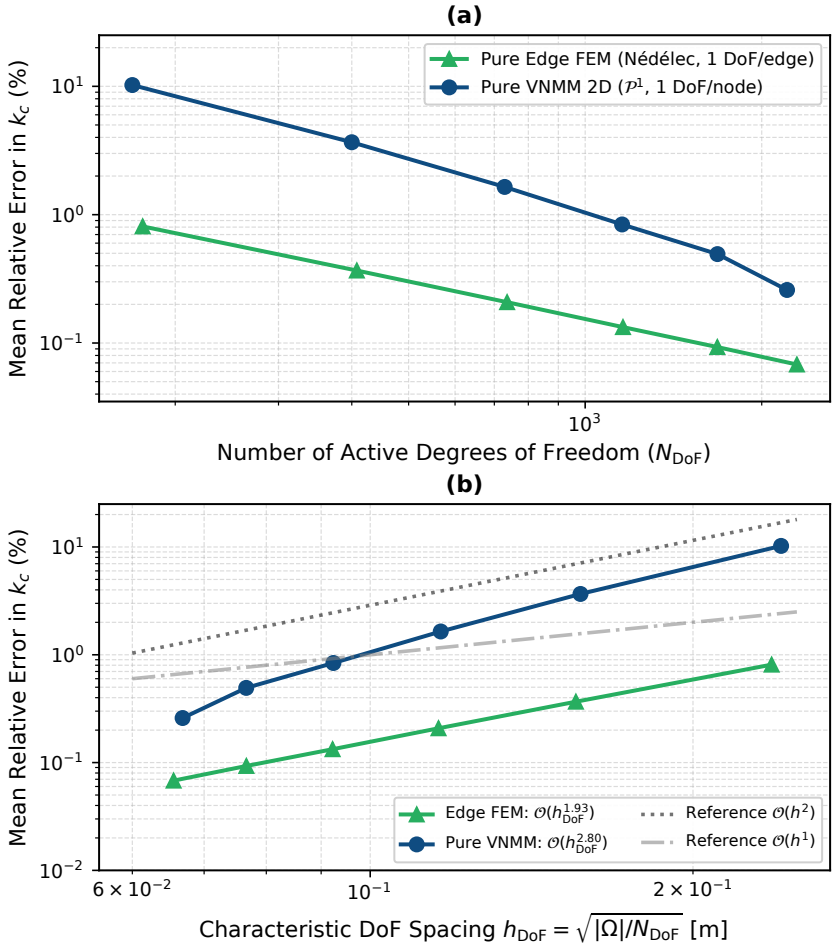


Figure 8: Comparative convergence study across 6 refinement levels ($N \in \{9, 13, 17, 21, 25, 29\}$): (a) Mean relative spectral error in cutoff wavenumber k_c (%) as a function of total active degrees of freedom N_{DoF}; (b) Convergence rate as a function of characteristic DoF spacing $h_{\mathrm{DoF}} = \sqrt{|\Omega|/N_{\mathrm{DoF}}}$. Pure Edge FEM exhibits an observed slope of $\mathcal{O}(h_{\mathrm{DoF}}^{1.93})$, while Pure VNMM exhibits an observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.80})$ over the tested refinement interval, demonstrating rapid error reduction without mesh connectivity.

5.5.2. Convergence and Accuracy Comparison

Table 7 and Fig. 8 report the convergence history across 6 successive mesh levels ($N \in \{9, 13, 17, 21, 25, 29\}$).

Inspection of the benchmark results reveals several noteworthy numerical properties:

Table 7: Comparative convergence of resonant wavenumber k_c for TE_z modes in a 2D PEC cavity over 6 mesh refinement levels between Edge FEM (Whitney 1st-kind) and Pure VNMM 2D ($\mathcal{P}^1$ complete basis, 3×3 Gauss quadrature, background cells $N_c = N_v - 1$, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$).

Mesh	**Edge FEM (Nédélec 1st-kind)**			**Pure VNMM 2D ($\mathcal{P}^1$ Complete Basis)**		
N	N_{DoF}	h_{DoF} [m]	Error (%)	N_{DoF}	h_{DoF} [m]	Error (%)
9	176	0.2368	0.810	169	0.2417	10.233
13	408	0.1555	0.367	400	0.1571	3.667
17	736	0.1158	0.208	729	0.1164	1.645
21	1160	0.0922	0.133	1156	0.0924	0.839
25	1680	0.0766	0.093	1681	0.0766	0.493
29	2296	0.0656	0.068	2209	0.0668	0.259

- **Strict Monotonicity and Quadrature Harmonization:** With the formulation of the scale-invariant dimensionless placement matrix $\widehat{\mathbf{A}}$ (Section 3.3.1), the support selection criterion $|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$ (operating at the stabilized value $\widehat{\text{Tol}} = 0.80$) naturally preserves compact nearest-neighbor stencils ($K_{\text{avg}} \approx 6.5$) on regular grids without requiring scale-dependent parameter tuning, preventing unphysical support dilation as $h \to 0$. Furthermore, employing node-conforming background cells ($N_c = N_v - 1$) with 3×3 Gauss points substantially reduces the quadrature-induced bias observed in the coarse $10 \times 10, 2 \times 2$ baseline of Section 5.4.4 (where $N = 21$ produced 1.00% error due to error cancellation). Under the refined 3×3 quadrature employed here, the corresponding level $N = 21$ ($N_v = 36$, 1156 DoFs, 35×35 cells) yields an error of 0.839%, and both Edge FEM and VNMM converge strictly monotonically toward the continuum limit as $h_{\text{DoF}} \to 0$.

- **Theoretical Error Hierarchy and Absolute Precision:** Throughout all six refinement levels, Edge FEM delivers lower absolute errors (ranging from 0.810% down to 0.068%, compared to 10.233% down to 0.259% for VNMM). This baseline accuracy advantage stems from exact closed-contour circulation cancellation via Stokes' theorem along conforming element boundaries in edge elements [7, 3].

- **Observed Convergence Rates and Pre-Asymptotic Behavior:** While Edge FEM consistently exhibits the canonical second-order rate across all levels ($\mathcal{O}(h_{\text{DoF}}^{1.93})$, with local consecutive slopes tightly bounded

between 1.88 and 2.02), the Pure VNMM solver displays an observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.80})$ over the six tested refinement levels, with an empirical slope of $\mathcal{O}(h_{\mathrm{DoF}}^{3.24})$ when restricted to the four finest meshes (consecutive two-level slopes progressing from 2.38 on $N = 9 \to 13$, to 2.67, 2.92, and 2.84).

Crucially, these steep slopes must be interpreted with caution and should *not* be construed as the theoretical asymptotic convergence order of the method. For conforming first-order affine polynomial representations $\mathcal{P}^1 \times \mathcal{P}^1$, standard conforming Galerkin variational theory bounds the asymptotic eigenvalue error by $\mathcal{O}(h^2)$ [3, 6]. In the present meshless setting, however, the global discrete space is non-conforming on unconstrained distributions ($V_h \not\subset H(\mathbf{curl}; \Omega)$) due to dynamic stencil switching, meaning that conforming spectral estimates cannot be directly invoked as predictive guarantees. The steeper empirical rate observed here reflects coupled pre-asymptotic and variational effects: (i) on coarse meshes, the div-curl penalty introduces an $\mathcal{O}(h)$ divergence error that acts as an artificial numerical stiffness, elevating coarse-grid errors (10.23% at $N = 9$); as $h \to 0$, pointwise divergence error vanishes ($\nabla \cdot \vec{E}^h \to 0$) and this penalty stiffening rapidly dissipates, producing an apparent steepening of the convergence curve; (ii) the six tested resolutions ($N = 9$–29, $h_{\mathrm{DoF}} \in [0.24, 0.07]$ m) capture a pre-asymptotic regime where overlapping support stencils dynamically settle; and (iii) the operational metric $h_{\mathrm{DoF}} = \sqrt{|\Omega|/N_{\mathrm{DoF}}}$ serves as a degree-of-freedom scaling proxy rather than a formal element diameter. Therefore, the values $\mathcal{O}(h_{\mathrm{DoF}}^{2.80})$ and $\mathcal{O}(h_{\mathrm{DoF}}^{3.24})$ represent empirical fitted slopes over the tested refinement interval rather than asymptotic convergence orders, illustrating how the meshless formulation overcomes coarse-grid penalty stiffening and closes the accuracy gap with Edge FEM as $h \to 0$.

- **Mesh-Free Discretization Advantage:** While Edge FEM maintains superior absolute accuracy for an identical DoF count on the evaluated regular geometry, the VNMM operates entirely without triangular mesh generation. This eliminates mesh quality degradation, element inversion, and remeshing bottlenecks in geometrically complex or topologically evolving electromagnetic domains.

It is important to emphasize the methodological consistency provided by the dimensionless matrix formulation: whereas any dimensional determinant

threshold would inherently couple stencil qualification to the physical mesh spacing via $\det(\mathbf{A}) \sim \mathcal{O}(h^4)$, the scale-invariant criterion $|\det(\widehat{\mathbf{A}})| \geq \widehat{\mathrm{Tol}}$ natively unifies regular grids and irregular point clouds under a single, transferable formulation. It protects stencils against degeneracies in perturbed clouds while simultaneously preserving compact supports on regular grids as $h \to 0$. Combined with the div-curl penalty regularization ($s_{\mathrm{div}} = 6.0$) investigated in Section 5.4, spurious gradient nullspace modes are shifted away from the operational spectrum in the tested cavity benchmarks, while the support selection strategy empirically bounds stencil conditioning across the tested regular grids and perturbed configurations.

5.6. Spectral Validation on Irregular Point Clouds with Arbitrary Vector Directors

While Sections 5.4 and 5.5 validated spectral accuracy and benchmarked convergence against canonical Edge FEM using regular Cartesian grids with alternating directors (Eq. (78)), assessing the practical utility of a meshless method demands solving the continuous vector boundary-value problem directly on **genuinely irregular point clouds with arbitrary unit vector directors**.

To appraise the numerical performance and robustness of the stabilized VNMM 2D ($\mathcal{P}^1$ complete basis, $s_{\mathrm{div}} = 6.0$) in the tested stochastic realization under simultaneous spatial and directional disorder, we discretize the square PEC resonant cavity $\Omega = [0, \pi]^2$ across six progressive refinement levels ($N \in \{9, 13, 17, 21, 25, 29\}$). To guarantee a strictly consistent comparison against the benchmark in Section 5.5, the point clouds are constructed using the identical edge-equivalent nodal resolution parameter $N_v = \mathrm{round}(\sqrt{3}\,(N-1)) + 1 \in \{15, 22, 29, 36, 43, 49\}$, corresponding to active interior degrees of freedom $N_{\mathrm{DoF}} = (N_v - 2)^2 \in \{169, 400, 729, 1156, 1681, 2209\}$ and characteristic DoF spacings $h_{\mathrm{DoF}} = \sqrt{|\Omega|/N_{\mathrm{DoF}}} \in [0.2417, 0.0668]\,\mathrm{m}$. The stochastic perturbation follows the protocol established in Section 5.3:

1. **Spatial Node Jitter (25%):** Interior node coordinates are perturbed as $\mathbf{x}_k \leftarrow \mathbf{x}_k + [\delta x_k, \delta y_k]^T$, with independent random displacements $\delta x_k, \delta y_k \sim \mathcal{U}(-0.25\,\Delta x, 0.25\,\Delta x)$, breaking Cartesian symmetry and producing a genuinely irregular spatial cloud.
2. **Arbitrary Director Orientations:** Interior nodes are assigned unit vector directors with fully random azimuthal angles $\theta_k \sim \mathcal{U}(0, 2\pi)$, setting $\vec{t}_k = [\cos\theta_k, \sin\theta_k]^T$.

3. **Boundary PEC Enforcement:** Boundary nodes strictly retain tangential alignment ($\vec{t}_k \parallel \partial\Omega$), enabling direct nodal collocation of homogeneous Dirichlet conditions ($\vec{E}^h(\mathbf{x}_k) \cdot \vec{t}_k = 0$) via $e_k = 0$.

To ensure complete numerical parity with the structured benchmarks of Section 5.5, numerical integration is executed under identical operational conditions: background cells aligned with the nodal spacing ($N_{cx} = N_{cy} = N_v - 1$), 3×3 Gauss-Legendre quadrature (9 points per cell), div-curl penalty $s_{\text{div}} = 6.0$, and scale-invariant dimensionless thresholding $|\det(\widehat{\mathbf{A}})| \geq \widehat{\text{Tol}}$ with $\widehat{\text{Tol}} = 0.80$.

Table 8 compiles the comparative convergence history across all six discretization levels between the structured Cartesian grid and the irregular point cloud, while Fig. 9 illustrates the convergence trajectory and scaling behavior as functions of N_{DoF} and h_{DoF}.

Table 8: Comparative spectral convergence of resonant wavenumber k_c for TE_z modes in a 2D PEC cavity over 6 refinement levels between Pure VNMM 2D on structured Cartesian grids (alternating directors) and on irregular point clouds (25% spatial jitter and random director orientations $\theta_k \in [0, 2\pi)$) under identical numerical conditions (3×3 Gauss quadrature, background cells $N_c = N_v - 1$, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$).

Mesh	**Regular Grid (Alternating Directors)**			**Irregular Cloud (25% Jitter, Random θ_k)**		
N	N_{DoF}	h_{DoF} [m]	Error (%)	N_{DoF}	h_{DoF} [m]	Error (%)
9	169	0.2417	10.233	169	0.2417	24.577
13	400	0.1571	3.667	400	0.1571	11.775
17	729	0.1164	1.645	729	0.1164	5.858
21	1156	0.0924	0.839	1156	0.0924	3.108
25	1681	0.0766	0.493	1681	0.0766	1.967
29	2209	0.0668	0.259	2209	0.0668	1.632

5.6.1. Convergence Behavior and Physical Discussion

Several critical findings emerge from the irregular point cloud eigenvalue benchmark:

1. **Strict Monotonic Convergence under Spatial Disorder:** As evidenced in Table 8 and Fig. 9a, the regularized VNMM converges strictly monotonically across all six refinement levels. The mean cutoff wavenumber error decreases from 24.58% ($N = 9$, $h_{\text{DoF}} = 0.2417\,\text{m}$) down to 1.63% ($N = 29$, $h_{\text{DoF}} = 0.0668\,\text{m}$). Across the six tested levels, linear regression yields an **observed fitted convergence slope** of

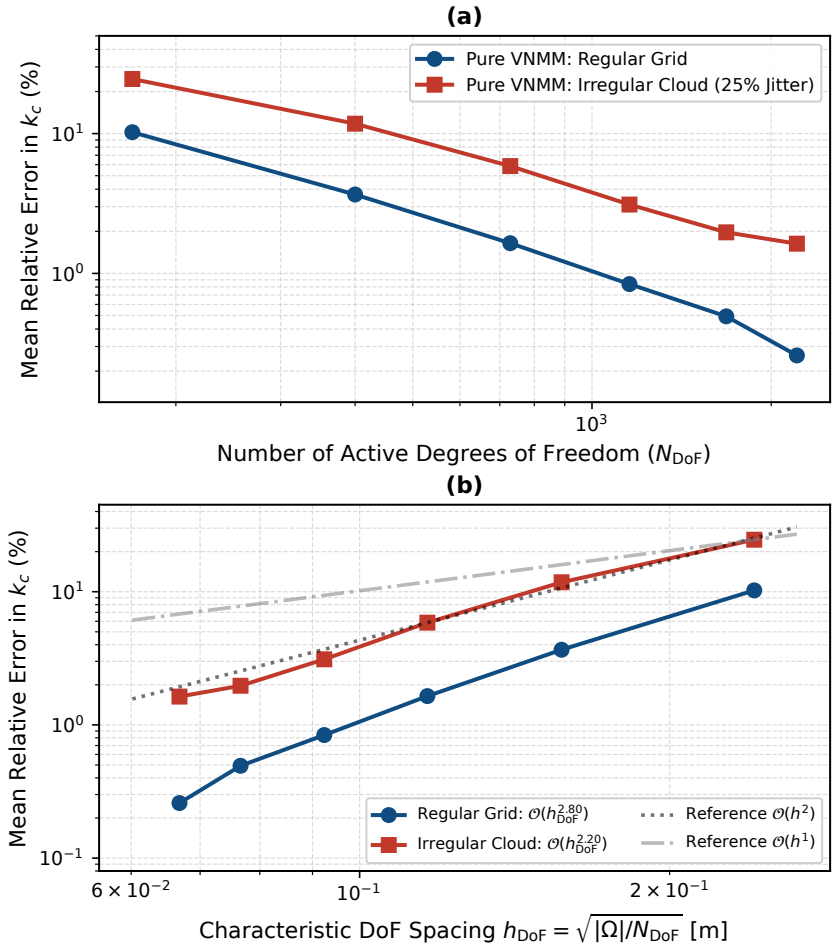


Figure 9: Comparative spectral convergence study across 6 refinement levels ($N \in \{9, 13, 17, 21, 25, 29\}$) under spatial and directional disorder: (a) Mean relative spectral error in cutoff wavenumber k_c (%) as a function of total active degrees of freedom N_{DoF}; (b) Convergence rate as a function of characteristic DoF spacing $h_{\mathrm{DoF}} = \sqrt{|\Omega|/N_{\mathrm{DoF}}}$. The regular grid exhibits an observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.80})$, while the irregular cloud (25% spatial jitter and random director orientations $\theta_k \in [0, 2\pi)$) exhibits an observed fitted slope of $\mathcal{O}(h_{\mathrm{DoF}}^{2.20})$ across the evaluated single-realization sequence, which is empirically compatible with the expected behavior of a complete linear polynomial approximation.

$\mathcal{O}(h_{\text{DoF}}^{2.20})$ as a function of the characteristic DoF spacing h_{DoF} (Fig. 9b). It is essential to emphasize that this value ($p_{\text{fitted}} = 2.20$) represents an observed fitted convergence slope obtained across six refinement levels on a single stochastic realization of a square cavity with 25% spatial jitter, random directors, $s_{\text{div}} = 6.0$, $\widehat{\text{Tol}} = 0.80$, and the specific node-aligned 3×3 Gauss quadrature scaffolding. Because the global discrete space is non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$) on arbitrary point clouds due to dynamically changing local supports, classical conforming Galerkin variational theory cannot be invoked to predict or mathematically confirm a universal asymptotic quadratic rate. Rather, this observed slope illustrates that the stabilized formulation remains numerically well-behaved and converges monotonically toward the continuum limit without spectral stagnation or algebraic locking under severe spatial-directional disorder, exhibiting an empirical rate compatible with the order expected of a complete linear polynomial representation.

2. **Error Hierarchy: Irregular vs. Regular Clouds:** Comparing the irregular cloud against the regular grid with alternating directors (Table 8 and Fig. 9) reveals an expected error hierarchy on coarse discretizations: at $N = 9$, the irregular cloud exhibits 24.58% error compared to 10.23% on the regular grid, and at $N = 21$ the error is 3.11% versus 0.84%. This discrepancy stems from two coupled physical effects: (i) on regular grids with alternating directors, symmetrical spatial arrangements induce partial cancellation of first-order Taylor truncation residuals, whereas irregular stencils lose this symmetry; and (ii) on coarse clouds, random director misalignments generate localized nonzero divergence residuals ($\nabla \cdot \vec{E}^h \sim \mathcal{O}(h)$) that, when multiplied by $s_{\text{div}} = 6.0$, introduce localized artificial numerical over-stiffening. Crucially, however, as $h_{\text{DoF}} \to 0$, this stiffening dissipates, and the error on the irregular cloud converges monotonically toward the continuum limit, reaching 1.63% at $N = 29$.
3. **Preservation of Spectral Purity without Spurious Modes:** To verify modal pairing under spatial-directional disorder, Table 9 details the first 10 computed physical TE_{nm} cavity modes for the fine irregular cloud ($N = 29$, $N_v = 49$, $N_{\text{total}} = 2401$ nodes, 2209 active DoFs). Despite the absence of grid symmetry and the completely random orientation of interior directors, the div-curl regularized weak form successfully shifts non-physical gradient nullspace modes outside the oper-

ational band ($\lambda \leq 10$). All 10 lowest eigenvalues correspond strictly to physical cavity modes in exact 1-to-1 order with analytical references, yielding a mean cutoff wavenumber error of 1.63% (with individual mode errors ranging from 0.09% for TE_{02} to 2.71% for TE_{03}, and individual eigenvalue errors ranging from 0.18% for TE_{02} to 5.50% for TE_{03}).

4. **Compact Support Locality and Search Efficiency:** Automated support selection maintains compact stencils across all irregular point clouds. Across the six evaluated refinement levels, the average number of candidate neighbors queried per Gauss point remains tightly bounded between $K_{\text{avg}} = 6.34$ and 6.71 (with the absolute maximum never exceeding $K_{\max} = 12$). Furthermore, the mean placement determinant scales quartically as $|\det(\mathbf{A})|_{\text{avg}} \sim \mathcal{O}(h^4)$, decaying smoothly from 2.00×10^{-3} at $N = 9$ to 1.56×10^{-5} at $N = 29$ without encountering rank deficiency. This demonstrates that the k-d tree candidate search identifies well-conditioned, regular sextets with minimal computational overhead in the tested realization even in the presence of severe spatial-directional perturbations.

Table 9: Discrete eigenvalue spectrum for the first 10 physical TE_{nm} cavity modes computed on an irregular point cloud with 25% spatial jitter and random director orientations on the finest tested resolution ($N = 29$, $N_v = 49$, $N_{\text{total}} = 2401$, 2209 active DoFs, $s_{\text{div}} = 6.0$).

Mode (TE_{nm})	$\lambda_{\text{analytical}}$	λ_{VNMM}	**Error** λ (%)	$k_{c,\text{analytical}}$	$k_{c,\text{VNMM}}$	**Error** k_c (%)
TE_{10}	1.00	0.9502	4.98%	1.000	0.975	**2.52%**
TE_{01}	1.00	0.9561	4.39%	1.000	0.978	**2.22%**
TE_{11}	2.00	1.9443	2.79%	1.414	1.394	**1.40%**
TE_{20}	4.00	3.9505	1.24%	2.000	1.988	**0.62%**
TE_{02}	4.00	3.9927	0.18%	2.000	1.998	**0.09%**
TE_{21}	5.00	5.0509	1.02%	2.236	2.247	**0.51%**
TE_{12}	5.00	5.1236	2.47%	2.236	2.264	**1.23%**
TE_{22}	8.00	8.4180	5.22%	2.828	2.901	**2.58%**
TE_{30}	9.00	8.5667	4.81%	3.000	2.927	**2.44%**
TE_{03}	9.00	9.4948	5.50%	3.000	3.081	**2.71%**
Global Mean	—	—	**3.26%**	—	—	**1.63%**

5.7. Limitations and Scope of Applicability

While the numerical experiments in Sections 5.1–5.6 provide evidence of convergence, conditioning stability, and spectral purity of the proposed VNMM 2D formulation within the tested configurations, a mature computational framework requires an explicit delineation of its operational scope, potential failure modes, and methodological limitations [11]. We identify seven primary boundaries of applicability:

1. **Background Quadrature Integration Dependency:** Although the local vector shape function construction is completely free of topological mesh connectivity, evaluating the global Ritz-Galerkin weak form (Section 4) relies on background quadrilateral integration cells with Gauss-Legendre quadrature (EFG-style [14, 44]). Because the point-based meshless basis functions change supports dynamically whenever the evaluation point shifts (occurring both across cell boundaries and internally within cells between adjacent quadrature points), the weak form integrands exhibit non-polynomial jumps, so standard Gauss-Legendre quadrature requires an adequate integration density to capture these non-polynomial terms accurately [44]. While the single-mesh baseline benchmark operates economically with 2×2 Gauss points on 10×10 cells ($N_g/N_{\mathrm{DoF}} \approx 1.11$, Section 5.4.4), our parametric operator analysis demonstrates that full operator convergence ($\|\mathbf{K} - \mathbf{K}_{\mathrm{ref}}\|_F / \|\mathbf{K}_{\mathrm{ref}}\|_F < 8\%$) requires cell refinement and higher-order rules (3×3 or 4×4), which unmasks the true spatial discretization floor (3.32%) previously masked by coarse under-integration softening. Consequently, systematic multi-level convergence studies (Sections 5.5 and 5.6) rely on background cells aligned with the nodal resolution ($N_{cx} = N_{cy} = N_v - 1$) and 3×3 Gauss quadrature to substantially reduce the quadrature-induced bias observed in the coarse baseline. Crucially, however, because support stencils can still change dynamically between integration points inside a cell, aligning cells with nodes does not guarantee that cells conform to regions where the integrand is smooth. For domains with highly complex, moving, or multi-scale boundaries, generating conforming background quadrature cells reintroduces a geometric pre-processing burden. Developing truly cell-free integration schemes—such as stabilized conforming nodal integration (SCNI) [40] or meshless local Petrov-Galerkin (MLPG) subdomains

[13]—remains an essential objective for fully unconstrained meshless vector solvers.

2. **Local $H(\mathbf{curl})$ Regularity versus Global Non-Conformity:** While each local vector shape function $\vec{N}_i(\mathbf{x})$ is smooth and belongs to $H(\mathbf{curl})$ within its local support domain, the assembled global discrete space is non-conforming on arbitrary point clouds ($V_h \not\subset H(\mathbf{curl}; \Omega)$). Unlike conforming Whitney edge elements, where shared edge circulations enforce exact tangential trace continuity across element facets ($\hat{n} \times [\vec{E}^h] = \mathbf{0}$), the meshless VNMM on unstructured clouds satisfies tangential continuity across dynamically changing support boundaries only in an approximate, weak sense. Consequently, the discrete approximation does not inherit the exactness of the discrete de Rham sequence; discrete gradient fields fail to maintain an identically vanishing curl under numerical quadrature ($\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$), leading to gradient nullspace leakage that requires variational div-curl regularization to prevent low-frequency spectral contamination.
3. **Pathological Stencils, Severe Anisotropy, and Support Starvation:** The automated support qualification algorithm (Section 3.3) assumes that the candidate search neighborhood contains adequate angular and spatial diversity. Local collocation matrix degeneracies in $\mathbf{A}_{6\times 6}$ manifest primarily through three geometric failure modes: (i) near-parallel directors (≥ 4 directors with mutual angles $< 5^\circ$), which eliminate linear independence among directional projection rows; (ii) near-collinear node distributions (≥ 5 nodes aligned along a line, or extreme boundary-layer aspect ratios $\Delta x / \Delta y \gg 10$), which collapse the 2D polynomial Vandermonde submatrix; and (iii) boundary and convex corner starvation, where domain truncation restricts the accessible angular half-space. Although the adaptive k-d tree candidate search pool ($K \in [8, 16]$) successfully recovers regular stencils in unconstrained clouds, severe directional clustering necessitates wider neighborhood sampling, resulting in localized support dilation and elevated condition numbers $\kappa_2(\mathbf{A})$.
4. **Scale Invariance and Transferability of the Dimensionless Matrix Criterion ($|\det(\widehat{\mathbf{A}})| \geq \widehat{\mathrm{Tol}}$):** As established in Section 3.3.1, formulating support selection via the dimensionless matrix $\widehat{\mathbf{A}}$ establishes strict scale invariance. Because local coordinates are normalized by the local nodal spacing ($\hat{x}_k = \Delta x_k / h_{\text{local}}$), the determinant

$|\det(\widehat{\mathbf{A}})| = |\det(\mathbf{A})|/h_{\text{local}}^4$ is strictly dimensionless, eliminating the quartic $\mathcal{O}(h^4)$ scaling dependence that couples algebraic qualification with grid spacing and avoiding scale-induced support dilation as $h \to 0$. While setting $\widehat{\text{Tol}} \in [0.05, 0.80]$ successfully stabilizes stencils across the tested regular and perturbed benchmark sequences, fully characterizing the optimal threshold $\widehat{\text{Tol}}$ for complex geometries with steep non-uniform nodal density gradients, local multiscale refinement, or strong directional clustering remains an important direction for future investigation.

5. **Div-Curl Penalty Sensitivity and Numerical Over-Stiffening:** The Ritz-Galerkin div-curl regularization shifts non-physical gradient nullspace modes to the high-frequency spectrum for penalty values $s_{\text{div}} \in [4.0, 10.0]$ with minimal perturbation to physical resonant frequencies ($R_{\text{div}} \leq 1\%$). However, because discrete divergence at Gauss quadrature points is only weakly enforced ($\nabla \cdot \vec{E}^h \sim \mathcal{O}(h)$), setting an excessively large penalty parameter ($s_{\text{div}} \gg 50$) exerts an artificial volumetric over-stiffening (locking) effect on solenoidal fields, deteriorating field convergence rates and inflating global matrix condition numbers. Conversely, taking $s_{\text{div}} \to 0$ allows spurious gradient modes to re-enter the physical operational bandwidth.
6. **High-Frequency Electromagnetics and Numerical Dispersion:** The eigenvalue benchmarks presented in Section 5.4 validate the spectral behavior across the lower cavity resonant modes ($\lambda \leq 20$). In high-frequency, short-wavelength regimes ($kh \gg 1$), all discrete wave solvers suffer from numerical dispersion and phase lag (the pollution effect) [1, 3]. For the VNMM, maintaining phase accuracy requires satisfying the standard spatial sampling criterion ($h \leq \lambda_{\text{EM}}/10$). Furthermore, for electrically large domains, the penalty parameter s_{div} may require frequency-dependent scaling to ensure consistent spectral separation between physical and non-physical modes.
7. **Dimensional Scope (2D Formulation) and Non-Linear Media:** The mathematical formulation and numerical experiments in this work are developed for two-dimensional electromagnetic vector fields (xy-plane, TE_z polarization). Extending the complete vector basis to three dimensions requires a 12-dimensional polynomial space $\mathcal{P}_1 \times \mathcal{P}_1 \times \mathcal{P}_1$ ($M_{\text{supp}} = 12$ support nodes per stencil) [30], which substantially expands the combinatorial candidate pool $\binom{K}{12}$ and introduces 3D surface-

integral boundary representations. Additionally, extending the regularized VNMM framework to non-linear eddy-current problems with ferromagnetic saturation (B–H curves) necessitates iterative Newton-Raphson linearizations and time-stepping schemes [46, 19].

6. Conclusion and Future Works

This work established a stabilized Vector Nodal Meshless Method (VNMM 2D) for electromagnetic vector field interpolation and eigenvalue analysis on unstructured point clouds without mesh connectivity. By resolving both the local polynomial incompleteness of early formulations and the global variational instabilities arising from mesh-free discretization, the proposed framework establishes the following core findings:

6.1. Summary of Findings and Methodological Conclusions

1. **Resolution of Local Incompleteness ($\mathcal{L}^1 \to \mathcal{P}^1$):** The $\mathcal{O}(1)$ curl error plateau in conventional solenoidal VNMM formulations was identified as a derivative aliasing mechanism caused by the incomplete $\mathcal{L}^1$ polynomial space under unconstrained point collocation. Replacing $\mathcal{L}^1$ with the complete linear vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ restores all four independent components of the local field Jacobian, eliminating derivative truncation and restoring consistent $\mathcal{O}(h)$ curl convergence on arbitrary point clouds.
2. **Scale-Invariant Support Selection ($\widehat{\mathbf{A}}$ + k-d tree):** To reject degenerate stencils on irregular clouds without topological meshing, we formulated a support qualification criterion based on the dimensionless placement matrix $\widehat{\mathbf{A}}$ ($|\det(\widehat{\mathbf{A}})| \geq \widehat{\mathrm{Tol}} > 0$). Normalizing local coordinates by the local nodal spacing h_{local} eliminates the quartic $\mathcal{O}(h^4)$ determinant decay, unifies support selection between regular and irregular grids, avoids scale-induced support dilation as $h \to 0$, and empirically bounds the local condition number ($\kappa_2(\widehat{\mathbf{A}}) \leq 5.9$) in the tested stochastic configuration.
3. **Global Non-Conformity and Gradient Leakage ($V_h \not\subset H(\mathbf{curl})$):** While VNMM is constructively equivalent to Whitney 1-forms on conforming simplices, dynamic support switching on arbitrary clouds produces an assembled space that is globally non-conforming ($V_h \not\subset H(\mathbf{curl}; \Omega)$). As a result, the approximation does not inherit the exactness of the discrete de Rham sequence ($\nabla_h \times \nabla_h \phi_h \neq \mathbf{0}$ at

quadrature points), causing irrotational gradient nullspace states to leak into the physical spectrum as spurious modes.

4. **Variational Div-Curl Regularization ($\mathbf{K}_{\text{curl}} + s_{\text{div}}\mathbf{K}_{\text{div}}$):** To suppress gradient leakage without auxiliary multipliers, we incorporated a div-curl regularized Ritz-Galerkin weak formulation. The divergence penalty selectively stiffens non-physical gradient modes, shifting them outside the operational spectral window while leaving solenoidal physical modes unperturbed ($R_{\text{div}} \ll 1$, decreasing approximately linearly with h). In the evaluated 2D PEC resonant cavity benchmarks, this restored clean modal ordering across all tested regular and perturbed configurations down to 1.63% error.
5. **Operational Scope and Validation Boundaries:** The present findings were demonstrated for 2D electromagnetic vector fields (TE_z polarization) on homogeneous PEC cavities across the evaluated regular and stochastically perturbed point-cloud realizations. While exact pointwise PEC boundary conditions are enforced directly at boundary nodes via unit tangent directors ($\vec{t}_k = \hat{\tau}(\mathbf{x}_k)$), continuous boundary traces and sharp material interface continuity remain approximate and require specialized treatment in generalized geometries.

6.2. Promising Directions for Future Research

Building upon this 2D stabilized vector formulation, several key avenues remain for future investigation:

1. **Extension to 3D Vector Formulations:** Extending the complete polynomial vector basis to three dimensions ($\mathcal{P}_1 \times \mathcal{P}_1 \times \mathcal{P}_1$, requiring $M_{\text{supp}} = 12$ nodes per stencil) to model 3D resonant cavities, waveguiding structures, and radiation problems [30].
2. **Cell-Free Quadrature and Complex Boundaries:** Replacing background integration cells with cell-free schemes—such as Stabilized Conforming Nodal Integration (SCNI) [40] or MLPG subdomains [13]—to eliminate background quadrature scaffolding and better accommodate moving, multi-scale, or curved domain boundaries.
3. **Subdomain-Partitioned Stencils for Material Interfaces:** Developing multi-tree spatial data structures [33] to restrict candidate stencils to homogeneous subdomains, preserving sharp constitutive parameter jumps ($\epsilon_{r1} \neq \epsilon_{r2}$) and weakly enforcing normal flux continuity across material discontinuities.

4. **Weak Trace Enforcement and Discontinuous Galerkin Frameworks:** Investigating interior penalty Discontinuous Galerkin (IP-DG) or hybridized trace formulations to penalize tangential trace jumps $[\hat{n} \times \vec{E}^h]$ across support boundaries, weakly restoring discrete de Rham exactness as an alternative to volumetric div-curl penalty regularization.
5. **Non-Linear Media and Motion Modeling:** Coupling regularized VNMM solvers with Newton-Raphson iterations and time-stepping schemes to handle ferromagnetic saturation (B–H curves) and electrical machine rotor motion without remeshing [19].

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used Google Antigravity / Gemini in order to assist with English language editing, LaTeX code formatting, script automation for numerical data consistency, and cross-referencing verification. After using this tool, the authors carefully reviewed, verified, and edited all resulting content and take full responsibility for the integrity and accuracy of the publication.

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